
Part One

Foundations of Value

Why Value Value?

The guiding principle of business value creation is a refreshingly simple construct: companies that grow and earn a return on capital that exceeds their cost of capital create value. Articulated as early as 1890 by Alfred Marshall,¹ the concept has proven to be both enduring in its validity and elusive in its application.

Nevertheless, managers, boards of directors, and investors sometimes ignore the foundations of value in the heat of competition or the exuberance of market euphoria. The tulip mania of the early 1600s, the dot-coms that soared spectacularly with the Internet bubble, only then to crash, and the mid-2000's real estate frenzy whose implosion touched off the financial crisis of 2007–2008 can all to some extent be traced to a misunderstanding or misapplication of this guiding principle.

At other moments, the system in which value creation takes place comes under fire. That happened at the turn of the twentieth century in the United States, when fears about the growing power of business combinations raised questions that led to more rigorous enforcement of antitrust laws. The Great Depression of the 1930s was another such moment, when prolonged unemployment undermined confidence in the ability of the capitalist system to mobilize resources, leading to a range of new policies in democracies around the world.

Today many people are again questioning the foundations of capitalism, especially shareholder-oriented capitalism. Challenges such as globalization, climate change, income inequality, and the growing power of technology titans have shaken public confidence in large corporations.² Politicians and commentators push for more regulation and fundamental changes in corporate

¹ A. Marshall, *Principles of Economics* (New York: Macmillan, 1890), 1:142.

² An annual Gallup poll in the United States showed that the percentage of respondents with little or no confidence in big business increased from 27 percent in 1997 to 34 percent in 2019, and those with “a great deal” or “quite a lot” of confidence in big business decreased by five percentage points over that period, from 28 percent to 23 percent. Conversely, those with “a great deal” or “quite a lot” of confidence in small business *increased* by five percentage points over the same period (from 63 percent in 1997 to 68 percent in 2019). For more, see Gallup, “Confidence in Institutions,” www.gallup.com.

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governance. Some have gone so far as to argue that “capitalism is destroying the earth.”³

Many business leaders share the view that change is needed to answer society’s call. In August 2019, Business Roundtable, an association of chief executives of leading U.S. corporations, released its Statement on the Purpose of a Corporation. The document’s 181 signers declared “a fundamental commitment to all⁴ of our stakeholders.”⁵ The executives affirmed that their companies have a responsibility to customers, employees, suppliers, communities (including the physical environment), and shareholders. “We commit to deliver value to all of them,” the statement concludes, “for the future success of our companies, our communities and our country.”

The statement’s focus on the future is no accident: issues such as climate change have raised concerns that today’s global economic system is short-changing the future. It is a fair critique of today’s capitalism. Managers too often fall victim to short-termism, adopting a focus on meeting short-term performance metrics rather than creating value over the long term. There also is evidence, including the median scores of companies tracked by McKinsey’s Corporate Horizon Index from 1999 to 2017, that this trend is on the rise. The roots of short-termism are deep and intertwined, so a collective commitment of business leaders to the long-term future is encouraging.

As business leaders wrestle with that challenge, not to mention broader questions about purpose and how best to manage the coalescing and colliding interests of myriad owners and stakeholders in a modern corporation, they will need a large dose of humility and tolerance for ambiguity. They’ll also need crystal clarity about the problems their communities are trying to solve. Otherwise, confusion about objectives could inadvertently undermine capitalism’s ability to catalyze progress as it has in the past, whether lifting millions of people out of poverty, contributing to higher literacy rates, or fostering innovations that improve quality of life and lengthen life expectancy.

As business leaders strive to resolve all of those weighty trade-offs, we hope this book will contribute by clarifying the distinction between creating shareholder value and maximizing short-term profits. Companies that conflate the two often put both shareholder value *and* stakeholder interests at risk. In the first decade of this century, banks that acted as if maximizing short-term profits would maximize value precipitated a financial crisis that ultimately destroyed billions of dollars of shareholder value. Similarly, companies whose short-term focus leads to environmental disasters destroy shareholder value by incurring cleanup costs and fines, as well as via lingering reputational damage. The best managers don’t skimp on safety, don’t make value-destroying decisions just

³ G. Monbiot, “Capitalism Is Destroying the Earth; We Need a New Human Right for Future Generations,” *Guardian*, March 15, 2019, www.guardian.com.

⁴ Emphasis added by Business Roundtable.

⁵ Kevin Sneader, the global managing partner of McKinsey & Company, is a signatory of the statement.

because their peers are doing so, and don't use accounting or financial gimmicks to boost short-term profits. Such actions undermine the interests of all stakeholders, including shareholders. They are the antithesis of value creation.

To dispel such misguided notions, this chapter begins by describing what value creation *does* mean. We then contrast the value creation perspective with short-termism and acknowledge some of the difficulties of value creation. We offer guidance on reconciling competing interests and adhering to principles that promote value creation. The chapter closes with an overview of the book's remaining topics.

WHAT DOES IT MEAN TO CREATE SHAREHOLDER VALUE?

Particularly at this time of reflection on the virtues and vices of capitalism, it's critical that managers and board directors have a clear understanding of what value creation means. For value-minded executives, creating value cannot be limited to simply maximizing today's share price. Rather, the evidence points to a better objective: maximizing a company's collective value to its shareholders, now and in the future.

If investors knew as much about a company as its managers do, maximizing its current share price might be equivalent to maximizing its value over time. But in the real world, investors have only a company's published financial results and their own assessment of the quality and integrity of its management team. For large companies, it's difficult even for insiders to know how financial results are generated. Investors in most companies don't know what's really going on inside a company or what decisions managers are making. They can't know, for example, whether the company is improving its margins by finding more efficient ways to work or by skimping on product development, resource management, maintenance, or marketing.

Since investors don't have complete information, companies can easily pump up their share price in the short term or even longer. One global consumer products company consistently generated annual growth in earnings per share (EPS) between 11 percent and 16 percent for seven years. Managers attributed the company's success to improved efficiency. Impressed, investors pushed the company's share price above those of its peers—unaware that the company was shortchanging its investment in product development and brand building to inflate short-term profits, even as revenue growth declined. Finally, managers had to admit what they'd done. Not surprisingly, the company went through a painful period of rebuilding. Its stock price took years to recover.

It would be a mistake, however, to conclude that the stock market is not "efficient" in the academic sense that it incorporates all public information. Markets do a great job with public information, but markets are not omniscient. Markets cannot price information they don't have. Think about the analogy of selling an older house. The seller may know that the boiler makes a weird sound every once in a while or that some of the windows are a bit

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drafty. Unless the seller discloses those facts, a potential buyer may have great difficulty detecting them, even with the help of a professional house inspector.

Despite such challenges, the evidence strongly suggests that companies with a long strategic horizon create more value than those run with a short-term mindset. Banks that had the insight and courage to forgo short-term profits during the last decade's real-estate bubble, for example, earned much better total shareholder returns (TSR) over the longer term. In fact, when we studied the patterns of investment, growth, earnings quality, and earnings management of hundreds of companies across multiple industries between 2001 and 2014, we found that companies whose focus was more on the long term generated superior TSR, with a 50 percent greater likelihood of being in the top decile or top quartile by the end of that 14-year period.⁶ In separate research, we've found that long-term revenue growth—particularly organic revenue growth—is the most important driver of shareholder returns for companies with high returns on capital.⁷ What's more, investments in research and development (R&D) correlate powerfully with long-term TSR.⁸

Managers who create value for the long term do not take actions to increase today's share price if those actions will damage the company down the road. For example, they don't shortchange product development, reduce product quality, or skimp on safety. When considering investments, they take into account likely future changes in regulation or consumer behavior, especially with regard to environmental and health issues. Today's managers face volatile markets, rapid executive turnover, and intense performance pressures, so making long-term value-creating decisions requires courage. But the fundamental task of management and the board is to demonstrate that courage, despite the short-term consequences, in the name of value creation for the collective interests of shareholders, now and in the future.

SHORT-TERMISM RUNS DEEP

Despite overwhelming evidence linking intrinsic investor preferences to long-term value creation,⁹ too many managers continue to plan and execute strategy—and then report their performance—against shorter-term measures, particularly earnings per share (EPS).

⁶ *Measuring the Economic Impact of Short-Termism*, McKinsey Global Institute, February 2017, www.mckinsey.com.

⁷ B. Jiang and T. Koller, "How to Choose between Growth and ROIC," *McKinsey on Finance*, no. 25 (Autumn 2007): 19–22, www.mckinsey.com. However, we didn't find the same relationship for companies with low returns on capital.

⁸ We've performed the same analyses for 15 and 20 years and with different start and end dates, and we've always found similar results.

⁹ R. N. Palter, W. Rehm, and J. Shih, "Communicating with the Right Investors," *McKinsey Quarterly* (April 2008), www.mckinsey.com. Chapter 34 of this book also examines the behaviors of intrinsic and other investor types.

As a result of their focus on short-term EPS, major companies often pass up long-term value-creating opportunities. For example, a relatively new CFO of one very large company has instituted a standing rule: every business unit is expected to increase its profits faster than its revenues, every year. Some of the units currently have profit margins above 30 percent and returns on capital of 50 percent or more. That's a terrific outcome if your horizon is the next annual report. But for units to meet that performance bar right now, they are forgoing growth opportunities that have 25 percent profit margins in the years to come. Nor is this an isolated case. In a survey of 400 chief financial officers, two Duke University professors found that fully 80 percent of the CFOs said they would reduce discretionary spending on potentially value-creating activities such as marketing and R&D in order to meet their short-term earnings targets.¹⁰ In addition, 39 percent said they would give discounts to customers to make purchases this quarter rather than next, in order to hit quarterly EPS targets. That's no way to run a railroad—or any other business.

As an illustration of how executives get caught up in a short-term EPS focus, consider our experience with companies analyzing a prospective acquisition. The most frequent question managers ask is whether the transaction will dilute EPS over the first year or two. Given the popularity of EPS as a yardstick for company decisions, you might think that a predicted improvement in EPS would be an important indication of an acquisition's potential to create value. However, there is no empirical evidence linking increased EPS with the value created by a transaction.¹¹ Deals that strengthen EPS and deals that dilute EPS are equally likely to create or destroy value.

If such fallacies have no impact on value, why do they prevail? The impetus for a short-term view varies. Some executives argue that investors won't let them focus on the long term; others fault the rise of activist shareholders in particular. Yet our research shows that even if short-term investors cause day-to-day fluctuations in a company's share price and dominate quarterly earnings calls, longer-term investors are the ones who align market prices with intrinsic value.¹² Moreover, the evidence shows that, on average, activist investors strengthen the long-term health of the companies they pursue—for example, challenging existing compensation structures that encourage short-termism.¹³ Instead, we often find that executives themselves or their boards are the source of short-termism. In one relatively recent survey of more than 1,000 executives and board members, most cited their own executive teams

¹⁰ J. R. Graham, C. R. Harvey, and S. Rajgopal, "Value Destruction and Financial Reporting Decisions," *Financial Analysts Journal* 62, no. 6 (2006): 27–39.

¹¹ R. Dobbs, B. Nand, and W. Rehm, "Merger Valuation: Time to Jettison EPS," *McKinsey Quarterly* (March 2005), www.mckinsey.com.

¹² Palter et al., "Communicating with the Right Investors."

¹³ J. Cyriac, R. De Backer, and J. Sanders, "Preparing for Bigger, Bolder Shareholder Activists," *McKinsey on Finance* (March 2014), www.mckinsey.com.

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and boards (rather than investors, analysts, and others outside the company) as the greatest sources of pressure for short-term performance.¹⁴

The results can defy logic. At a company pursuing a major acquisition, we participated in a discussion about whether the deal's likely earnings dilution was important. One of the company's bankers said he knew any impact on EPS would be irrelevant to value, but he used it as a simple way to communicate with boards of directors. Elsewhere, we've heard company executives acknowledge that they, too, doubt the importance of impact on EPS but use it anyway, "for the benefit of Wall Street analysts." Investors also tell us that a deal's short-term impact on EPS is not that important. Apparently, everyone knows that a transaction's short-term impact on EPS doesn't matter. Yet they all pay attention to it.

The pressure to show strong short-term results often builds when businesses start to mature and see their growth begin to moderate. Investors continue to bay for high profit growth. Managers are tempted to find ways to keep profits rising in the short term while they try to stimulate longer-term growth. However, any short-term efforts to massage earnings that undercut productive investment make achieving long-term growth even more difficult, spawning a vicious circle.

Some analysts and some short-term-oriented investors will always clamor for short-term results. However, even though a company bent on growing long-term value will not be able to meet their demands all the time, this continuous pressure has the virtue of keeping managers on their toes. Sorting out the trade-offs between short-term earnings and long-term value creation is part of a manager's job, just as having the courage to make the right call is a critical personal quality. Perhaps even more important, it is up to corporate boards to investigate and understand the economics of the businesses in their portfolio well enough to judge when managers are making the right trade-offs and, above all, to protect managers when they choose to build long-term value at the expense of short-term profits.

Improving a company's corporate governance proposition might help. In a 2019 McKinsey survey, an overwhelming majority of executives (83 percent) reported that they would be willing to pay about a 10 percent median premium to acquire a company with a positive reputation for environmental, regulatory, and governance (ESG) issues over one with a negative reputation.

¹⁴ Commissioned by McKinsey & Company and by the Canada Pension Plan Investment Board, the online survey, "Looking toward the Long Term," was in the field from April 30 to May 10, 2013, and garnered responses from 1,038 executives representing the full range of industries and company sizes globally. Of these respondents, 722 identified themselves as C-level executives and answered questions in the context of that role, and 316 identified themselves as board directors and answered accordingly. To adjust for differences in response rates, the data are weighted by the contribution of each respondent's nation to global gross domestic product (GDP). For more, see J. Bailey, V. Bérubé, J. Godsall, and C. Kehoe, "Short-termism: Insights from Business Leaders," FCLTGlobal, January 2014, <https://www.fcltglobal.org>.

Investors seem to agree; one recent report found that global sustainable investment topped \$30 trillion in 2018, rising 34 percent over the previous two years.¹⁵

Board members might also benefit from spending more time on their board activities, so they have a better understanding of the economics of the companies they oversee and the strategic and short-term decisions managers are making. In a survey of 20 UK board members who had served on the boards of both exchange-listed companies and companies owned by private-equity firms, 15 of 20 respondents said that private-equity boards clearly added more value. Their answers suggested two key differences. First, private-equity directors spend on average nearly three times as many days on their roles as do those at listed companies. Second, listed-company directors are more focused on risk avoidance than value creation.¹⁶

Changes in CEO evaluation and compensation might help as well. The compensation of many CEOs and senior executives is still skewed to short-term accounting profits, often by formula. Given the complexity of managing a large multinational company, we find it odd that so much weight is given to a single number.

SHAREHOLDER CAPITALISM CANNOT SOLVE EVERY CHALLENGE

Short-termism is a critical affliction, but it isn't the only source of today's crisis of trust in corporate capitalism. Imagine that short-termism were magically cured. Would other foundational problems suddenly disappear as well? Of course not. Managers struggle to make many trade-offs for which neither a shareholder nor a stakeholder approach offers a clear path forward. This is especially true when it comes to issues affecting people who aren't immediately involved with the company—for example, a company's carbon emissions affecting parties that may be far away and not even know what the company is doing. These so-called externalities can be extremely challenging for corporate decision making, because there is no objective basis for making trade-offs among parties.

Consider how this applies to climate change. One natural place to look for a solution is to reduce coal production used to make electricity, among the largest human-made sources of carbon emissions.¹⁷ How might the managers of a coal-mining company assess the trade-offs needed to begin solving environmental problems? If a long-term shareholder focus led them to anticipate

¹⁵ 2018 *Global Sustainable Investment Review*, Global Sustainable Investment Alliance, 2018, www.gsi-alliance.org.

¹⁶ V. Acharya, C. Kehoe, and M. Reyner, "The Voice of Experience: Public versus Private Equity," *McKinsey on Finance* (Spring 2009): 16–21.

¹⁷ In 2011, coal accounted for 44 percent of the global CO₂ emissions from energy production. CO₂ Emissions from Fuel Combustion online data service, International Energy Agency, 2013, www.iea.org.

potential regulatory changes, they would modify their investment strategies accordingly; they might not want to open new mines, for example.

With perfect knowledge a decade or even five years ago, a coal company could have reduced production dramatically or even closed mines in accordance with the decline in demand from U.S. coal-fired power plants. But perfect information is a scarce resource indeed, sometimes even in hindsight, and the timing of production changes and, especially, mine closures, would inevitably be abrupt. Further, closures would result in significant consequences even if the choice is the “right” one.

In the case of mine closures, not only would the company’s shareholders lose their entire investment, but so would its bondholders, who are often pension funds. All the company’s employees would be out of work, with magnifying effects on the entire local community. Second-order effects would be unpredictable. Without concerted action among all coal producers, another supplier could step up to meet demand. Even with concerted action, power plants might be unable to produce electricity, idling workers and causing electricity shortages that undermine the economy. What objective criteria would any individual company use to weigh the economic and environmental trade-offs of such decisions—whether they’re privileging shareholders or stakeholders?

That’s not to say that business leaders should just dismiss externalities as unsolvable or a problem to solve on a distant day. Putting off such critical decisions is the essence of short-termism. With respect to the climate, some of the world’s largest energy companies, including BP and Shell, are taking bold measures right now toward carbon reduction, including tying executive compensation to emissions targets.

Still, the obvious complexity of striving to manage global threats like climate change that affect so many people, now and in the future, places bigger demands on governments. Trading off different economic interests and time horizons is precisely what people charge their governments to do. In the case of climate change, governments can create regulations and tax and other incentives that encourage migration away from polluting sources of energy. Ideally, such approaches would work in harmony with market-oriented approaches, allowing creative destruction to replace aging technologies and systems with cleaner and more efficient sources of power. Failure by governments to price or control the impact of externalities will lead to a misallocation of resources that can stress and divide shareholders and other stakeholders alike.

Institutional investors such as pension funds, as stewards of the millions of men and women whose financial futures are often at stake, can play a critical supporting role. Already, longer-term investors concerned with environmental issues such as carbon emissions, water scarcity, and land degradation are connecting value and long-term sustainability. In 2014, heirs to the Rockefeller Standard Oil fortune decided to join Stanford University’s board of trustees in a campaign to divest shares in coal and other fossil fuel companies.

Long-term-oriented companies must be attuned to long-term changes that investors and governments will demand. This enables executives to adjust their strategies over a 5-, 10-, or 20-year time horizon and reduce the risk of holding still-productive assets that can't be used because of environmental or other issues. For value-minded executives, what bears remembering is that a delicate chemistry will always exist between government policy and long-term investors, and between shareholder value creation and the impact of externalities.

CAN STAKEHOLDER INTERESTS BE RECONCILED?

Much recent criticism of shareholder-oriented capitalism has called on companies to focus on a broader set of stakeholders beyond just its shareholders. It's a view that has long been influential in continental Europe, where it is frequently embedded in corporate governance structures. It's gaining traction in the United States as well, with the rise of public-benefit corporations, which explicitly empower directors to consider the interests of constituencies other than shareholders.

For most companies anywhere in the world, pursuing the creation of long-term shareholder value requires satisfying other stakeholders as well. You can't create long-term value by ignoring the needs of your customers, suppliers, and employees. Investing for sustainable growth should and often does result in stronger economies, higher living standards, and more opportunities for individuals.

Many corporate social-responsibility initiatives also create shareholder value.¹⁸ Consider Alphabet's free suite of tools for education, including Google Classroom, which equips teachers with resources to make their work easier and more productive. As the suite meets that societal need, it also familiarizes students around the world with Google applications—especially in underserved communities, where people might otherwise not have access to meaningful computer science education at all. Nor is Alphabet reticent about choosing *not* to do business in instances the company deems harmful to vulnerable populations; the Google Play app store now prohibits apps for personal loans with an annual percentage rate of 36 percent or higher, an all too common feature of predatory payday loans.¹⁹

Similarly, Lego's mission to "play well"—to use the power of play to inspire "the builders of tomorrow, their environment and communities"—has led to a program that unites children in rural China with their working parents.

¹⁸ S. Bonini, T. Koller, and P. H. Mirvis, "Valuing Social Responsibility Programs," *McKinsey Quarterly* (July 2009), www.mckinsey.com.

¹⁹ Y. Hayashi, "Google Shuts Out Payday Loans with App-Store Ban," *Wall Street Journal*, October 13, 2019, www.wsj.com.

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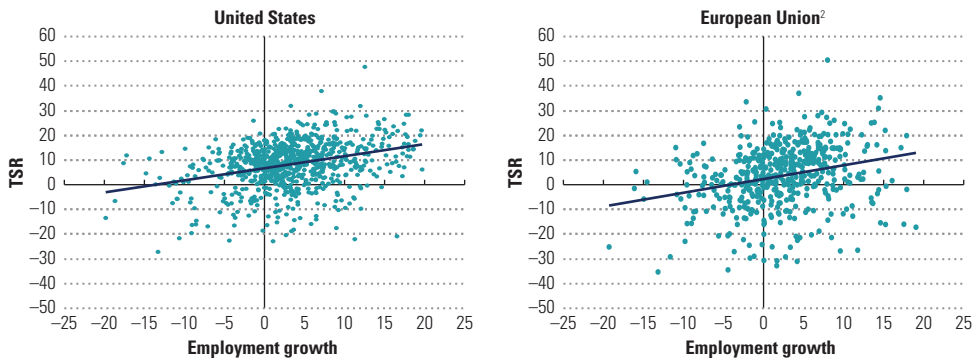
Programs such as these no doubt play a role in burnishing Lego's brand throughout communities and within company walls, where it reports that employee motivation and satisfaction levels beat 2018 targets by 50 percent. Or take Sodexo's efforts to encourage gender balance among managers. Sodexo says the program has not only increased employee retention by 8 percent, but also increased client retention by 9 percent and boosted operating margins by 8 percent.

Inevitably, though, there will be times when the interests of a company's stakeholders are not entirely complementary. Strategic decisions involve trade-offs, and the interests of different groups can be at odds with one another. Implicit in the Business Roundtable's 2019 statement of purpose is concern that business leaders have skewed some of their decisions too much toward the interests of shareholders. As a starting point, we'd encourage leaders, when trade-offs must be made, to prioritize long-term value creation, given the advantages it holds for resource allocation and economic health.

Consider employee stakeholders. A company that tries to boost profits by providing a shabby work environment, underpaying employees, or skimping on benefits will have trouble attracting and retaining high-quality employees. Lower-quality employees can mean lower-quality products, reduced demand, and damage to the brand reputation. More injury and illness can invite regulatory scrutiny and step up friction with workers. Higher turnover will inevitably increase training costs. With today's mobile and educated workforce, such a company would struggle in the long term against competitors offering more attractive environments.

If the company earns more than its cost of capital, it might afford to pay above-market wages and still prosper; treating employees well can be good business. But how well is well enough? A focus on long-term value creation suggests paying wages that are sufficient to attract quality employees and keep them happy and productive, pairing those wages with a range of non-monetary benefits and rewards. Even companies that have shifted manufacturing of products like clothing and textiles to low-cost countries with weak labor protection have found that they need to monitor the working conditions of their suppliers or face a consumer backlash.

Similarly, consider pricing decisions. A long-term approach would weigh price, volume, and customer satisfaction to determine a price that creates sustainable value. That price would have to entice consumers to buy the products not just once, but multiple times, for different generations of products. Any adjustments to the price would need to weigh the value of a lower price to buyers against the value of a higher price to shareholders and perhaps other stakeholders. A premium price that signals prestige for a luxury good can contribute long-term value. An obvious instance of going too far—or more accurately, not looking far enough ahead—is Turing Pharmaceuticals. In 2015, the company acquired the rights to a medication commonly used to treat

EXHIBIT 1.1 **Correlation between Total Shareholder Returns and Employment Growth**Compound annual growth rate,¹ 2007–2017, %¹ Samples include companies with real revenues greater than \$500 million and excludes outliers with more than 20% employment growth.² Sample includes companies in the core 15 EU member states.

AIDS-related illnesses and then raised the price per pill by more than 5,000 percent. The tactic prompted outrage and a wave of government investigations. The CEO was even derided as “the most hated man in America.”²⁰

But far more often, the lines between creating and destroying value are gray. Companies in mature, competitive industries, for example, grapple with whether they should keep open high-cost plants that lose money, just to keep employees working and prevent suppliers from going bankrupt. To do so in a globalizing industry would distort the allocation of resources in the economy, notwithstanding the significant short-term local costs associated with plant closures.²¹ At the same time, politicians pressure companies to keep failing plants open. The government may even be a major customer of the company’s products or services.

In our experience, not only do managers carefully weigh bottom-line impact, they agonize over decisions that have pronounced consequences on workers’ lives and community well-being. But consumers benefit when goods are produced at the lowest possible cost, and the economy benefits when operations that become a drain on public resources are closed and employees move to new jobs with more competitive companies. And while it’s true that employees often can’t just pick up and relocate, it’s also true that value-creating companies create more jobs. When examining employment, we found that the U.S. and European companies that created the most shareholder value from

²⁰ Z. Thomas and T. Swift, “Who Is Martin Shkreli—‘the Most Hated Man in America’?” BBC News, August 4, 2017, www.bbc.com.

²¹ Some argue that well-functioning markets also need well-functioning governments to provide the safety nets and retraining support to make essential restructuring processes more equitable.

2007 to 2017—measured as total shareholder returns—have shown stronger employment growth (see Exhibit 1.1).²²

CONSEQUENCES OF FORGETTING VALUE-CREATION PRINCIPLES

When companies forget the simple value-creation principles, the negative consequences to the economy can be huge. Two recent examples of many executives failing in their duty to focus on true value creation are the Internet bubble of the 1990s and the financial crisis of 2008.

During the Internet bubble, managers and investors lost sight of what drives return on invested capital (ROIC); indeed, many forgot the importance of this ratio entirely. Multiple executives and investors either forgot or threw out fundamental rules of economics in the rarefied air of the Internet revolution. The notion of “winner take all” led companies and investors to believe that all that mattered was getting big fast, on the assumption that they could wait until later to worry about creating an effective business model. The logic of achieving ever-increasing returns was also mistakenly applied to online pet supplies and grocery delivery services, even though these firms had to invest (unsustainably, eventually) in more drivers, trucks, warehouses, and inventory when their customer base grew. When the laws of economics prevailed, as they always do, it was clear that many Internet businesses did not have the unassailable competitive advantages required to earn even modest returns on invested capital. The Internet has revolutionized the economy, as have other innovations, but it did not and could not render obsolete the rules of economics, competition, and value creation.

Shortsighted focus can breed dishonorable dealing, and sometimes the consequences can shake confidence in capitalism to its foundations. In 2008, too many financial institutions ignored core principles. Banks lent money to individuals and speculators at low teaser rates on the assumption that housing prices would only increase. Banks packaged these high-risk debts into long-term securities and sold them to investors who used short-term debt to finance the purchase, thus creating a long-term risk for whoever lent them the money. When the home buyers could no longer afford the payments, the real estate market crashed, pushing the values of many homes below the values of the loans taken out to buy them. At that point, homeowners could neither make the required payments nor sell their homes. Seeing this, the banks that had issued short-term loans to investors in securities backed by mortgages became unwilling to roll over those loans, prompting the investors to sell all such securities at once. The value of the securities plummeted. Finally, many of the large banks themselves owned these securities, which they, of course, had also financed with short-term debt they could no longer roll over.

²² We’ve performed the same analyses for 15 and 20 years and with different start and end dates, and we’ve always found similar results.

THIS BOOK

This book is a guide to how to measure and manage the value of a company. The faster companies can increase their revenues and deploy more capital at attractive rates of return, the more value they create. The combination of growth and return on invested capital (ROIC), relative to its cost, is what drives cash flow and value. Anything that doesn't increase ROIC or growth at an attractive ROIC doesn't create value. This category can include steps that change the ownership of claims to cash flows, and accounting techniques that may change the timing of profits without actually changing cash flows.

This guiding principle of value creation links directly to competitive advantage, the core concept of business strategy. Only if companies have a well-defined competitive advantage can they sustain strong growth and high returns on invested capital. To the core principles, we add the empirical observation that creating sustainable value is a long-term endeavor, one that needs to take into account wider social, environmental, technological, and regulatory trends.

Competition tends to erode competitive advantages and, with them, returns on invested capital. Therefore, companies must continually seek and exploit new sources of competitive advantage if they are to create long-term value. To that end, managers must resist short-term pressure to take actions that create illusory value quickly at the expense of the real thing in the long term. Creating value is not the same as, for example, meeting the analysts' consensus earnings forecast for the next quarter. Nor is it ignoring the effects of decisions made today that may create greater costs down the road, from environmental cleanup to retrofitting plants to meet future pollution regulations. It means balancing near-term financial performance against what it takes to develop a healthy company that can create value for decades ahead—a demanding challenge.

This book explains both the economics of value creation (for instance, how competitive advantage enables some companies to earn higher returns on invested capital than others) and the process of measuring value (for example, how to calculate return on invested capital from a company's accounting statements). With this knowledge, companies can make wiser strategic and operating decisions, such as what businesses to own and how to make trade-offs between growth and return on invested capital. Equally, this knowledge will enable investors to calculate the risks and returns of their investments with greater confidence.

Applying the principles of value creation sometimes means going against the crowd. It means accepting that there are no free lunches. It means relying on data, thoughtful analysis, a deep understanding of the competitive dynamics of your industry, and a broad, well-informed perspective on how society continually affects and is affected by your business. We hope this book provides readers with the knowledge to help them throughout their careers to make and defend decisions that will create value for investors and for society at large.

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REVIEW QUESTIONS

1. What benefits does a long-term perspective on value creation offer for companies? For the economy?
2. Why is maximizing current share price not equivalent to maximizing long-term value?
3. When managers and boards of directors evaluate firm performance, how might focusing exclusively on corporate earnings lead them astray?
4. Give examples of situations where shareholders' and other stakeholders' interests are complementary. Give examples of situations where these interests are not complementary. If interests conflict, what should management do?
5. What are some of the common features of the 2007–2008 stock market crash and previous market crashes—for example, the tulip mania of the early 1600s or the Internet bubble around the turn of the millennium?
6. If growth is a significant value driver, does getting bigger necessarily translate into creating value? Explain.
7. What more could boards of directors and shareholders do to ensure that managers pursue long-term value creation?

Finance in a Nutshell

Companies create value when they earn a return on invested capital (ROIC) greater than their opportunity cost of capital.¹ If the ROIC is at or below the cost of capital, growth may not create value. Companies should aim to find the combination of growth and ROIC that drives the highest discounted value of their cash flows. In so doing, they should consider that performance in the stock market may differ from intrinsic value creation, generally as a result of changes in investors' expectations.

To illustrate how value creation works, this chapter uses a simple story. Our heroes are Lily and Nate, who start out as the owners of a small chain of trendy clothing stores. Success follows. Over time, their business goes through a remarkable transformation. They develop the idea of Lily's Emporium and convert their stores to the new concept. To expand, they take their company public to raise additional capital. Encouraged by the resulting gains, they develop more retail concepts, including Lily's Furniture and Lily's Garden Supplies. In the end, Lily and Nate are faced with the complexity of managing a multibusiness retail enterprise.

THE EARLY YEARS

When we first met Lily and Nate, their business had grown from a tiny boutique into a small chain of trendy, midpriced clothing stores called Lily's Dresses. They met with us to find out how they could know if they were achieving attractive financial results. We told them they should measure their business's return on invested capital: after-tax operating profits divided by the capital invested in working capital and property, plant, and equipment.

¹ A simple definition of return on invested capital is after-tax operating profit divided by invested capital (working capital plus fixed assets). ROIC's calculation from a company's financial statements is explained in detail in Chapters 10 and 11.

Then they could compare the ROIC with what they could earn if they invested their capital elsewhere—for example, in the stock market.

Lily and Nate had invested \$10 million in their business, and in 2020 they earned about \$1.8 million after taxes, with no debt. So they calculated their return on invested capital as 18 percent. They asked what a reasonable guess would be for the rate they could earn in the stock market, and we suggested they use 10 percent. They easily saw that their money was earning 8 percent more than what we were assuming they could earn by investing elsewhere, so they were pleased with their business's performance.

We commented that growth is also important to consider in measuring financial performance. Lily told us that the business was growing at about 5 percent per year. Nate added that they discovered growth can be expensive; to achieve that growth, they had to invest in new stores, fixtures, and inventory. To grow at 5 percent and earn 18 percent ROIC on their growth, they reinvested about 28 percent of their profits back into the business each year. The remaining 72 percent of profits was available to withdraw from the business. In 2020, then, they generated cash flow of about \$1.30 million.

Lily and Nate were satisfied with 5 percent growth and 18 percent ROIC until Lily's cousin Logan told them about his aggressive expansion plans for his own retail business, Logan's Stores. Based on what Logan had said, Lily and Nate compared the expected faster growth in operating profit for Logan's Stores with their own company's 5 percent growth, as graphed in Exhibit 2.1. Lily and Nate were concerned that Logan's faster-growing profits signaled a defect in their own vision or management.

"Wait a minute," we said. "How is Logan getting all that growth? What about his ROIC?" Lily and Nate checked and returned with the data shown in Exhibit 2.2. As we had suspected, Logan was achieving his growth by

EXHIBIT 2.1 Expected Profit Growth at Logan's Stores Outpacing Lily's Dresses

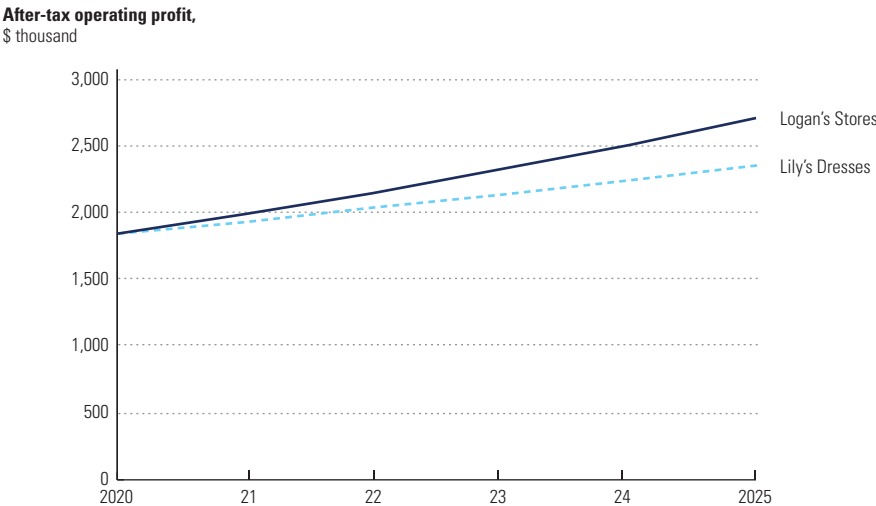
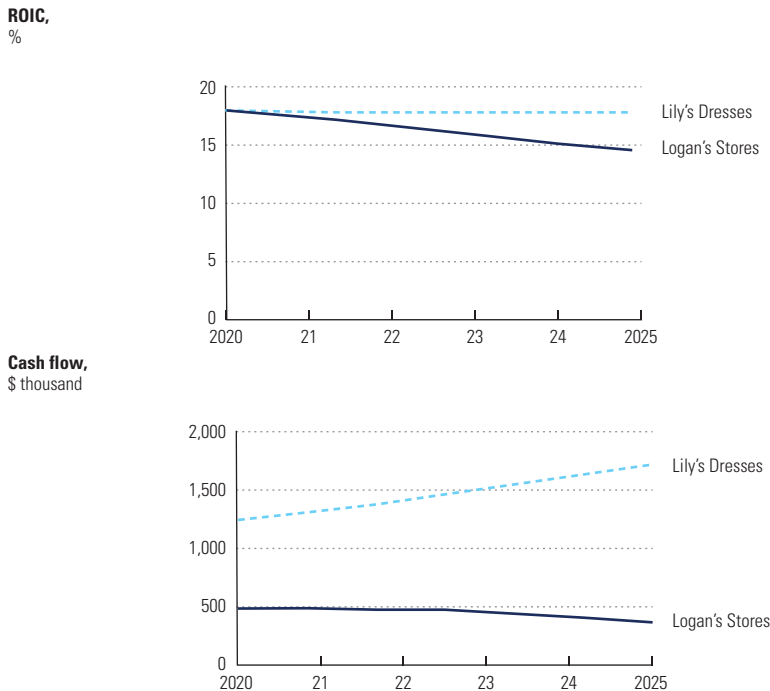


EXHIBIT 2.2 **Lily's Dresses Outperforming in Return on Invested Capital (ROIC) and Cash Flow**



investing heavily. Despite all the growth in operating profit, his company's ROIC was declining significantly, so cash flow was slipping downward.

We asked the two why they thought their stores earned higher returns on capital than Logan's. Nate said one reason was that their products were unique and cutting-edge fashion, so their customers were willing to pay higher prices for their dresses than for the products at many other dress shops. Lily added that each of their stores attracted more customers, so their sales per square foot (a proxy for fixed costs) were greater than Logan's. As they saw it, Logan's products were not much different from those of his competitors, so he had to match his prices to theirs and had less customer traffic in his stores. This discussion helped Nate and Lily appreciate that it was beneficial to consider ROIC along with growth.

A NEW CONCEPT

Several years later, Lily and Nate called us with a great idea. They wanted to develop a new concept, which they called Lily's Emporium. Lily's Emporium would operate larger stores carrying a wider assortment of clothes and accessories that their talented designers were working on. But when they looked at the projected results (they now had a financial-analysis department), they found that all the new capital investment to convert their stores would reduce ROIC and cash flow for four years, even though revenue and profits would be

growing faster, as shown in Exhibit 2.3. After four years, cash flow would be greater, but they didn't know how to trade off the short-term decline in ROIC and cash flow against the long-term improvement.

We affirmed that these were the right questions and explained that answering them would require more sophisticated financial tools. We advised them to use discounted cash flow (DCF), a measure that is also known as present value. DCF is a way of collapsing the future performance of the company into a single number. Lily and Nate needed to forecast the future cash flow of the company and discount it back to the present at the same opportunity cost of capital we had used for our earlier comparisons.

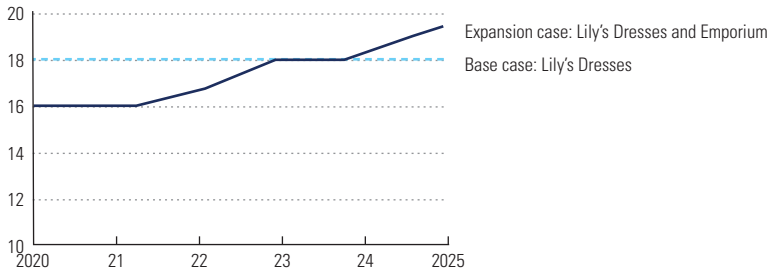
We helped Lily and Nate apply DCF to their new concept, discounting the projected cash flows at 10 percent. We showed them that the DCF value of their company would be \$53 million if they did not adopt the new concept. With the new concept, the DCF value would be greater: \$62 million. (Actually, on our spreadsheet, we rounded to the nearest thousand: \$61,911,000.) These numbers gave them confidence in their idea for Lily's Emporium.

SHOULD LILY AND NATE TRY TO MAXIMIZE ROIC?

As they saw how these financial measures could help them build a more valuable business, Lily and Nate began to formulate more questions about measuring value. Lily asked if their strategy should be to maximize their return on

EXHIBIT 2.3 Expansion's Impact on ROIC and Cash Flow

ROIC,
%



Cash flow,
\$ thousand

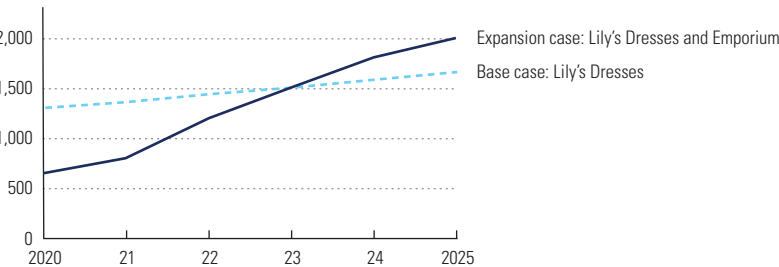


EXHIBIT 2.4 **Economic Profit Is Higher with Lower-Performing Stores in the Mix**

	ROIC, %	Cost of capital, %	Spread, %	Invested capital, \$ thousand	Economic profit, \$ thousand
Entire company	18	10	8	12,000	960
Without lower-performing stores	19	10	9	9,500	855

invested capital. She pointed to the fact that some stores outperformed others. For example, some were earning an ROIC of only 14 percent. If the business closed those lower-performing stores, they could increase their average return on invested capital.

Our advice was to focus not on the ROIC itself, but on the combination of ROIC (versus cost of capital) and the amount of capital. A tool for doing that is called economic profit. We showed them how economic profit applies to their business, using the measures in Exhibit 2.4.

We defined economic profit as the spread between ROIC and cost of capital multiplied by the amount of invested capital. In Lily and Nate's case, their economic profit forecast for 2024 would be the 8 percent spread by \$12 million in invested capital, or \$960,000. If they closed their low-returning stores, their average ROIC would increase to 19 percent, but their economic profit would decline to \$855,000. This is because even though some stores earn a lower ROIC than others do, the lower-earning stores are still earning more than the cost of capital. Using this example, we made the case that Lily and Nate should seek to maximize economic profit, not ROIC, over the long term.

For Nate, though, this analysis raised a practical concern. With different methods available, it wasn't obvious which one to use. He asked, "When do we use economic profit, and when do we use DCF?"

"Good question," we said. "In fact, they're the same." We prepared Exhibit 2.5 to show Nate and Lily a comparison, using the DCF we had previously estimated for their business: \$61,911,000. To apply the economic-profit method, we discounted the future economic profit at the same cost of capital we had used with the DCF. Then we added the discounted economic profit to the amount of capital invested today. The results for the two approaches are the same—exactly, to the penny.²

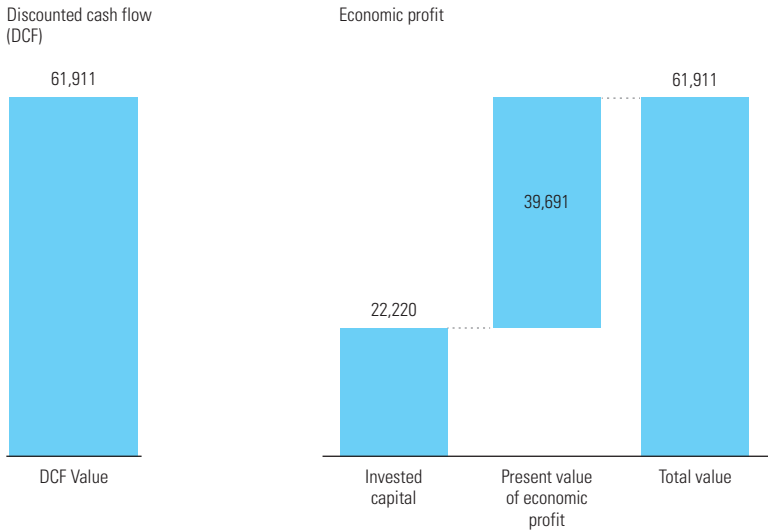
GOING PUBLIC

Now Lily and Nate had a way to make important strategic decisions over multiple time periods. Lily's Emporium was successful, and the next time they called us, they talked excitedly about new ambitions. "We need more

² See Chapter 10 for a detailed discussion of these two valuation approaches.

EXHIBIT 2.5 **Identical Results from DCF and Economic-Profit Valuation**

Valuation, by method, \$ thousand



capital to build more stores more quickly,” Nate said. “Besides, we want to provide an opportunity for some of our employees to become owners. So we’ve decided to go public.” They asked us to help them understand how going public would affect their financial decision making.

“Well,” we said, “now’s the time to learn what the distinction is between financial markets and real markets and how they are related to each other. You’ll want to understand that good performance in one market does not necessarily mean good performance in another.”

Up until now, we’d been talking with Lily and Nate about the real market. How much profit and cash flow were they earning relative to the investments they were making? Were they maximizing their economic profit and cash flow over time? In the real market, the decision rule is simple: choose strategies or make operational decisions that maximize the present value of future cash flow or future economic profit.

When a company enters the capital market, the decision rules for the real market remain essentially unchanged. But life gets more complicated, because management must simultaneously deal with the financial market.

When a company goes public and sells shares to a wide range of investors who can trade those shares in an organized market, the interaction (or trading activity) between investors and even market speculators sets a price for those shares. The price of the shares is based on what investors think those shares are worth. Each investor decides what he or she thinks the value of the shares should be and makes trades based on whether the current price is above or below that estimate of the intrinsic value.

This intrinsic value is based on the future cash flows or earnings power of the company. This means, essentially, that investors are paying for the performance they expect the company to achieve in the future, not what the company has done in the past (and certainly not the cost of the company's assets).

Lily asked us how much their company's shares would be worth. "Let's assume," we said, "that the market's overall assessment of your company's future performance is similar to what you think your company will do. The first step is to forecast your company's performance and discount the future expected cash flows. Based on this analysis, the intrinsic value of your shares is \$20 per share."

"That's interesting," said Nate, "because the amount of capital we've invested is only \$7 per share." We told them that this difference meant the market should be willing to pay their company a premium of \$13 over the invested capital for the future economic profit the company would earn.

"But," Lily asked, "if they pay us this premium up front, how will the investors make any money?"

"They may not," we said. "Let's see what will happen if your company performs exactly as you and the market expect. Let's value your company five years into the future. If you perform exactly as expected over the next five years and if expectations beyond five years don't change, your company's value will be \$32 per share. Let's assume that you have not paid any dividends. An investor who bought a share for \$20 per share today could sell the share for \$32 in five years. The annualized return on the investment would be 10 percent, the same as the discount rate we used to discount your future performance. The interesting thing is that as long as you perform as expected, the return for your shareholders will be just their opportunity cost. But if you do better than expected, your shareholders will earn more than 10 percent. And if you do worse than expected, your shareholders will earn less than 10 percent."

"So," said Lily, "the return that investors earn is driven not by the performance of our company, but by its performance relative to expectations."

"Exactly!" we said.

Lily paused and reflected on the discussion. "That means we must manage our company's performance in the real markets and the financial markets at the same time."

We agreed and explained that if they were to create a great deal of value in the real market—say, by earning more than their cost of capital and growing fast—but didn't do as well as investors expected, the investors would be disappointed. Managers have a dual task: to maximize the intrinsic value of the company and to properly manage the expectations of the financial market.

"Managing market expectations is tricky," we added. "You don't want investor expectations to be too high or too low. We've seen companies convince the market that they will deliver great performance and then not deliver on those promises. Not only does the share price drop when the market realizes

that the company won't be able to deliver, but regaining credibility may take years. Conversely, if the market's expectations are too low and you have a low share price relative to the opportunities the company faces, you may be subject to a hostile takeover."

After exploring these issues, Lily and Nate felt prepared to take their company public. They went forward with an initial public offering and raised the capital they needed.

EXPANSION INTO RELATED FORMATS

Lily and Nate's business was successful, growing quickly and regularly beating the expectations of the market, so their share price was a top performer. They were comfortable that their management team would be able to achieve high growth in their Emporium stores, so they decided next to try some new concepts they had been thinking about: Lily's Furniture and Lily's Garden Supplies. But they grew concerned about managing the business as it became more and more complex. They had always had a good feel for the business, but as it expanded and they had to delegate more decision making, they were less confident that things would be managed well.

They met with us again and told us that their financial people had put in place a planning and control system to closely monitor the revenue growth, ROIC, and economic profit of every store and each division overall. Their team set revenue and economic-profit targets annually for the next three years, monitored progress monthly, and tied managers' compensation to economic profit against these targets. Yet they told us they weren't sure the company was on track for the long-term performance that they and the market expected.

"You need a planning and control system that incorporates forward-looking measures besides looking backward at financial measures," we told them.

"Tell us more," Nate said.

"As you've pointed out," we said, "the problem with any financial measure is that it cannot tell you how your managers are doing at building the business for the future. For example, in the short term, managers could improve their financial results by cutting back on customer service, such as by reducing the number of employees available in the store to help customers, by cutting into employee training, or by deferring maintenance costs or brand-building expenditures. You need to make sure that you build in measures related to customer satisfaction or brand awareness—measures that let you know what the future will look like, not just what the current performance is."

Lily and Nate both nodded, satisfied. The lessons they so quickly absorbed and applied have placed their company on a solid foundation. The two of them still come to see us from time to time, but only for social visits. Sometimes they bring flowers from their garden supplies center.

SOME LESSONS

While we have simplified the story of Lily and Nate's business, it highlights the core ideas around value creation and its measurement:

1. In the real market, you create value by earning a return on your invested capital greater than the opportunity cost of capital.
2. The more you can invest at returns above the cost of capital, the more value you create. That is, growth creates more value as long as the return on invested capital exceeds the cost of capital.
3. You should select strategies that maximize the present value of future expected cash flows or economic profit. The answer is the same regardless of which approach you choose.
4. The value of a company's shares in the stock market equals the intrinsic value based on the market's expectations of future performance, but the market's expectations of future performance may not be same as the company's.
5. The returns that shareholders earn depend on changes in expectations as much as on the actual performance of the company.

In the next chapter, we develop a more formal framework for understanding and measuring value creation.

REVIEW QUESTIONS

1. How can Lily's Dresses assess whether its business is creating value?
2. Why was Logan's Stores' superior growth in profits over Lily's Dresses not translating into a higher ROIC?
3. In deciding whether to expand Lily's Dresses to create Lily's Emporium, how do Lily and Nate trade off the short-term hit to ROIC with the long-term increase in ROIC?
4. Why should Lily and Nate keep the stores with lower ROIC open, even if closing them would raise their firm's overall ROIC?
5. Which method provides a better assessment of the value of the expanded firm—the discounted cash flow (DCF) or economic profit (EP) method?
6. What is the difference between the share price of Lily's Emporium and its intrinsic value? When would investors want to buy Lily's stock?
7. Suppose that an investor bought shares in Lily's Emporium and held them for five years. If the firm generated ROICs above their cost of capital, does this mean that the investor had returns above the cost of capital?
8. As Lily's Emporium expands into additional lines of business, how does it ensure that it continues to create long-term value?

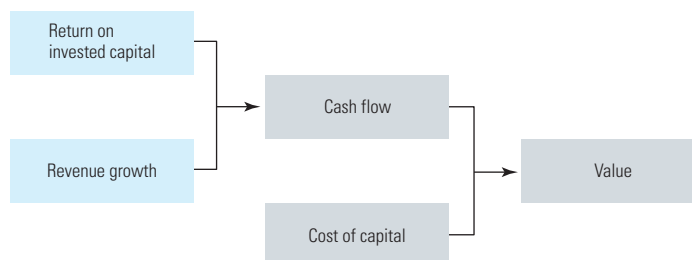
Fundamental Principles of Value Creation

Companies create value for their owners by investing cash now to generate more cash in the future. The amount of value they create is the difference between cash inflows and the cost of the investments made, adjusted to reflect the fact that tomorrow's cash flows are worth less than today's because of the time value of money and the riskiness of future cash flows. As we illustrated in Chapter 2, the conversion of revenues into cash flows—and earnings—is a function of a company's return on invested capital (ROIC) and its revenue growth. That means the amount of value a company creates is governed ultimately by its ROIC, revenue growth, and ability to sustain both over time. Keep in mind that a company will create value only if its ROIC is greater than its cost of capital.¹ Moreover, only if ROIC exceeds the cost of capital will growth increase a company's value. Growth at lower returns actually reduces a company's value. Exhibit 3.1 illustrates this core principle of value creation.²

Following these principles helps managers decide which strategies and investments will create the most value for shareholders in the long term. The principles also help investors assess the potential value of companies they might consider investing in. This chapter explains the relationships that tie together

¹ The cost of capital is an opportunity cost for the company's investors, not a cash cost. See Chapter 4 for a more detailed explanation.

² In its purest form, *value* is the sum of the present values of future expected cash flows—a point-in-time measure. *Value creation* is the change in value due to company performance (changes in growth and ROIC). Sometimes we refer to value and value creation based on explicit projections of future growth, ROIC, and cash flows. At other times, we use the market price of a company's shares as a proxy for value, and total shareholder returns (share price appreciation plus dividends) as a proxy for value creation.

EXHIBIT 3.1 **Growth and ROIC Drive Value**

growth, ROIC, cash flows, and value, and it introduces the way managers can use these relationships to decide among different investments or strategies. For example, we will show that high-ROIC companies typically create more value by focusing on growth, while lower-ROIC companies create more value by increasing ROIC. We'll also explore the principle, often forgotten by executives, that anything that doesn't increase cash flows, such as noncash accounting charges or changes in accounting methods, won't create value. And we'll introduce a simple equation that captures the essence of valuation in practice.

One might expect universal agreement on a notion as fundamental as value, but this isn't the case: many executives, boards, and financial media still treat accounting earnings and value as one and the same and focus almost obsessively on improving earnings. However, while earnings and cash flow are often correlated, earnings don't tell the whole story of value creation. Focusing too much on earnings or earnings growth often leads companies to stray from a value-creating path.

For example, earnings growth alone can't explain why investors in discount retailer Costco, the fourth-largest retailer in the United States, with sales of \$126 billion in 2017, and Brown-Forman, the producer of Jack Daniels and other alcoholic beverages, with sales of \$4 billion the same year, earned similar shareholder returns (dividends plus appreciation in the share price) between 1996 and 2017. These two successful companies had very different growth rates. During the period, after-tax operating profits for Costco grew 11 percent per year, while those of Brown-Forman grew 7 percent annually. This means that profits for Costco in 2017 were nine times larger than in 1996, while profits at Brown-Forman were only four times larger. Costco was one of the fastest-growing companies in the United States during this time; its average annual shareholder returns were 15 percent. Brown-Forman was growing much more slowly, yet its annual shareholder returns were also 15 percent. The reason Brown-Forman could create the same value as Costco, despite much slower growth, was that Brown-Forman earned a 29 percent ROIC (excluding the impact of acquisitions), while Costco's ROIC was 13 percent.

To be fair, if all companies in an industry earned the same ROIC, then earnings growth *would* be the differentiating metric, because then only growth and

not ROIC would determine differences in companies' cash flow. For reasons of simplicity, analysts and academics have sometimes made this assumption. But as Chapter 8 demonstrates, returns on invested capital can vary considerably, not only across industries but also between companies within the same industry and across time.

THE RELATIONSHIP OF GROWTH, ROIC, AND CASH FLOW

Disaggregating cash flow into revenue growth and ROIC helps illuminate the underlying elements that power a company's performance. Say a company's cash flow was \$100 last year and will be \$115 next year. This doesn't tell us much about its economic performance, since the \$15 increase in cash flow could come from many sources, including revenue growth, a reduction in capital spending, or a reduction in marketing expenditures. But if we told you that the company was generating revenue growth of 7 percent per year and would earn a return on invested capital of 15 percent, then you would be able to evaluate its performance. You could, for instance, compare the company's growth rate with the growth rate of its industry or the economy, and you could analyze its ROIC relative to peers, its cost of capital, and its own historical performance.

Growth, ROIC, and cash flow are mathematically linked. To see how, consider two companies, Value Inc. and Volume Inc., whose projected earnings, investment, and resulting cash flows are displayed in Exhibit 3.2. Earnings, in this illustration, are expressed as net operating profit after taxes (NOPAT), a term we use throughout the book. Both companies earned NOPAT of \$100 million in year 1 and are expected to increase their revenues and earnings at 5 percent per year, so their projected earnings are identical. If the popular view that value depends only on earnings were true, the two companies' values also would be the same. But this simple example demonstrates how wrong that view can be.

EXHIBIT 3.2 **Tale of Two Companies: Same Earnings, Different Cash Flows**

\$ million

Value Inc.	Year 1	Year 2	Year 3	Year 4	Year 5
NOPAT ¹	100	105	110	116	122
Investment	(25)	(26)	(28)	(29)	(31)
Cash flow	75	79	82	87	91
Volume Inc.	Year 1	Year 2	Year 3	Year 4	Year 5
NOPAT ¹	100	105	110	116	122
Investment	(50)	(53)	(55)	(58)	(61)
Cash flow	50	52	55	58	61

¹ Net operating profit after taxes.

Almost all companies need to invest in plant, equipment, or working capital to grow. Free cash flow is what's left over for investors once investments have been subtracted from earnings. Value Inc. generates higher free cash flows with the same earnings because it invests only 25 percent of its profits—its investment rate—to achieve the same profit growth as Volume Inc., which invests 50 percent of its profits. Value Inc.'s lower investment rate results in 50 percent higher cash flows each year than Volume Inc. sees while generating the same level of profits.

We can value the two companies by discounting their future free cash flows at a discount rate that reflects what investors expect to earn from investing in the companies—that is, their cost of capital. For both companies, we assumed their growth and investment rates were perpetual, and we discounted each year's cash flow to the present at a 10 percent cost of capital. So, for example, Value Inc.'s year 1 cash flow of \$75 million has a present value of \$68 million today (see Exhibit 3.3). We summed each year's results to derive a total present value of all future cash flows: \$1,500 million for Value Inc. and \$1,000 million for Volume Inc.

The companies' values can also be expressed as price-to-earnings ratios (P/E_s). Divide each company's value by its first-year earnings of \$100 million. Value Inc.'s P/E is 15, while Volume Inc.'s is only 10. Despite identical earnings and growth rates, the companies have different earnings multiples because their cash flows are so different. Value Inc. generates higher cash flows because it doesn't have to invest as much as Volume Inc. does.

Differences in ROIC—defined here as the incremental NOPAT earned each year relative to the prior year's investment—are what drives difference in investment rates. In this case, Value Inc. invested \$25 million in year

EXHIBIT 3.3 **Value Inc.: DCF Valuation**

\$ million

	Value Inc.						Sum
	Year 1	Year 2	Year 3	Year 4	Year 5	Year X	
NOPAT ¹	100	105	110	116	122	...	
Investment	(25)	(26)	(28)	(29)	(31)	...	
Cash flow	75	79	82	87	91	...	
Value today	68	65	62	59	56	...	1,500



*Present value of 75
discounted at 10% for
1 year*



*Present value of 87
discounted at 10% for
4 years*

¹ Net operating profit after taxes.

1 to increase its profits by \$5 million in year 2. Its return on new capital is 20 percent (\$5 million of additional profits divided by \$25 million of investment).³ In contrast, Volume Inc.'s return on invested capital is 10 percent (\$5 million in additional profits in year 2 divided by an investment of \$50 million).

Growth, ROIC, and cash flow (as represented by the investment rate) are tied together mathematically in the following relationship:

$$\text{Growth} = \text{ROIC} \times \text{Investment Rate}$$

Applying the formula to Value Inc.:

$$5\% = 20\% \times 25\%$$

Applying it to Volume Inc.:

$$5\% = 10\% \times 50\%$$

As you can see, Volume Inc. needs a higher investment rate to achieve the same growth.

Another way to look at this comparison is in terms of cash flow:

$$\text{Cash Flow} = \text{Earnings} \times (1 - \text{Investment Rate})$$

In this equation, the investment rate is equal to growth divided by ROIC:

$$\text{Cash Flow} = \text{Earnings} \times (1 - \text{Growth} / \text{ROIC})$$

For Value Inc.:

$$\begin{aligned} \$75 &= \$100 \times (1 - 5\% / 20\%) \\ &= \$100 \times (1 - 25\%) \end{aligned}$$

For Volume Inc.:

$$\begin{aligned} \$50 &= \$100 \times (1 - 5\% / 10\%) \\ &= \$100 \times (1 - 50\%) \end{aligned}$$

Since the three variables are tied together mathematically, you can describe a company's performance with any two variables. We generally describe a company's performance in terms of growth and ROIC because, as mentioned earlier, you can analyze growth and ROIC across time and versus peers.

³ We assumed that all of the increase in profits is due to the new investment, with the return on Value Inc.'s existing capital remaining unchanged.

EXHIBIT 3.4 Translating Growth and ROIC into Cash Flow Available for Distribution

% of NOPAT

Growth	9%	-14	0	31	64
	6%	14	33	54	76
	3%	57	67	77	88
		7%	9%	13%	25%
		ROIC			

Exhibit 3.4 shows how different combinations of growth and ROIC generate different levels of cash flow that can be paid out to investors. The numbers in the boxes represent cash flow as a percentage of NOPAT, which represents the profits available for distribution to investors. You can see that as growth slows at any level of ROIC, the cash generated per dollar of NOPAT increases. That explains why even maturing companies experiencing slowing growth can pay out much larger amounts of their earnings to investors. Note also that companies with high ROIC tend to generate lots of cash flow as long as they are growing modestly. This explains why mature tech and pharma companies with high returns on capital pay out so much of their earnings to investors. They don't really have a choice, because they typically generate much more cash flow than they can reinvest at attractive returns on capital.

Note that near-term cash flow by itself may not be a meaningful performance indicator. Consider what would happen if Value Inc. were to find more investment opportunities at a 25 percent ROIC and be able to increase its growth to 8 percent per year. Exhibit 3.5 shows the projected NOPAT and cash flow. Because it would be growing faster, Value Inc. would need to invest more of its earnings each year, so its cash flow at 8 percent growth would be lower than at 5 percent growth until year 9. However, its value, which at 5 percent growth would be \$1.5 billion, would double at 8 percent growth to \$3 billion, because its cash flows would be higher in the long term.

EXHIBIT 3.5 **Value Inc.: Lower Initial Cash Flow at Higher Growth Rate**

\$ million

5% growth

	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8	Year 9	Year 10	Year 11	Year 12
NOPAT	100	105	110	116	122	128	138	141	148	155	163	171
Net investment	(25)	(26)	(28)	(29)	(30)	(32)	(34)	(35)	(37)	(39)	(41)	(43)
Cash flow	75	79	83	87	91	96	101	106	111	116	122	128

8% growth

	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8	Year 9	Year 10	Year 11	Year 12
NOPAT	100	108	117	126	136	147	159	171	185	200	216	233
Net investment	(40)	(43)	(47)	(50)	(54)	(59)	(63)	(69)	(74)	(80)	(86)	(93)
Cash flow	60	65	70	76	82	88	95	103	111	120	130	140

Higher growth rate initially generates less cash flow

If you simplify some assumptions—for example, that a company grows at a constant rate and maintains a constant ROIC—you can reduce the discounted cash flow to a simple formula. We call this the value driver formula. Here, NOPAT represents the net operating profit after taxes, g is the growth rate of the company, and WACC is the cost of capital.

$$\text{Value} = \frac{\text{NOPAT}_{t=1} \left(1 - \frac{g}{\text{ROIC}} \right)}{\text{WACC} - g}$$

Using this equation, you can see that value is driven by growth, ROIC, and the cost of capital, just as we described in the example. In practice, we rarely use this formula by itself, because of its assumption of constant growth and ROIC forever. Still, we find it useful as a reminder of the elements that drive value. Note that improving ROIC, for any level of growth, always increases value because it reduces the investment required for growth. The impact of growth, however, is ambiguous, as it appears in both the numerator and the denominator. In the next section, we'll show that faster growth increases value only when a company's ROIC is greater than its cost of capital. At the end of this chapter, we'll also show how this equation is derived.

BALANCING ROIC AND GROWTH TO CREATE VALUE

It is possible to create a matrix that shows how different combinations of growth and ROIC translate into value (Exhibit 3.6). Each cell in the matrix represents the present value of future cash flows under each of the assumptions of growth and ROIC, discounted at the company’s cost of capital. This case assumes a 9 percent cost of capital and a company that earns \$100 in the first year.⁴

Observe that for any level of growth, value increases with improvements in ROIC. In other words, when all else is equal, a higher ROIC is always good, because it means that the company doesn’t have to invest as much to achieve a given level of growth. The same can’t be said of growth. When ROIC is high, faster growth increases value. But when ROIC is lower than the company’s cost of capital, faster growth destroys value. When return on capital is lower than the cost of capital, growing faster means investing more at a value-destroying return. Where ROIC equals the cost of capital, we can draw the dividing line between creating and destroying value through growth. On that line, value is neither created nor destroyed, regardless of how fast the company grows. It’s as if management were on a treadmill. They’re working hard, but after their workout, they are right where they started.

From the exhibit, you can also see that a company with high ROIC and low growth may have a similar or higher valuation multiple than a company with

EXHIBIT 3.6 Translating Growth and ROIC into Value

Value,¹ \$

Growth	9%	400	1,100	1,900	2,700
	6%	600	1,100	1,600	2,100
	3%	800	1,100	1,400	1,600
		7%	9%	13%	25%
ROIC					

¹ Present value of future cash flows, assuming year 1 earnings of \$100 and a 9% cost of capital. After 15 years, all scenarios grow at 4.5%.

⁴ We made explicit cash flow forecasts for the first 15 years and assumed that growth after that point converges on 4.5 percent in all scenarios. If a company grew faster than the economy forever, it would eventually overtake the entire world economy.

higher growth but low ROIC. For example, at the end of 2017, Brown-Forman and Costco were both valued with a ratio of enterprise value to pretax operating profits in the range of 19 to 20 times. Yet Costco had been growing at 7 percent per year over the prior three years, while Brown-Forman had grown at less than 2 percent per year. Again, Brown-Forman made up for its lower growth with a higher ROIC of 30 percent in 2017, versus 15 percent for Costco (which is good for a capital-intensive, low-margin retailer).

We sometimes hear the argument that even low-ROIC companies should strive for growth. The logic is that if a company grows, its ROIC will naturally increase. However, we find this is true only for young, start-up businesses. Most often in mature companies, a low ROIC indicates a flawed business model or unattractive industry structure. Don't fall for the trap that growth will lead to scale economies that automatically increase a company's return on capital. It almost never happens for mature businesses.

SOME EXAMPLES

The logic laid out in this section reflects the way companies perform in the stock market. Recall the earlier explanation of why shareholder returns for Costco and Brown-Forman were the same even though earnings for Costco grew much faster. Another example of the relative impact of growth and ROIC on value is Rockwell Automation, which provides integrated systems to monitor and control automation in factories. Rockwell's total shareholder returns (TSR) from 1995 to 2018 were 19 percent per year, placing it in the top quartile of industrial companies. During this period, Rockwell's revenues actually shrank from \$13 billion in 1995 to \$7 billion in 2018 as it divested its aviation and power systems divisions. The major factor behind its high TSR was its success in increasing ROIC, from about 12 percent in the mid-1990s to about 35 percent in 2018 (including goodwill). After spinning off its aviation business (now known as Rockwell Collins) in 2001, Rockwell focused on its core industrial-automation business and improved ROIC significantly. While this was partially accomplished by divesting lower-margin ancillary businesses, the majority of the improvement came from operational improvement in industrial automation. The company publicly reiterated its focus on cost and capital productivity many times during the period.

Clearly, the core valuation principle applies at the company level. We have found that it applies at the sector level, too. Consider companies as a whole in the consumer packaged-goods sector. Even though well-known names in the sector such as Procter & Gamble and Colgate-Palmolive aren't high-growth companies, the market values them at average or higher earnings multiples because of their high returns on invested capital.

The typical large packaged-goods company increased its revenues 1.2 percent a year from 2014 to 2019, slower than the median of about 4.5 percent for

all Standard & Poor's (S&P) 500 companies, excluding financial institutions. Yet at the end of 2018, the median P/E of consumer packaged-goods companies was about 15, almost exactly the same as the median S&P 500 company. The valuations of companies in this sector rested on their high ROICs—in aggregate above 40 percent, compared with an aggregate ROIC of 22 percent for the S&P 500 in 2018.

To test whether the core valuation principle also applies at the level of countries and the aggregate economy, we compared large companies based in Europe and the United States. The median trailing P/E ratio for large U.S. companies was 15.5 times, versus 12.8 for large European companies. The difference in valuation relative to invested capital is even more extreme. The median enterprise value to invested capital for U.S. companies was 5.4, versus 3.2 for European companies. Some executives assume the reason is that investors are simply willing to pay higher prices for shares of U.S. companies (an assumption that has prompted some non-U.S. companies to consider moving their share listings to the New York Stock Exchange in an attempt to increase their value). But the real reason U.S. companies trade at higher multiples is that they typically earn higher returns on invested capital. The median large U.S. company earned a 30 percent ROIC (before goodwill and intangibles) in 2018, while the median large European company earned 19 percent. A large part of the difference is a different industry mix; the United States has many more high-ROIC pharmaceutical, medical-device, and technology companies. These broad comparisons also hide the fact that some European companies—for example, Robert Bosch in auto parts and Reckitt Benckiser in consumer packaged goods—outperform many of their U.S. counterparts.

More evidence showing that ROIC and growth drive value appears in Chapter 7.

IMPLICATIONS FOR MANAGERS

We'll dive deeper into the managerial dimensions of ROIC and growth in Chapters 8 and 9, respectively. For now, we outline several lessons managers should learn for strategic decision making.

Start by referring back to Exhibit 3.6, because it contains the most important strategic insights for managers concerning the relative impact that changes in ROIC and growth can have on a company's value. In general, companies already earning a high ROIC can generate more additional value by increasing their rate of growth, rather than their ROIC. For their part, low-ROIC companies will generate relatively more value by focusing on increasing their ROIC.

For example, Exhibit 3.7 shows that a typical high-ROIC company, such as a branded consumer packaged-goods company, can increase its value by

EXHIBIT 3.7 **Increasing Value: Impact of Higher Growth and ROIC**

Change in value, %



10 percent if it increases its growth rate by one percentage point, while a typical moderate-ROIC company, such as the average retailer, will increase its value by only 5 percent for the same increase in growth. In contrast, the moderate-ROIC company gets a 15 percent bump in value from increasing its return on invested capital by one percentage point, while the high-ROIC company gets only a 6 percent bump from the same increase in return on invested capital.

The general lesson is that high-ROIC companies should focus on growth, while low-ROIC companies should focus on improving returns before growing. Of course, this analysis assumes that achieving a one-percentage-point increase in growth is as easy as achieving a one-percentage-point increase in ROIC, everything else being constant. In reality, achieving either type of increase poses different degrees of difficulty for different companies in different industries, and the impact of a change in growth and ROIC will also vary between companies. However, every company needs to conduct the analysis to set its strategic priorities.

Until now, we have assumed that all growth earns the same ROIC and therefore generates the same value, but this is clearly unrealistic: different types of growth earn different returns on capital, so not all growth is equally value-creating. Each company must understand the pecking order of growth-related value creation that applies to its industry and company type.

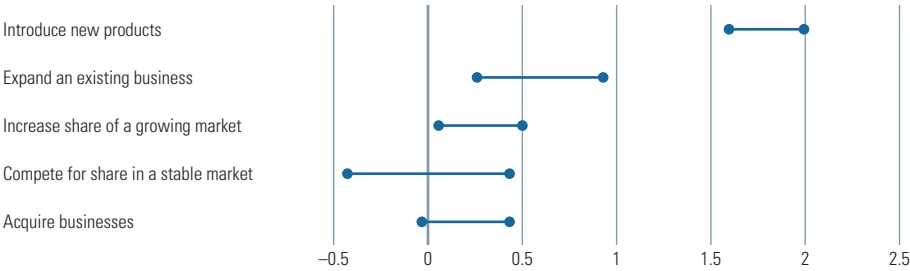
Exhibit 3.8 shows the value created from different types of growth for a typical consumer products company.⁵ These results are based on cases with which we are familiar, not on a comprehensive analysis. Still, we believe they reflect the broader reality.⁶ The results are expressed in terms of value created for \$1.00 of incremental revenue. For example, \$1.00 of additional revenue from a new product creates \$1.75 to \$2.00 of value. The most important implication of this chart is the rank order. New products typically create more value for shareholders, while acquisitions typically create the least. The key to the difference between these extremes is differences in returns on capital for the different types of growth.

⁵ This exhibit will look different for different industries.

⁶ We identified examples for each type of growth and estimated their impact on value creation. For instance, we obtained several examples of the margins and capital requirements for new products.

EXHIBIT 3.8 Value Creation by Type of Growth

Shareholder value created for incremental \$1.00 of revenue, \$¹



¹ Value for a typical consumer packaged goods company.

Growth strategies based on organic new-product development frequently have the highest returns because they don't require much new capital; companies can add new products to their existing factory lines and distribution systems. Furthermore, the investments to produce new products are not all required at once. If preliminary results are not promising, future investments can be scaled back or canceled.

Acquisitions, by contrast, require that the entire investment be made up front. The amount of up-front payment reflects the expected cash flows from the target plus a premium to stave off other bidders. So even if the buyer can improve the target enough to generate an attractive ROIC, the rate of return is typically only a small amount higher than its cost of capital.

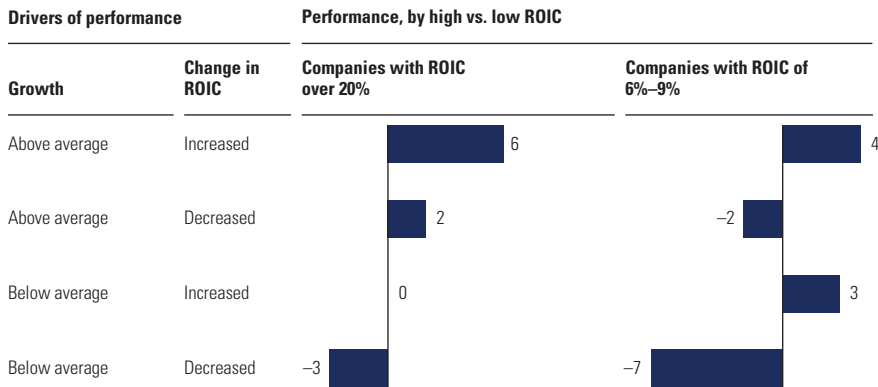
To be fair, this analysis doesn't reflect the risk of failure. Most product ideas fail before reaching the market, and the cost of failed ideas is not reflected in the numbers. By contrast, acquisitions typically bring existing revenues and cash flows that limit the downside risk to the acquirer. But including the risk of failure would not change the pecking order of investments from a value-creation viewpoint.

The interaction between growth and ROIC is a key factor to consider when assessing the likely impact of a particular investment on a company's overall ROIC. For example, we've found that some very successful, high-ROIC companies in the United States are reluctant to invest in growth if it will reduce their returns on capital. One technology company had a 30 percent operating margin and ROIC of more than 50 percent, so it didn't want to invest in projects that might earn only 25 percent returns, fearing this would dilute its average returns. But as the first principle of value creation would lead you to expect, even an opportunity with a 25 percent return would still create value as long as the cost of capital was lower, despite the resulting decline in average ROIC.

The evidence backs this up. We examined the performance of 157 companies with high (greater than 20 percent) ROIC over two time periods: 1996–2005

EXHIBIT 3.9 Impact of Growth and ROIC on High- and Low-ROIC Companies

Median annualized TSR vs. S&P 500, 1996–2005 and 2010–2017, %



Source: B. Jiang and T. Koller, "How to Choose between Growth and ROIC," *McKinsey on Finance*, no. 25 (Autumn 2007): 19–22. Updated to include 2010–2017 data by the authors of this book.

and 2010–2017.⁷ Not surprisingly, the companies that created the most value, measured by total shareholder returns, were those that grew fastest and maintained their high ROICs (see Exhibit 3.9). But the second-highest value creators within this group were those that grew fastest even though they experienced moderate declines in their ROICs. They created more value than companies that increased their ROICs but grew slowly.

We've also seen companies with low returns pursue growth on the assumption that this will also improve their profit margins and returns, reasoning that growth will increase ROIC by spreading fixed costs across more revenues. As mentioned earlier in this chapter, however, except at small start-up companies, faster growth rarely fixes a company's ROIC problem. Low returns usually indicate a poor industry structure (as is the case with airlines in Europe and Asia),⁸ a flawed business model, or weak execution. If a company has a problem with ROIC, the company shouldn't grow until the problem is fixed.

The evidence backs this up as well. We examined the performance of 110 low-ROIC companies (the right column in Exhibit 3.9). The companies that

⁷ B. Jiang and T. Koller, "How to Choose between Growth and ROIC," *McKinsey on Finance*, no. 25 (Autumn 2007): 19–22. Updated to include 2010–2017 data by the authors of this book.

⁸ Airlines have traditionally suffered from overcapacity and lack of differentiation, leading to price competition and low returns. Recently, U.S. airlines, after a wave of consolidation, have been disciplined about adding capacity and creating ways to charge for services, like checking bags, with the result that returns on capital are higher than in the past.

EXHIBIT 3.10 **Impact on Value of Improving Margin vs. Capital Productivity**

Increase in value from improving ROIC by 1 percentage point ¹ % change			
ROIC, %	Through margin improvement	Through capital productivity	Ratio of margin impact to capital productivity impact
10	20.0	13.5	1.2x
20	6.7	2.9	2.3x
30	4.0	1.2	3.4x
40	2.9	0.6	4.6x

¹ For a company with a 9% cost of capital.

had low growth but increased their ROICs outperformed the faster-growing companies that did not improve their ROICs.

One final factor for management to consider is the method by which it chooses to improve ROIC. A company can increase ROIC by either improving profit margins or improving capital productivity. With respect to future growth, it doesn't matter which of these paths a company emphasizes. But for current operations, at moderate ROIC levels, a one-percentage-point increase in ROIC through margin improvement will have a moderately higher impact on value relative to improving capital productivity. At high levels of ROIC, though, improving ROIC by increasing margins will create much more value than an equivalent ROIC increase by improving capital productivity. Exhibit 3.10 shows how this works for a company that has a 9 percent cost of capital.

The reason for this relationship is best explained by an example. Consider a company with zero growth, \$1,000 of revenues, \$100 of profits, and \$500 of invested capital (translating to a 10 percent margin, a 50 percent ratio of invested capital to revenues, and ROIC of 20 percent). One way to increase ROIC by one percentage point is to increase the profit margin to 10.5 percent, increasing profits by \$5. Since the company is not growing, the \$5 of extra profits translates to \$5 of cash flow each year going forward. Discounting at a 10 percent cost of capital, this represents a \$50 increase in value. The company could also increase ROIC by reducing working capital. If it reduced working capital by \$24, ROIC would increase to 21 percent (\$100 divided by \$476). The company's value would increase only by the \$24 one-time cash inflow from reducing working capital. Future cash flows would not be affected.

ECONOMIC PROFIT COMBINES ROIC AND SIZE

You can also measure a company's value creation using economic profit, a measure that combines ROIC and size into a currency metric (here we use the

U.S. dollar). Economic profit measures the value created by a company in a single period and is defined as follows:

$$\text{Economic Profit} = \text{Invested Capital} \times (\text{ROIC} - \text{Cost of Capital})$$

In other words, economic profit is the spread between the return on invested capital and the cost of capital times the amount of invested capital. Value Inc.'s economic profit for year 1 is \$50 (Value Inc. must have \$500 of starting capital if it earns \$100 at a 20 percent return in year 1):

$$\begin{aligned}\text{Economic Profit} &= \$500 \times (20\% - 10\%) \\ &= \$500 \times 10\% \\ &= \$50\end{aligned}$$

Volume Inc.'s economic profit in year 1 is zero (Volume Inc. must have \$1,000 of starting capital if it earns \$100 at a 10 percent return in year 1):

$$\begin{aligned}\text{Economic Profit} &= \$1,000 \times (10\% - 10\%) \\ &= \$1,000 \times 0\% \\ &= \$0\end{aligned}$$

You can also value a company by discounting its projected economic profit at the cost of capital and adding the starting invested capital. Value Inc. starts with \$500 of invested capital. Its economic profit in year 1 is \$50, which grows at 5 percent. Discounting the growing economic profit at a 10 percent discount rate gives a present value of economic profit of \$1,000.⁹ Use these amounts to solve for value:

$$\begin{aligned}\text{Value} &= \text{Starting Invested Capital} + \text{PV}(\text{Projected Economic Profit}) \\ &= \$500 + \$1,000 \\ &= \$1,500\end{aligned}$$

The value of Value Inc. using the economic-profit approach is \$1,500, exactly the same as with the discounted-cash-flow (DCF) approach.

Economic profit is also useful for comparing the value creation of different companies or business units. Consider Value Inc.'s economic profit of \$50. Suppose Big Inc. had \$5,000 in invested capital but earned only a 15 percent return on capital (and assume it doesn't have investment opportunities with

⁹ The present value of economic profit for a growing perpetuity is economic profit in year 1 divided by the cost of capital minus the growth rate. For Value Inc., the present value of economic profit is therefore $\$50 / (10\% - 5\%)$.

higher returns on capital). Its economic profit would be \$250. Clearly, creating \$250 of economic profit is preferable to creating \$50.

Finally, measuring performance in terms of economic profit encourages a company to undertake investments that earn more than their cost of capital, even if their return is lower than the current average return. Suppose Value Inc. had the opportunity to invest an extra \$200 at a 15 percent return. Its average ROIC would decline from 20 percent to 18.6 percent, but its economic profit would increase from \$50 to \$60.

CONSERVATION OF VALUE

A corollary of the principle that discounted cash flow (DCF) drives value is the conservation of value: anything that doesn't increase cash flows doesn't create value. That means value is conserved, or unchanged, when a company changes the ownership of claims to its cash flows but doesn't change the total available cash flows—for example, when it substitutes debt for equity or issues debt to repurchase shares. Similarly, changing the appearance of the cash flows without actually changing the cash flows—say, by changing accounting techniques—doesn't change the value of a company.¹⁰ While the validity of this principle is obvious, it is worth emphasizing because executives, investors, and pundits so often forget it, as when they hope that one accounting treatment will lead to a higher value than another or that some fancy financial structure will turn a mediocre deal into a winner.

The battle over how companies should account for executive stock options illustrates the extent to which executives continue to believe (erroneously) that the stock market is unaware of the conservation of value. Even though there is no cash effect when executive stock options are issued, they reduce the cash flow available to existing shareholders by diluting their ownership when the options are exercised. Under accounting rules dating back to the 1970s, companies could exclude the implicit cost of executive stock options from their income statements. In the early 1990s, as options became more material, the Financial Accounting Standards Board (FASB) proposed a change to the accounting rules, requiring companies to record an expense for the value of options when they are issued. A large group of executives and venture capitalists thought investors would be spooked if options were brought onto the income statement. Some claimed that the entire venture capital industry would be decimated because young start-up companies that provide much of their compensation through options would show low or negative profits.

The FASB issued its new rules in 2004,¹¹ more than a decade after taking up the issue and only after the bursting of the dot-com bubble. Despite dire

¹⁰ In some cases, a company can increase its value by reducing its cost of capital by using more debt in its capital structure. However, even in this case, the underlying change is to reduce taxes, but the overall pretax cost of capital doesn't change. See Chapter 33 for further discussion.

¹¹ Financial Accounting Standard 123R, released in December 2004, effective for periods beginning after June 15, 2005.

predictions, the stock prices of companies didn't change when the new accounting rules were implemented, because the market already reflected the cost of the options in its valuations of companies.¹² One respected analyst told us, "I don't care whether they are recorded as an expense or simply disclosed in the footnotes. I know what to do with the information."

In this case, the conservation of value principle explains why executives didn't need to worry about any effects that changes in stock option accounting would have on their share price. The same applies to questions such as whether an acquisition creates value simply because reported earnings increase, whether a company should return cash to shareholders through share repurchases instead of dividends, or whether financial engineering creates value. In every circumstance, executives should focus on increasing cash flows rather than finding gimmicks that merely redistribute value among investors or make reported results look better. Executives should also be wary of proposals that claim to create value unless they're clear about how their actions will materially increase the size of the pie. If you can't pinpoint the tangible source of value creation, you're probably looking at an illusion, and you can be sure that's what the market will think, too.

Conserving Value: A Brief History

The value conservation principle is described in the seminal textbook *Principles of Corporate Finance*, by Richard Brealey, Stewart Myers, and Franklin Allen.¹³ One of the earliest applications of the principle can be found in the pioneering work of Nobel Prize winners Franco Modigliani and Merton Miller, financial economists who in the late 1950s and early 1960s questioned whether managers could use changes in capital structure to increase share prices. In 1958, they showed that the value of a company shouldn't be affected by changing the structure of the debt and equity ownership unless the overall cash flows generated by the company also change.¹⁴

Imagine a company that has no debt and generates \$100 of cash flow each year before paying shareholders. Suppose the company is valued at \$1,000. Now suppose the company borrows \$200 and pays it out to the shareholders. Our knowledge of the core valuation principle and the value conservation principle tells us that the company would still be worth \$1,000, with \$200 for the creditors and \$800 for the shareholders, because its cash flow available to pay the shareholders and creditors is still \$100.

¹² D. Aboody, M. Barth, and R. Kasznik, "Firms' Voluntary Recognition of Stock-Based Compensation Expense," *Journal of Accounting Research* 42, no. 2 (December 2004): 251–275; D. Aboody, M. Barth, and R. Kasznik, "SFAS No. 123 Stock-Based Compensation Expense and Equity Market Values," *Accounting Review* 79, no. 2 (2004): 251–275; M. Semerdzhian, "The Effects of Expensing Stock Options and a New Approach to the Valuation Problem" (working paper, May 2004, SSRN).

¹³ R. Brealey, S. Myers, and F. Allen, *Principles of Corporate Finance*, 12th ed. (New York: McGraw-Hill/Irwin, 2017).

¹⁴ F. Modigliani and M. H. Miller, "The Cost of Capital, Corporation Finance and the Theory of Investment," *American Economic Review* 48, no. 3 (1958): 261–297.

In most countries, however, borrowing money does change cash flows because interest payments are tax deductible. The total taxes paid by the company are lower, thereby increasing the cash flow available to pay both shareholders and creditors. In addition, having debt may induce managers to be more diligent (because they must have cash available to repay the debt on time) and, therefore, increase the company's cash flow. On the downside, having debt could make it more difficult for managers to raise capital for attractive investment opportunities, thereby reducing cash flow. The point is that what matters isn't the substitution of debt for equity in and of itself; it matters only if the substitution changes the company's cash flows through tax reductions or if associated changes in management decisions change cash flows.

In a similar vein, finance academics in the 1960s developed the idea of efficient markets. While the meaning and validity of efficient markets are subjects of continuing debate, especially after the bursting of the dot-com and real estate bubbles, one implication of efficient-market theory remains: the stock market isn't easily fooled when companies undertake actions to increase reported accounting profit without increasing cash flows. One example is the market's reaction to changes in accounting for employee stock options, as described in the previous section of this chapter. And when the FASB eliminated goodwill amortization effective in 2002 and the International Accounting Standards Board (IASB) did the same in 2005, many companies reported increased profits, but their underlying values and stock prices didn't change, because the accounting change didn't affect cash flows. The evidence is overwhelming that the market isn't fooled by actions that don't affect cash flow, as we will show in Chapter 7.

A Tool for Managers

The conservation of value principle is so useful because it tells us what to look for when analyzing whether some action will create value: the cash flow impact and nothing else. This principle applies across a wide range of important business decisions, such as accounting policy, acquisitions (Chapter 31), corporate portfolio decisions (Chapter 28), dividend payout policy (Chapter 33), and capital structure (also Chapter 33).

This section provides three examples where applying the conservation of value principle can be useful: share repurchases, acquisitions, and financial engineering.

Share Repurchases Share repurchases have become a popular way for companies to return cash to investors (see Chapter 33 for more detail). Until the early 1980s, more than 90 percent of the total distributions by large U.S. companies to shareholders were dividends, and less than 10 percent were share

repurchases. But since 1998, about 50 percent of total distributions have been share repurchases.¹⁵

While buying back shares is often a good thing for management to do, a common fallacy is that share repurchases create value simply because they increase earnings per share (EPS).¹⁶ For example, assume that a company with \$700 of earnings and 1,000 shares outstanding borrows \$1,000 to repurchase 10 percent of its shares. For every \$1,000 of shares repurchased, the company will pay, say, 5 percent interest on its new debt. After tax savings of 25 percent, its total earnings would decline by \$37.50, or 5.4%. However, the number of shares has declined by 10 percent, so earnings per share (EPS) would increase by about 5 percent.

A 5 percent increase in EPS without working very hard sounds like a great deal. Assuming the company's P/E ratio doesn't change, its market value per share also will increase by 5 percent. In other words, you can get something for nothing: higher EPS with a constant P/E.

Unfortunately, this doesn't square with the conservation of value, because the total cash flow of the business has not increased. While EPS has increased by 5 percent, the company's debt has increased as well. With higher leverage, the company's equity cash flows will be more volatile, and investors will demand a higher return. This will bring down the company's P/E, offsetting the increase in EPS.

Moreover, you must consider where the company could have invested the cash rather than returning it to shareholders. If the return on capital from the investment exceeded the company's cost of capital, it's likely that the longer-term EPS would be higher from the investment than from the share repurchases. Share repurchases increase EPS immediately, but possibly at the expense of lower long-term earnings.¹⁷

However, even if cash flow isn't increased by a buyback, some have rightly argued that repurchasing shares can reduce the likelihood that management will invest the cash at low returns. If this is true and it is likely that management would otherwise have invested the money unwisely, then you have a legitimate source of value creation, because the operating cash flows of the company would increase. Said another way, when the likelihood of investing cash at low returns is high, share repurchases make sense as a tactic for avoiding value destruction. But they don't in themselves create value.

Some argue that management should repurchase shares when the company's shares are undervalued. Suppose management believes that the current

¹⁵ T. Koller, "Are Share Buybacks Jeopardizing Future Growth?," *McKinsey on Finance*, no. 56 (October 2015), www.mckinsey.com.

¹⁶ O. Ezekoye, T. Koller, and A. Mittal, "How Share Repurchases Boost Earnings without Improving Returns," *McKinsey on Finance*, no. 58 (April 2016), www.mckinsey.com.

¹⁷ *Ibid.*

share price of the company doesn't reflect its underlying potential, so it buys back shares today. One year later, the market price adjusts to reflect management's expectations. Has value been created? Once again, the answer is no, value has not been created; it has only been shifted from one set of shareholders (those who sold) to the shareholders who did not sell. So while the holding shareholders may have benefited, the shareholders as a whole were not affected. Buying back shares when they are undervalued may be good for the shareholders who don't sell, but studies of share repurchases have shown that companies aren't very good at timing share repurchases, often buying when their share prices are high, not low.¹⁸

Executives as a rule need to exercise caution when presented with transactions like share repurchases that appear to create value by boosting EPS. Always ask, "Where is the source of the value creation?" Some research and development (R&D)-intensive companies, for example, have searched for ways to capitalize R&D spending through complex joint ventures, hoping to lower expenses that reduce EPS. But does the joint venture create value by increasing short-term EPS? No, and in fact it may destroy value because the company now transfers upside potential—and risk, of course—to its partners.

Acquisitions Chapter 31 covers acquisitions in more detail, but for now we can say that acquisitions create value only when the combined cash flows of the two companies increase due to cost reductions, accelerated revenue growth, or better use of fixed and working capital.

To give you a sense of how a good transaction might work, we'll use the example of United Rentals' purchase of RSC (another equipment rental company) for \$1.9 billion in 2011. Within several years, they had achieved more than \$250 million of annual cost savings. We conservatively estimated that the cost savings were worth over \$1.5 billion in present value. That's equivalent to about 80 percent of the purchase price.

A revenue acceleration example comes from Johnson & Johnson, which in 1994 acquired Neutrogena, a maker of skin-care products, for \$924 million. Over the next eight years, management introduced 20 new products within existing product categories and launched an entire line of men's care products. It also accelerated the brand's presence outside the United States. As a result, J&J increased Neutrogena's sales from \$281 million to \$778 million by 2002.

¹⁸ B. Jiang and T. Koller, "The Savvy Executive's Guide to Buying Back Shares," *McKinsey on Finance*, no. 41 (Autumn 2011): 14–17. The results here are counter to academic studies that were based on earlier samples and included many small companies. Our study included only companies in the S&P 500. When share buybacks were rare, announcements were made with great fanfare and often provided strong signals of management's concern for capital discipline. Most of the fanfare has faded, as companies regularly repurchase shares, so announcements aren't a surprise to the market anymore.

The common element of both these acquisitions was radical performance improvement, not marginal change. But sometimes we have seen acquisitions justified by what could only be called magic.

Assume, for example, that Company A is worth \$100 and Company B is worth \$50, based on their respective expected cash flows. Company A buys Company B for \$50, issuing its own shares. For simplicity, assume that the combined cash flows are not expected to increase. What is the new Company AB worth?

Immediately after the acquisition, the two companies are the same as they were before, with the same expected cash flows, and the original shareholders of the two companies still own the shares of the combined company. So Company AB should be worth \$150, and the original A shareholders' shares of AB should be worth \$100, while the original B shareholders' shares of AB should be worth \$50.

As simple as this seems, some executives and financial professionals will still see some extra value in the transaction. Assume that Company A is expected to earn \$5 next year, so its P/E is 20 times. Company B is expected to earn \$3 next year, so its P/E is 16.7 times. What then will be the P/E of Company AB? A straightforward approach suggests that the value of Company AB should remain \$150. Its earnings will be \$8, so its P/E will be about 18.8, between A's and B's P/Es. But here's where the magic happens. Many executives and bankers believe that once A buys B, the stock market will apply A's P/E of 20 to B's earnings. In other words, B's earnings are worth more once they are owned by A. By this thinking, the value of Company AB would be \$160, a \$10 increase in the combined value.

There are even terms for this: *multiple expansion* in the United States and *rerating* in the United Kingdom. The notion is that the multiple of Company B's earnings expands to the level of Company A's because the market doesn't recognize that perhaps the new earnings added to A are not as valuable. This must be so, because B's earnings will now be all mixed up with A's, and the market won't be able to tell the difference.

Another version of the multiple-expansion illusion works the other way around. Now suppose Company B purchases Company A. We've heard the argument that since a company with a lower price-to-earnings (P/E) ratio is buying a higher-P/E company, it must be getting into higher-growth businesses. Higher growth is generally good, so another theory postulates that because B is accelerating its growth, its P/E will increase.

If multiple expansion were true, all acquisitions would create value because the P/E on the lower-P/E company's earnings would rise to that of the company with the higher P/E, regardless of which was the buyer or seller. But no data exist that support this fallacy. Multiple expansion may sound great, but it is an entirely unsound way of justifying an acquisition that doesn't have tangible benefits.

Every corporate leader must know this. So why are we discussing such obvious fallacies? The answer is that companies often do justify acquisitions using this flawed logic. Our alternative approach is simple: if you can't point to specific sources of increased cash flow, the stock market won't be fooled.

Financial Engineering Another area where the value conservation principle is important is financial engineering, which unfortunately has no standard definition. For our purposes, we define financial engineering as the use of financial instruments or structures other than straight debt and equity to manage a company's capital structure and risk profile.

Financial engineering can include the use of derivatives, structured debt, securitization, and off-balance-sheet financing. While some of these activities can create real value, most don't. Even so, the motivation to engage in non-value-added financial engineering remains strong because of its short-term, illusory impact.

Consider that many of the largest hotel companies in the United States don't own most of the hotels they operate. Instead, the hotels themselves are owned by other companies, often structured as partnerships or real estate investment trusts (REITs). Unlike corporations, partnerships and REITs don't pay U.S. income taxes; only their owners do. Therefore, an entire layer of taxation is eliminated by placing hotels in partnerships and REITs in the United States. This method of separating ownership and operations lowers total income taxes paid to the government, so investors in the ownership and operating companies are better off as a group, because their aggregate cash flows are higher. This is an example of financial engineering that adds real value by increasing cash flows.

In contrast, sale-leaseback transactions rarely create value for investment-grade companies.¹⁹ In a sale-leaseback transaction, a company sells an asset that it owns but wants to continue to use, such as an office building, to a buyer who then leases it back to the company. Often, the company structures the lease so that it is treated as a sale for accounting purposes, and then removes the asset from the company's balance sheet. It can also use the sale proceeds to pay down debt. Now it appears that the company has fewer assets and less debt. Rental expense replaces future depreciation and interest expense (though rental expense is typically higher than the sum of depreciation and interest expense).

For larger investment-grade companies, the implied interest rate on the lease is often higher than the company's regular borrowing rate, because the lessor uses the creditworthiness of the lessee to finance its purchase. In addition, the company buying the asset must cover its cost of equity and its operating costs.

¹⁹ Both the FASB and IASB changed the lease accounting rules effective for the 2019 calendar year. Under the new rules, all leases greater than one year must be capitalized.

If the company intends to use the asset for its remaining life (by renewing the lease as it expires), then it has created no value, even though the company appears to be less capital-intensive and to have lower debt. In fact, it has destroyed value because the cost of the lease is higher than the cost of borrowing. The company also incurs its own transaction costs and may have to pay taxes on any gain from the sale of the asset. What's more, other creditors and rating agencies will often treat the lease as a debt equivalent anyway.

The transaction may create value if the company wants the ability to stop using the asset before its remaining life expires and wants to eliminate the risk that the value of the asset will be lower when it decides to stop using the asset.

Sale-leaseback transactions may also create value if the lessor is better able to use the tax benefits associated with owning the asset, such as accelerated depreciation. This does not violate the conservation of value principle, because the total cash flows to the companies involved have increased—at the expense of the government.

THE MATH OF VALUE CREATION

Earlier in this chapter, we introduced the value driver formula, a simple equation that captures the essence of valuation. For readers interested in the technical math of valuation, this section will show how we derive the formula. Let's begin with some terminology that we will use throughout the book (Part Two defines the terms in detail):

- *Net operating profit after taxes (NOPAT)* represents the profits generated from the company's core operations after subtracting the income taxes related to those core operations.
- *Invested capital* represents the cumulative amount the business has invested in its core operations—primarily property, plant, and equipment and working capital.
- *Net investment* is the increase in invested capital from one year to the next:

$$\text{Net Investment} = \text{Invested Capital}_{t+1} - \text{Invested Capital}_t$$

- *Free cash flow (FCF)* is the cash flow generated by the core operations of the business after deducting investments in new capital:

$$\text{FCF} = \text{NOPAT} - \text{Net Investment}$$

- *Return on invested capital (ROIC)* is the return the company earns on each dollar invested in the business:

$$\text{ROIC} = \frac{\text{NOPAT}}{\text{Invested Capital}}$$

ROIC can be defined in two ways: as the return on all capital or as the return on new, or incremental, capital. For now, we assume that both returns are the same.

- *Investment rate (IR)* is the portion of NOPAT invested back into the business:

$$IR = \frac{\text{Net Investment}}{\text{NOPAT}}$$

- *Weighted average cost of capital (WACC)* is the rate of return that investors expect to earn from investing in the company and therefore the appropriate discount rate for the free cash flow. WACC is defined in detail in Chapter 15.
- *Growth (g)* is the rate at which the company's NOPAT and cash flow grow each year.

Assume that the company's revenues and NOPAT grow at a constant rate and the company invests the same proportion of its NOPAT in its business each year. Investing the same proportion of NOPAT each year also means that the company's free cash flow will grow at a constant rate.

Since the company's cash flows are growing at a constant rate, we can begin by valuing a company using the well-known cash-flow perpetuity formula:

$$\text{Value} = \frac{FCF_{t=1}}{WACC - g}$$

This formula is well established in the finance and mathematics literature.²⁰

Next, define free cash flow in terms of NOPAT and the investment rate:

$$\begin{aligned} FCF &= \text{NOPAT} - \text{Net Investment} \\ &= \text{NOPAT} - (\text{NOPAT} \times IR) \\ &= \text{NOPAT}(1 - IR) \end{aligned}$$

Earlier, we developed the relationship between the investment rate (IR), the company's projected growth in NOPAT (g), and the return on investment (ROIC):²¹

$$g = \text{ROIC} \times IR$$

²⁰ For the derivation, see T. E. Copeland and J. Fred Weston, *Financial Theory and Corporate Policy*, 3rd ed. (Reading, MA: Addison-Wesley, 1988), Appendix A.

²¹ Technically, we should use the return on new, or incremental, capital, but for simplicity we assume that the ROIC and incremental ROIC are equal.

Solving for IR, rather than g , leads to:

$$IR = \frac{g}{ROIC}$$

Now build this into the definition of free cash flow:

$$FCF = NOPAT \left(1 - \frac{g}{ROIC} \right)$$

Substituting for free cash flow in the cash-flow perpetuity formula gives the key value driver formula:²²

$$\text{Value} = \frac{NOPAT_{t=1} \left(1 - \frac{g}{ROIC} \right)}{WACC - g}$$

This formula underpins the discounted-cash-flow (DCF) approach to valuation, and a variant of the equation lies behind the economic-profit approach. Chapter 10 describes in depth these two mathematically equivalent valuation techniques. You might go so far as to say that this formula represents all there is to valuation. Everything else is mere detail.

Substituting the forecast assumptions given for Value Inc. and Volume Inc. in Exhibit 3.2 into the key value driver formula results in the same values we came up with when we discounted their cash flows:

Company	$NOPAT_{t=1}$, \$	Growth, %	ROIC, %	WACC, %	Value, \$
Value Inc.	100	5	20	10	1,500
Volume Inc.	100	5	10	10	1,000

In most cases, we do not use this formula in practice. The reason is that in most situations, the model is overly restrictive, as it assumes a constant ROIC and growth rate going forward. For companies whose key value drivers are expected to change, we need a model that is more flexible in its forecasts. Nevertheless, while we do not use this formula in practice, it is extremely useful as a means to maintain focus on what drives value.

Until now, we have concentrated on how ROIC and growth drive the DCF valuation. It is also possible to use the key value driver formula to show that ROIC and growth determine the multiples commonly used to analyze company

²² Technically, this formula should use the return on new invested capital (RONIC), not the company's return on all invested capital (ROIC). For convenience throughout this book, we frequently use ROIC to denote both the return on all capital and the return on new invested capital.

valuation, such as price-to-earnings and market-to-book ratios. To see this, divide both sides of the key value driver formula by NOPAT:

$$\frac{\text{Value}}{\text{NOPAT}_{t=1}} = \frac{\left(1 - \frac{g}{\text{ROIC}}\right)}{\text{WACC} - g}$$

As the formula shows, a company's earnings multiple is driven by both its expected growth and its return on invested capital.

You can also turn the formula into a value-to-invested-capital formula. Start with the identity:

$$\text{NOPAT} = \text{Invested Capital} \times \text{ROIC}$$

Substitute this definition of NOPAT into the key value driver formula:

$$\text{Value} = \frac{\text{Invested Capital} \times \text{ROIC} \times \left(1 - \frac{g}{\text{ROIC}}\right)}{\text{WACC} - g}$$

Divide both sides by invested capital:²³

$$\frac{\text{Value}}{\text{Invested Capital}} = \text{ROIC} \left(\frac{1 - \frac{g}{\text{ROIC}}}{\text{WACC} - g} \right)$$

Now that we have explained the logic behind the DCF approach to valuation, you may wonder why analysts' reports and investment-banking pitches so often use earnings multiples, rather than valuations based on DCF analysis. The answer is partly that earnings multiples are a useful shorthand for communicating values to a wider public. A leading sell-side analyst told us that he uses discounted cash flow to analyze and value companies but typically communicates his findings in terms of implied multiples. For example, an analyst might say Company X deserves a higher multiple than Company Y because it is expected to grow faster, earn higher margins, or generate more cash flow. Earnings multiples are also a useful sanity check for your valuation. In practice, we always compare a company's implied multiple based on our valuation with those of its peers to see if we can explain why its multiple is higher or lower in terms of its ROIC or growth rates. See Chapter 18 for a discussion of how to analyze earnings multiples.

²³ If total ROIC and incremental ROIC are not the same, then this equation becomes:

$$\frac{\text{Value}}{\text{Invested Capital}} = \text{ROIC} \left(\frac{1 - \frac{g}{\text{RONIC}}}{\text{WACC} - g} \right)$$

where ROIC equals the return on the company's current capital and RONIC equals the return on new invested capital.

SUMMARY

This chapter has explored how expected cash flows, discounted at a cost of capital, drive value. Cash flow, in turn, is driven by expected returns on invested capital and revenue growth. Companies create value only when ROIC exceeds their cost of capital. Further, higher-ROIC companies should typically prioritize growth over further improving ROIC, as growth is a more powerful value driver for them. In contrast, lower-ROIC companies should prioritize improving ROIC, as it is a stronger value driver for them.

A corollary of this is the conservation of value: anything that doesn't increase cash flows doesn't create value. So changing the appearance of a company's performance through, say, accounting changes or write-ups or write-downs, without changing cash flows, won't change a company's value. Risk enters into valuation both through the company's cost of capital and in the uncertainty of future cash flows. Because investors can diversify their portfolios, the only risk that affects the cost of capital is the risk that investors cannot diversify, a topic we take up in Chapters 4 and 15.

REVIEW QUESTIONS

1. How does return on invested capital (ROIC) affect a company's cash flow? Explain the relationship between ROIC, growth, and cash flow.
2. If value is based on discounted cash flows, why should a company or investor analyze growth and ROIC?
3. Under what circumstances does growth destroy value?
4. Which type of business, a pharmaceutical firm or an electric utility, would benefit more from improving ROIC than from increasing growth? Why?
5. Describe how different types of organic growth might create different amounts of value. Why does organic growth often create more value than growth from acquisitions does?
6. What is economic profit? How is it useful in assessing firm performance?
7. What is the conservation of value principle? Provide some examples of where it might apply.
8. Under what circumstances would changing a company's capital structure affect its value?
9. What is financial engineering? When does it create value?
10. Apply the conservation of value principle to acquisitions.

Risk and the Cost of Capital

In valuing companies or projects, the subjects of risk and the cost of capital are essential, inseparable, and fraught with misconceptions. These misconceptions can lead to damaging strategic mistakes. For example, when a company borrows money to finance an acquisition and applies only the cost of debt to the target's cash flows, it might easily overestimate by two times the target's value. Conversely, when a company adds an arbitrary risk premium to a target's cost of capital in an emerging market, it could underestimate the value of the target by half.

A company's cost of capital is critical for determining value creation and for evaluating strategic decisions. It is the rate at which you discount future cash flows for a company or project. It is also the rate you compare with the return on invested capital to determine if the company is creating value. The cost of capital incorporates both the time value of money and the risk of investment in a company, business unit, or project.

In this chapter, we'll explain why the cost of capital is *not* a cash cost, but an opportunity cost. The opportunity cost is based on what investors could earn by investing their money elsewhere at the same level of risk. This is always an option for publicly listed companies.¹ Only certain types of risks—those that cannot be diversified—affect a company's cost of capital. Other risks, which can be diversified, should only be reflected in the cash flow forecast using multiple cash flow scenarios.

We'll also discuss how much cash flow risk to take on. Companies should take on all investments that have a positive expected value,² regardless of

¹ As a reminder from Chapter 2, the amount of value that companies create is the amount they earn above their cost of capital. That is, companies create value only when they can invest funds at higher returns than their investors can earn themselves.

² This is often referred to as net present value (NPV); we prefer the term *expected value* because it emphasizes the riskiness of underlying cash flows.

their risk profile, unless the projects are so large that failure would threaten the viability of the entire company. Most executives are reluctant to take on smaller risky projects even if the returns are very high. By aggregating projects into portfolios, rather than assessing them individually, executives can often overcome excessive loss aversion.

Our focus in this chapter will be on key principles. Chapter 15 provides detail on how to measure the cost of capital.

COST OF CAPITAL IS AN OPPORTUNITY COST

The cost of capital is not a cash cost. It is an opportunity cost. To illustrate, when one company acquires another company, the alternative might have been to return that cash to shareholders, who could then reinvest it in other companies. So the cost of capital for the acquiring company is the price investors charge for bearing risk—what they could have earned by reinvesting the proceeds in other investments with similar risk.³ Similarly, when valuing individual business units or projects for strategic decision making, the correct cost of capital is what a company's investors could expect to earn in other similarly risky projects, not necessarily the whole company. The core principle is that the cost of capital is driven by investors' opportunity cost, because the executives leading the company are the investors' agents and have a fiduciary responsibility to the company's investors.⁴ That's why the cost of capital is also referred to as the investors' required return or expected return. The meaning of these terms may differ in academia, but for the most part you can use cost of capital, required return, and expected return interchangeably.

Chapter 15 describes in detail how to estimate a company's opportunity cost of capital. Most practitioners use a weighted average cost of capital (WACC), meaning the weighted average of the cost of equity capital and the cost of debt capital.⁵ For now, it's enough to say that a company's cost of equity capital is what investors could earn by investing in a broad portfolio of

³ To be more precise, the cost of capital is the return investors can earn from investing in a well-diversified, "efficient" portfolio of investments with similar risk.

⁴ In some countries, executives also have a duty to the "company," but that concept is typically vaguely defined and does not provide executives with much guidance. For the most part, even in those countries, the opportunity cost for investors is the best calculation to make. In the United States, a recent innovation is the "benefit" corporation, whose charter includes additional objectives that executives can weigh against the interest of shareholders, including positive impact on society, workers, communities, and the environment. The concept is relatively new; not many large listed companies are benefit corporations, the conversion to which requires a shareholder vote.

⁵ The use of WACC is a practical solution. In theory, the opportunity cost of capital is independent of capital structure (a company's amount of debt versus equity) except for the tax benefit of debt. An alternative is to estimate the opportunity cost of capital as the company's cost of equity (what equity investors expect to earn) if it had no debt, adjusted directly for the tax benefit of debt. In theory, the two approaches should yield the same result.

companies (say, the S&P 500), adjusted for the riskiness of the company relative to the average of all companies.

Within a company, individual business units can have different costs of capital if their risk profiles differ. The company's overall cost of capital is simply a weighted average of its business units' costs of capital. In banking, for example, risky trading operations carry much higher costs of capital than more stable retail banking units.

Executives often fail to adequately incorporate the idea of opportunity cost in thinking about their cost of capital. Sometimes they mix up the opportunity cost of capital by associating different funding streams with different investments. For example, when one company acquires another, the buyer might raise enough debt to pay for the entire company. It is tempting to say that the cost of capital for the acquisition is the cost of the debt. But this would be a mistake, because the risk of the target's free cash flows does not equal the risk of the bondholders' cash flows.

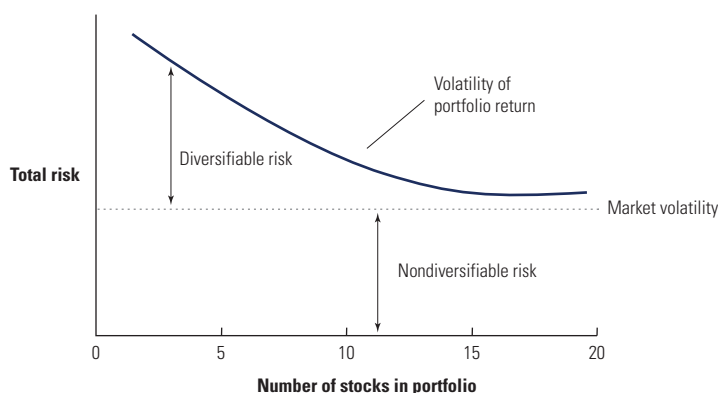
To illustrate, say Company A is considering buying Company B. Both operate in the same product area with similar risk. Company A has no debt and an opportunity cost of capital of 8 percent. Suppose Company A can borrow at 4 percent after taxes. For a target company growing at 3 percent with \$1 billion in earnings and a 15 percent return on capital, the value of the target would be \$80 billion at a 4 percent cost of capital and \$20 billion at an 8 percent cost of capital. To get a sense of how absurd it would be to use the 4 percent cost of capital, consider that the implied price-to-earnings ratio (P/E) at 4 percent is 80, compared with 20 at an 8 percent cost of capital. Companies growing at 3 percent don't trade at a P/E of 80.

In addition, if you apply the cost of debt to the acquisition, you end up with a perverse situation: Company A's existing businesses are assigned an 8 percent cost of capital, and the acquired business is assigned a 4 percent cost of capital. In addition, the only reason Company A can borrow 100 percent of the cost of the acquisition is that it has unused debt capacity in its existing businesses. And don't forget, the cost of capital is determined by the acquired company's riskiness, not that of the parent company (although their risk profiles are likely to be the same if they are in the same industry).

COMPANIES HAVE LITTLE CONTROL OVER THEIR COST OF CAPITAL

It might be surprising to learn that the cost of capital for a company with steady revenues, like Procter & Gamble, isn't that different from a company like LyondellBasell, a chemical company in an industry known for having more variable earnings and cash flows. In 2019, most large companies' WACC fell in the range of 7 to 9 percent. The range is small because investors purposely avoid putting all their eggs in one basket. The ability of investors to diversify their portfolios means that only nondiversifiable risk affects the cost

EXHIBIT 4.1 Volatility of Portfolio Return Declines with Diversification



of capital. Furthermore, because nondiversifiable risk also generally affects all companies in the same industry in the same way, a company's industry is what primarily drives its cost of capital. Companies in the same industry will have similar costs of capital.

Stock market investors, especially institutional investors, may hold hundreds of different stocks in their portfolios. Even the most concentrated investors have at least 50. As a result, their exposure to any single company is limited. It is possible to show how the total risk of a portfolio of stocks declines as more shares are added to the portfolio (Exhibit 4.1). The risk declines because companies' cash flows are not perfectly correlated. Over any period of time, some will increase while others decline.

One of the durable tenets of academic finance concerns the effect of diversification on the cost of capital. If diversification reduces risk to investors and it is not costly to diversify, then investors will not demand a higher return for any risks that can be eliminated through diversification. They require compensation only for risks they cannot diversify away.

The risks they cannot diversify away are those that affect all companies—for example, exposure to economic cycles. However, since most of the risks that companies face are, in fact, diversifiable, most risks don't affect a company's cost of capital. One way to see this in practice is to note the relatively narrow range of P/Es for large companies. Most large companies have P/Es between 12 and 20. If the cost of capital varied from 5 to 15 percent instead of 7 to 9 percent, many more companies would have P/Es below 8 and above 25.

Whether a company's cost of capital is 7 percent or 9 percent or somewhere in between is a question of great dispute (as we explore in Chapter 15). For decades, the standard model for measuring differences in costs of capital has been the capital asset pricing model (CAPM). The CAPM has been challenged

by academics and practitioners, but so far, no practical competing model has emerged.⁶ At any rate, when returns on capital across companies vary from less than 5 percent to more than 30 percent (sometimes even within the same sector), a one-percentage-point difference in the cost of capital seems hardly worth arguing about.

The unique risks that any company faces—say, product obsolescence and new competition—are not priced into the cost of capital. That does not mean a company's value is immune to these risks; they do affect expected cash flows and therefore expected value. Companies certainly do need to worry about the effects of such risks, as we discuss later in this chapter.

It is a common misconception that the cost of capital is company-specific, rather than a function of the industries in which a company operates and the specific investments it makes. For the most part, companies have scant influence over the cost of capital of their individual business units or their company as a whole. There are some theoretical examples of how companies could reduce their cost of capital. For example, a company could outsource production to lower fixed costs and therefore reduce the volatility of cash flows. If you can achieve lower volatility than your peers', your cost of capital will be slightly lower. But it's unlikely that the change in the cost of capital will be large enough relative to other strategic considerations of outsourcing manufacturing. Some companies have shortened the duration of their debt to try to reduce their cost of capital. What these companies fail to recognize is that this increases their risk because of the possibility that interest rates will be higher when the shorter-term debt is rolled over or that the company may have difficulty refinancing the debt at all.

CREATE BETTER FORECASTS, NOT AD HOC RISK PREMIUMS

Certain projects carry what many investors see as high risk.⁷ These include large capital projects in politically unstable countries (common among companies in the mining and oil and gas sectors), speculative R&D projects in

⁶ Many in the academic community use the Fama-French three-factor model, but mostly for capital market research rather than business valuation. With this model, a stock's excess returns are regressed on excess market returns (like the CAPM), the excess returns of small stocks minus big stocks (SMB), and the excess returns of high book-to-market stocks minus low book-to-market stocks (HML). In 2015, the authors expanded the model to five factors, adding operating profitability and investment. See E. Fama and K. French, "The Cross-Section of Expected Stock Returns," *Journal of Finance* (June 1992): 427–465; E. Fama and K. French, "Common Risk Factors in the Returns on Stocks and Bonds," *Journal of Financial Economics* 33 (1993): 3–56; and E. Fama and K. French, "A Five-Factor Asset Pricing Model," *Journal of Financial Economics* 116 (2015): 1–22.

⁷ This section is adapted from R. Davies, M. Goedhart, and T. Koller, "Avoiding a Risk Premium That Unnecessarily Kills Your Project," *McKinsey on Finance*, no. 44 (Summer 2012).

high tech and pharmaceuticals, and acquisitions of unproven technologies or businesses in a wide range of industries. The potential returns for such investments are alluring, but what if the projects or companies fail? The answer is not to ignore these risks, but to explicitly include them in cash flow forecasts, not the cost of capital. The preferred way is to develop multiple cash flow scenarios.

It's not unusual for companies to bump up the assumed cost of capital to reflect the uncertainty of risky projects. In doing so, however, they often unwittingly set these rates at levels that even substantial underlying risks would not justify—and end up rejecting good investment opportunities as a result.⁸ What many don't realize is that assumptions of discount rates that are only three to five percentage points higher than the cost of capital can significantly reduce estimates of expected value. Adding just three percentage points to an 8 percent cost of capital for an acquisition, for example, can reduce its present value by 30 to 40 percent.

Moreover, increasing the discount rate embeds into the valuation process opaque risk assumptions that are often based on little more than a gut sense that the risk is higher. The problem arises because companies take shortcuts when they estimate cash flows. To calculate expected value, project analysts should discount the expected cash flows at an appropriate cost of capital. In many cases, though, they use only estimates of cash flow that assume everything goes well. Managers, realizing this, increase the discount rate to compensate for the possibility that cash flows are overstated.

A better approach for determining the expected value of a project is to develop multiple cash flow scenarios, value them at the unadjusted cost of capital, and then apply probabilities for the value of each scenario to estimate the expected value of the project or company. Exhibit 4.2 provides an example. For simplicity, we assume just two scenarios, one with a present value of \$1,000 and the other with a present value of \$1,667, based on each scenario's expected cash flow. Assuming a 50 percent probability for each scenario leads to an expected value of \$1,333.

EXHIBIT 4.2 **Scenario Approach to Incorporating Nondiversifiable Risk**

Expected net present value (NPV), \$	Probability of scenario	NPV at 8% WACC, \$	Cash flows, \$			
			Year 1	Year 2	Year 3	...
1,333 ←	Base case: 50%	1,667	100	102	104	
	Downside case: 50%	1,000	60	61	62	

⁸ M. Goedhart and P. Haden, "Are Emerging Markets as Risky as You Think?" *McKinsey on Finance*, no. 7 (Spring 2003).

Using scenarios has several advantages:

- It provides decision makers with more information. Rather than looking at a project with a single-point estimate of expected value (say, \$100 million), decision makers know that there is a 20 percent chance that the project's value is $-\$20$ million and an 80 percent chance it is \$120 million. Making implicit risk assumptions explicit encourages dialogue about the risk of the project.
- It encourages managers to develop strategies to mitigate specific risks, because it explicitly highlights the impact of failure or less than complete success. For example, executives might build more flexibility into a project by providing options for stepwise investments—scaling up in case of success and scaling down in case of failure. Creating such options can significantly increase the value of projects.
- It acknowledges the full range of possible outcomes. When project advocates submit a single scenario, they need it to reflect enough upside to secure approval but also be realistic enough that they can commit to its performance targets. These requirements often produce a poor compromise. If advocates present multiple scenarios, they can show a project's full upside potential and realistic project targets they can truly commit to while also fully disclosing a project's potential downside risk.

Managers applying the scenario approach should be wary of overly simplistic assumptions—say, a 10 percent increase or decrease to the cash flows. A good scenario analysis will often lead to a highly successful case that is many multiples of the typical base case. It will often also include a scenario with a negative value. In addition, there may not be a traditional base case. For many projects, there is only big success or failure, with low likelihood that a project will just barely earn more than the cost of capital.

Consider an extreme example. Project A requires an up-front investment of \$2,000. If everything goes well with the project, the company earns \$1,000 per year forever. If not, the company gets zero. (Such all-or-nothing projects are not unusual.) To value project A, finance theory directs you to discount the expected cash flow at the cost of capital. But what is the expected cash flow in this case? If there is a 60 percent chance of everything going well, the expected cash flows would be \$600 per year. At a 10 percent cost of capital, the project would be worth \$6,000 once completed. Subtracting the \$2,000 investment, the net value of the project before the investment is made is \$4,000.

But the project will never generate \$600 per year. It will generate annual cash flows of either \$1,000 or zero. That means the present value of the discounted cash flows will be either \$10,000 or nothing, making the project net of the initial investment worth either \$8,000 or $-\$2,000$. The probability of it being worth the expected value of \$4,000 (that is, \$6,000 less the investment) is zero. Rather than

trying to identify the expected value, managers would be better off knowing that the project carries a 60 percent chance of being worth \$8,000 and a 40 percent risk of losing \$2,000. Managers can then examine the scenarios under which each outcome prevails and decide whether the upside compensates for the downside, whether the company can comfortably absorb the potential loss, and whether they can take actions to reduce the magnitude or risk of loss. The theoretical approach of focusing on expected values, while mathematically correct, hides some important information about the range and exclusivity of particular outcomes.

Moreover, some companies don't apply the expected-value approach correctly. Few companies discuss multiple scenarios, preferring a single-point forecast on which to base a yes-or-no decision. Most companies would simply represent the expected cash flows from this project as being \$1,000 per year, the amount if everything goes well, and allow for uncertainty in the cash flow by arbitrarily increasing the discount rate. While you can get to the right answer with this approach, it has two flaws. First, there is no easy way to determine the cost of capital that gives the correct value. In this case, using a 16.7 percent cost of capital instead of 10 percent results in a project value of \$6,000 before the investment and \$4,000 after the investment. But the only way to know that this is the correct value would be to conduct a thorough scenario analysis. Companies sometimes arbitrarily add a risk premium to the cost of capital, but there is no way for them to know whether the amount they add is even reasonably accurate. Second, the decision makers evaluating a project with cash flows of \$1,000 per year and a 16.7 percent cost of capital are still not thinking through the 40 percent risk that it might generate no cash at all.

If for some reason you must use a single cash flow scenario, you can analytically estimate the equivalent risk premium for different probability levels of failure, as in Exhibit 4.3. The exhibit shows the amounts by which you would increase the cost of capital instead of using cash flow scenarios for different combinations of the probability of failure, as represented on the vertical axis, and the size of loss relative to the base case, as represented by the horizontal axis. For example, if there was a 50 percent chance of failure in which case the project would be worth 40 percent less than expected, the equivalent risk premium would be 1.5 percent. Notice in this exhibit that the risk premiums are small relative to what most people expect. To get close to a 3 percent risk premium, for example, you'd have to believe there was a 50 percent chance of failure and a 60 percent reduction in cash flows associated with failure. To get to a 5 percent risk premium, you'd have to believe there is a 50 percent chance of failure and more than an 80 percent reduction in cash flows.⁹

Adding ad hoc risk premiums is a crude way to include project-specific uncertainty in a valuation. Scenario-based approaches have the dual appeal of better answers and more transparency on the assumptions embedded within them.

⁹ Note that the risk premium will be determined not just by the probability and magnitude of loss, but also by the duration of the project or company and pattern of cash flows.

EXHIBIT 4.3 **Example of Equivalent Risk Premiums for Different Probability Levels of Failure**

		Size of cash flow reduction, %					Risk premium, %
		20	40	60	80	100	
Probability of lower cash flow, %	10	0.1	0.2	0.4	0.5	0.7	
	20	0.2	0.5	0.8	1.1	1.5	
	30	0.4	0.8	1.3	1.9	2.6	
	40	0.5	1.1	1.9	2.8	4.0	
	50	0.7	1.5	2.6	4.0	6.0	

A 1.5% risk premium is required, assuming even odds that an investment will lose 40% of its value

Note: This particular example is for a company with an indefinite life, assuming a smooth cash flow profile, 8% weighted average cost of capital, and 2% terminal growth. The cost of capital adjustments would be larger for a project with a short life.

Source: R. Davis, M. Goedhart, and T. Koller, "Avoiding a Risk Premium That Unnecessarily Kills Your Project," *McKinsey Quarterly* (August 2012).

DECIDE HOW MUCH CASH FLOW RISK TO TAKE ON

Now let's turn to cash flow risk. When we talk about total cash flow risk, we mean the uncertainty that a company faces about its future cash flows, whether for the company as a whole, a business unit, or a single project. Finance theory provides guidance on pricing the nondiversifiable part of cash flow risk in the cost of capital. In theory, a company should take on all projects or growth opportunities that have positive expected values even if there is high likelihood of failure, as long as the project is small enough that failure will not put the company in financial distress. In practice, we've found that companies overweight the impact of losses from smaller projects, thereby missing value creation opportunities.

For instance, how should a company think through whether to undertake a project—let's call it project A—with a 60 percent chance of earning \$8,000, a 40 percent chance of losing \$2,000, and an expected value of \$4,000? Theory says to take on all projects with a positive expected value, regardless of the upside-versus-downside risk. A company is likely to have many small projects like this example, so for small projects, it should take on all projects with positive expected value, regardless of risk.

But what if the company instead has one large project where the downside possibility would bankrupt the company? Consider an electric power company with the opportunity to build a nuclear power facility for \$15 billion (a realistic amount for a facility with two reactors). Suppose the company has \$25 billion in existing debt and \$25 billion in equity market capitalization. If the plant is successfully constructed and brought on line, there is an 80 percent

chance it will be worth \$28 billion, for a net value of \$13 billion. But there is a 20 percent chance it will fail to receive regulatory approval and be worth zero, leading to a loss of \$15 billion. The expected value is \$7 billion net of investment.¹⁰ Failure will bankrupt the company, because the cash flow from the company's existing plants would be insufficient to cover its existing debt plus the debt on the failed plant. In this case, the economics of the nuclear plant spill over onto the value of the rest of the company. Failure would wipe out all the equity of the company, not just the \$15 billion invested in the plant.

The implication is that a company should not take on a risk that will put the rest of the company in danger. In other words, don't do anything that has large negative spillover effects on the rest of the company. This caveat would be enough to guide managers in the earlier example of deciding whether to go ahead with project A. If a \$2,000 loss would endanger the company as a whole, management should forgo the project, despite its 60 percent likelihood of success. But by the same token, companies should not avoid risks that don't threaten their ability to operate normally.

Executives making decisions for their companies should think about the company's risk profile, not their own.¹¹ After all, that's the job of corporations; they are designed to take risks and overcome the natural loss aversion of individuals. The earliest corporations were the British and Dutch East India shipping companies. With those, if a ship sank, all shareholders would lose a tolerable amount instead of having one ship owner lose his entire fortune.

Professors Daniel Kahneman and Amos Tversky have demonstrated that most people place greater weight on the potential economic losses from their decisions than on the potential equivalent gains. In a McKinsey survey of 1,500 global executives across many industries,¹² we presented the executives with the following scenario: You are considering making a \$10 million investment that has some chance of returning, in present value, \$40 million over three years, with some chance of losing the entire investment in the first year. What is the highest loss you would tolerate and still proceed with the investment?

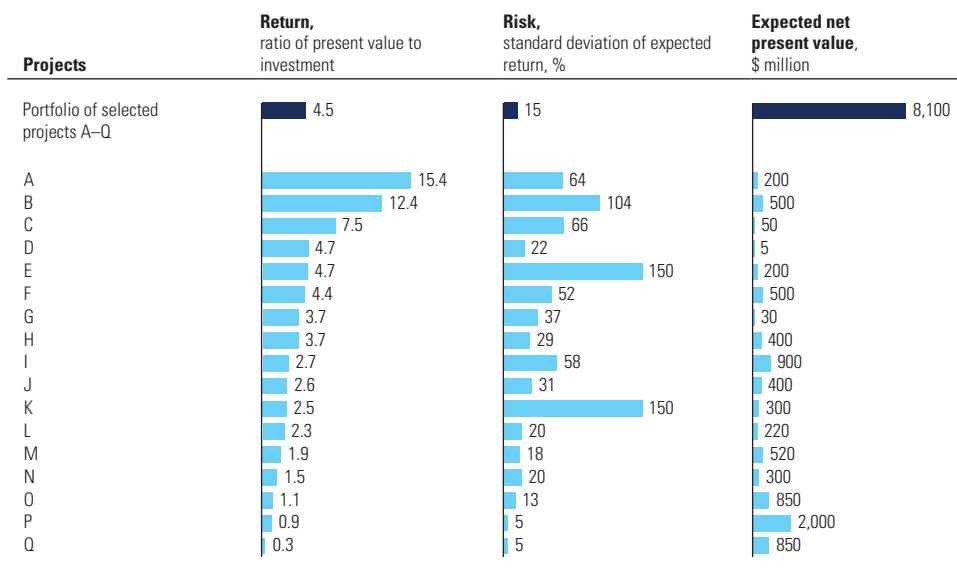
A risk-neutral executive would be willing to accept a 75 percent chance of loss and a 25 percent chance of gain. One-quarter of \$40 million is \$10 million, which is the initial investment, so a 25 percent chance of gain creates an expected risk-neutral value of zero. But most survey respondents demonstrated extreme loss aversion; they were willing to accept only a 19 percent chance of loss to make this investment, nowhere near the risk-neutral answer of 75 percent. In fact, only 9 percent of respondents were willing to accept a 40 percent

¹⁰ The calculation is $(\$13 \text{ billion} \times 80\%) + (-\$15 \text{ billion} \times 20\%)$.

¹¹ "The remainder of this section is adapted from D. Lovo, T. Koller, R. Uhlman, and D. Kahneman, "Your Company Is Too Risk-Averse," *Harvard Business Review* (March–April 2020), hbr.org.

¹² T. Koller, D. Lovo, and Z. Williams, "Overcoming a Bias against Risk," McKinsey & Company (August 2012), <https://www.mckinsey.com/business-functions/strategy-and-corporate-finance/our-insights/overcoming-a-bias-against-risk>.

EXHIBIT 4.4 Aggregating Projects Reduces Risk While Achieving High Expected Returns



or greater chance of loss. Informally, we've asked groups of executives the same question at even lower levels of investment and found similar results. Our findings echo those from Professor Ralph O. Swalm, going back to 1966.¹³

This phenomenon has serious consequences for hierarchical organizations. Executives are just as loss-averse when the bets are small as they are when the gambles are large, even though small gambles do not raise the same issues of survival or ruin that provide a rationale for aversion to large risks. What's more, small gambles offer opportunities for the risk-reducing effects of aggregation.

To overcome loss aversion and make better investment decisions, individuals and organizations must learn to frame choices in the context of the entire company's success, not the individual project's performance. In practice, this means looking at projects as a portfolio by aggregating them, rather than focusing on the risk of individual projects.

One technology company successfully used a portfolio approach to assess its projects. First, executives estimated the expected return of each project proposal (measured as expected present value divided by investment) and the risks associated with each (measured as the standard deviation of projected returns). Executives then built portfolios of projects and identified combinations that would deliver the best balance between return and risk. When they viewed portfolios of projects in the aggregate (Exhibit 4.4), executives could

¹³ These results build on a 1966 *Harvard Business Review* article, "Utility Theory: Insights into Risk Taking," by Ralph O. Swalm. He studied executives with varying levels of spending authority and found that risk-preference profiles were very similar for executives at different levels of the organization.

see that portfolios of projects had higher returns than most of the individual projects and much lower risk compared with most of the individual projects.

It's worth pointing out that even though a portfolio of projects has lower risk, the use of portfolios does not lower a company's cost of capital. That's because the portfolio, by definition, cannot reduce the nondiversifiable risk, which is the risk embedded in the cost of capital.

DECIDE WHICH TYPES OF RISK TO HEDGE

There are also risks that investors are eager for companies to take. For example, investors in gold-mining companies and oil production companies buy those stocks to gain exposure to often-volatile gold or oil prices. If gold and oil companies attempt to hedge their revenues, that effort merely complicates life for their investors, who then must guess how much price risk is being hedged and how and whether management will change its policy in the future. Moreover, hedging may lock in today's prices for two years, the time horizon within which it is possible to hedge those commodities, but a company's present value includes the cash flows from subsequent years at fluctuating market prices. So while hedging may reduce the short-term cash flow volatility, it will have little effect on the company's valuation based on long-term cash flows.

Some risks, like the commodity price risk in this example, can be managed by shareholders themselves. Other, similar-looking risks—for example, some forms of currency risk—are harder for shareholders to manage. The general rule is to avoid hedging the first type of risk but hedge the second if possible.

Consider the effect of U.S. dollar currency risk on Heineken, the global brewer. For the U.S. market, Heineken produces its flagship brand, Heineken, in the Netherlands, and ships it to America. In most other markets, it produces and sells in the same country. So, for most markets, an exchange rate change affects only the translation of local profits into their reporting currency. For example, for most markets, a 1 percent change in the value of the local currency relative to the euro translates into a 1 percent change in revenues and a 1 percent change in profits as well. Note that the effect on revenues and profits is the same, because all the revenues and costs are in the same currency. There is no change in operating margin.

The U.S. market is different. When the dollar/euro exchange rate changes, Heineken's revenues in euros are affected, but its costs are not. If the dollar declines by 1 percent, Heineken's euro revenues also decline by 1 percent. But since its costs are in euros, those don't change. Assuming a 10 percent margin to begin with, a 1 percent decline in the dollar will reduce Heineken's margin to 9 percent, and its profits reported in euros will decline by a whopping 10 percent.

Because Heineken's production facilities are in a different country and it is unable to pass on cost increases because it is competing with locally produced products, its currency risk is larger for its U.S. business than for its other markets. Hedging might be much more important for Heineken's U.S. business than for other markets, because a rise or fall in the dollar/euro exchange rate has a much greater impact on its business.

SUMMARY

To avoid unfavorable strategic decisions, executives must understand well the dynamic relationship between the cost of capital and risk. Risk enters valuation both through the company's cost of capital (an opportunity cost) and through the uncertainty surrounding future cash flows. Because investors can diversify their portfolios, a company's cost of capital is for the most part determined by the industry in which it operates.

Valuations that use multiple cash flow scenarios better reflect diversifiable risks than those that adjust the cost of capital. Executives tend to shy away from risky projects even if the potential return is high. This excessive loss aversion can be overcome by examining portfolios of projects, rather than individual ones.

REVIEW QUESTIONS

1. What broadly determines a firm's cost of capital?
2. Suppose Company A plans to acquire a competitor, Company T, by issuing debt to fund the acquisition. If Company A's cost of debt is 4 percent and its cost of capital is 8 percent, what is the appropriate discount rate to use in valuing the cash flows expected from the acquisition? Why?
3. Can management have a significant influence on its cost of capital? Why?
4. Suppose that you are evaluating three R&D projects within your division of a technology firm. Two projects involve follow-on investments that will use proven technology, while the third involves an unproven technology. How should you evaluate the funding decisions for these three projects?
5. What are some advantages to using a scenario expected cash flow approach over an ad hoc risk premium approach when evaluating high-risk projects?
6. How should managers decide on how much cash flow risk that they should take on for their firms?
7. According to current research, do managers generally follow the guidance given in the above question? Why or why not?
8. Which types of risks should managers hedge? Why?

The Alchemy of Stock Market Performance

A commonly used measure for evaluating the performance of a company and its management is total shareholder returns (TSR), defined as the percent increase in share price plus the dividend yield over a period of time.¹ In fact, in the United States, the Securities and Exchange Commission requires that companies publish in their annual reports their TSR relative to a set of peers over the last five years. That sounds like a good idea: if managers focus on improving TSR to win performance bonuses, then their interests and the interests of their shareholders should be aligned. The evidence shows that this is indeed true over long periods—at a minimum, 10 to 15 years. But TSR measured over shorter periods may not reflect the actual performance of a company, because TSR is heavily influenced by changes in investors' expectations, not just the company's performance.

Earning a high TSR is much harder for managers leading an already-successful company than for those leading a company with substantial room for improvement. That's because a company performing above its peers will attract investors expecting more of the same, pushing up the share price. Managers then must pull off herculean feats of real performance improvement to exceed those expectations and outperform on TSR. We call their predicament the "expectations treadmill." For high-performing companies, TSR in isolation can unfairly penalize their high performance. Another drawback is that using TSR by itself, without understanding its components, doesn't help executives or their boards understand how much of the TSR comes from operating performance, nonoperating items, and changes in expectations.

¹ Later in this chapter, we'll show that we also need to consider the impact of share repurchases as a significant source of cash distributions.

The widespread use of TSR over short periods as a measure of management performance can create perverse incentives. Managers running full tilt on the expectations treadmill may be tempted to pursue ideas that give an immediate bump to their TSR at the expense of longer-term investments that will create more value for shareholders over a longer horizon. In addition, TSR may rise or fall across the board for all companies because of external factors beyond managers' control, such as changing inflation rates. Strictly speaking, such factors should play no part in managers' compensation.

This chapter starts by explaining the expectations treadmill. It then shows an approach to analyzing TSR that isolates how much TSR comes from revenue growth and improvements in return on invested capital (ROIC)—the factors that drive long-term value creation—versus changes in expectations and nonoperating items. Managers, boards of directors, and investors can learn much more about company performance from this granular breakdown of TSR.

WHY SHAREHOLDER EXPECTATIONS BECOME A TREADMILL

As we described in Chapter 2, the return on capital that a company earns is not the same as the return earned by every shareholder. Suppose a company can invest \$1,000 in a factory and earn \$200 a year, which it pays out in dividends to its shareholders. The first investors in the company pay \$1,000 in total for their shares, and if they hold the shares, they will earn 20 percent per year (\$200 divided by \$1,000).

Suppose that after one year, all the investors decide to sell their shares, and they find buyers who pay \$2,000 for the lot. The buyers will earn only 10 percent per year on their investment (\$200 divided by \$2,000). The first investors will earn a 120 percent return (\$200 dividends plus \$1,000 gain on their shares versus their initial investment of \$1,000). The company's return on capital is 20 percent, while one group of investors earns 120 percent, and the other group earns 10 percent. All the investors collectively will earn, on a time-weighted average, the same return as the company. But individual groups of investors will earn very different returns, because they pay different prices for the shares, based on their expectations of future performance.

One way of understanding the effects of this dynamic is through the analogy of a treadmill, the speed of which represents the expectations built into a company's share price. If the company beats expectations, and if the market believes the improvement is sustainable, the company's stock price goes up, in essence capitalizing the future value of this incremental improvement. But it also means that managers must run even faster just to maintain

the new stock price,² let alone improve it further: the speed of the treadmill quickens as performance improves. So a company with low expectations of success among shareholders at the beginning of a period may have an easier time outperforming the stock market simply because low expectations are easier to beat.

The treadmill analogy is useful because it describes the difficulty of continuing to outperform the stock market. At some point, it becomes almost impossible for management to deliver on accelerating expectations without faltering, just as anyone would eventually stumble on a treadmill that keeps moving faster.

Consider the case of Terry Turnaround, a fictional character based on the experience of many CEOs. Terry has just been hired as the CEO of Prospectus, a company with below-average returns on capital and growth relative to competitors. Because of this past performance, the market doesn't expect much, so the value of Prospectus is low relative to competitors. Terry hires a top-notch team and gets to work. After two years, Prospectus is gaining ground on its peers in margins and return on capital, and its market share is rising. Prospectus's stock price rises twice as fast as its peers' because the market wasn't expecting the company's turnaround.

Terry and her team continue their hard work. After two more years, Prospectus has become the industry leader in operating performance, with the highest return on capital. Because of its low starting point, the company's share price has risen at four times the rate of the industry average. Given Prospectus's new trajectory and consistent performance, the market expects continued above-average returns on capital and revenue growth.

As time goes by, Prospectus maintains its high return on capital and leading market share. But two years later, Terry notes with frustration that her company's shares are now doing no better than those of its peers, even though the company has outperformed rivals. At this point, Terry is trapped on the expectations treadmill: she and her team have done such a good job that the expectation of continued high performance is already incorporated into the company's share price. As long as Prospectus delivers results in line with the market's expectations, its share price performance will be no better or worse than average.

This explains why extraordinary managers may deliver only ordinary TSR: even for the extraordinary manager, it can be extremely difficult to keep beating high expectations. It also explains why managers of companies with low performance expectations might easily earn a high TSR, at

² Theoretically, if a company's performance exactly matches expectations, its TSR will equal the cost of equity. In practice, however, with continual changes in interest rates, inflation, and economic activity, comparison to the broader market is sometimes preferable.

least for a short time. They can create a higher TSR by delivering performance that raises shareholder expectations to the level of expectations for their peers in the sector.

The danger for companies whose shareholders already have high expectations is that in their quest to achieve above-peer TSR, they may resort to misguided actions, such as pushing for unrealistic earnings growth or pursuing big, risky acquisitions. Consider the electric power boom at the end of the 1990s and in the early 2000s. Deregulation led to high hopes for power-generation companies, so deregulated energy producers were spun off from their regulated parents at extremely high valuations. Mirant, for instance, was spun off from Southern Company in October 2000 with a combined equity and debt capitalization of almost \$18 billion, a multiple of about 30 times earnings before interest, taxes, and amortization (EBITA)—quite extraordinary for a power-generation company. To justify its value, Mirant expanded aggressively, as did similar companies, investing in power plants in the Bahamas, Brazil, Chile, the United Kingdom, Germany, China, and the Philippines, as well as 14 U.S. states. The debt burden from these investments quickly became too much for Mirant to handle, and the company filed for bankruptcy in July 2003. The expectations treadmill pushed Mirant into taking enormous risks to justify its share price, and it paid the ultimate price.

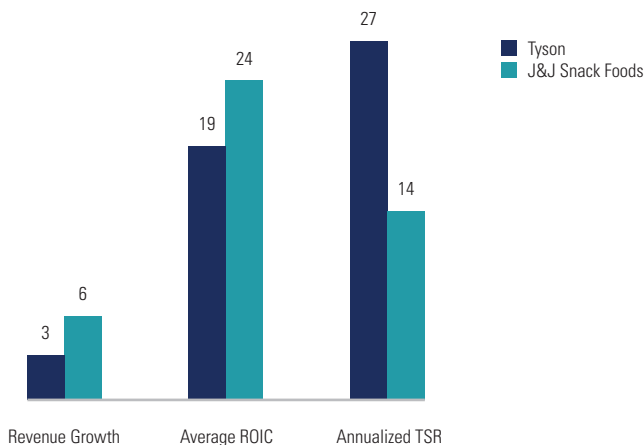
The expectations treadmill is the dynamic behind the adage that a good company and a good investment may not be the same. In the short term, good companies may not be good investments, because future great performance might already be built into the share price. Smart investors may prefer weaker-performing companies, because they have more upside potential, as the expectations expressed in their lower share prices are easier to beat.

THE TREADMILL'S REAL-WORLD EFFECTS

Tyson Foods and J&J Snack Foods are two U.S. branded-food processors. Tyson is one of the largest in the world, with brands such as Hillshire Farm and Sara Lee. Its revenues in 2017 were \$40 billion. J&J Snack Foods is smaller, at just over \$1 billion of revenues in 2017, with brands such as Icee and Auntie Anne's. Not surprisingly, given its smaller size and more snack-oriented products, J&J grew its revenues faster, at 6 percent per year from 2013 to 2017, while Tyson grew only 3 percent (Exhibit 5.1). J&J also outperformed on ROIC, with after-tax ROIC (before goodwill and intangibles) averaging about 24 percent over the period, compared with Tyson's 19 percent. Yet Tyson's shareholders earned almost twice the TSR: 27 percent versus 14 percent annualized.

EXHIBIT 5.1 **Tyson Foods vs. J&J Snack Foods: Growth, Return on Invested Capital (ROIC), and Total Shareholder Returns (TSR)**

Dec 2014–Dec 2017, %



The expectations treadmill explains the mismatch between TSR and the underlying value created by the two companies. Using the ratio of enterprise value (EV) to net operating profit after taxes (NOPAT) as a proxy for market expectations, J&J's EV/NOPAT started the period at 23 times, while Tyson started at 13 times. This means that J&J's treadmill was already running fast, with high expectations already built into the share price. Tyson's EV/NOPAT was below average, reflecting modest performance expectations. The EV/NOPAT for both companies increased during the period—J&J from 23 times to 29 times, and Tyson from 13 times to 17 times.

Another source of the difference in TSR was changes in ROIC, driven primarily by changes in margins. Tyson's adjusted EBITA/revenues increased from 4 percent to 9 percent, while J&J's remained flat at about 12 percent. Similarly, Tyson's ROIC (excluding goodwill) increased from 12 percent to 22 percent, while J&J's declined from 25 percent to 21 percent.³

Which company did a better job? You can make arguments for either one: Tyson succeeded in outperforming its expectations, and J&J Snack Foods succeeded in delivering against high expectations. TSR might have been a fair measure of the performance of Tyson's managers, but it would not have reflected what a great job the J&J team did. For TSR to provide deeper insight into a company's true performance, we need a finer-grained look inside this measure.

³J&J Snack Foods' ROIC declined while its EBITA margin went up because it used more capital (working capital and net property, plant, and equipment) to generate each dollar of revenues in 2017 versus 2013.

DECOMPOSING TSR

We recommend analyzing TSR by decomposing it and quantifying its components in the manner outlined in this section. The effort serves two purposes. First, when managers, boards of directors, and investors understand the sources of TSR, they are better able to evaluate management. For example, it's important to know that J&J's TSR, though lower than Tyson's, reflects strong underlying performance against high expectations. Second, decomposing TSR can help with setting future targets. For example, it may be challenging for Tyson's managers to repeat their high TSR, because that would probably require raising profit margins and earnings multiples much higher.

The traditional approach to analyzing total shareholder returns is mathematically correct, but it does not link TSR to the true underlying sources of value creation. The decomposition we recommend gives managers a clearer understanding of the elements of TSR they can change, those that are beyond their control, and the speed at which their particular expectations treadmill is running. This information helps managers focus on creating lasting value and communicate to investors and other stakeholders how their plans are likely to affect TSR in the short and long terms.

Decomposition of TSR begins with its definition as the percent change in a company's market value plus its dividend yield (for simplicity, we assume the company has no debt and distributes all its excess cash flow as dividends each year):

$$\text{TSR} = \text{Percent Change in Market Value} + \text{Dividend Yield}$$

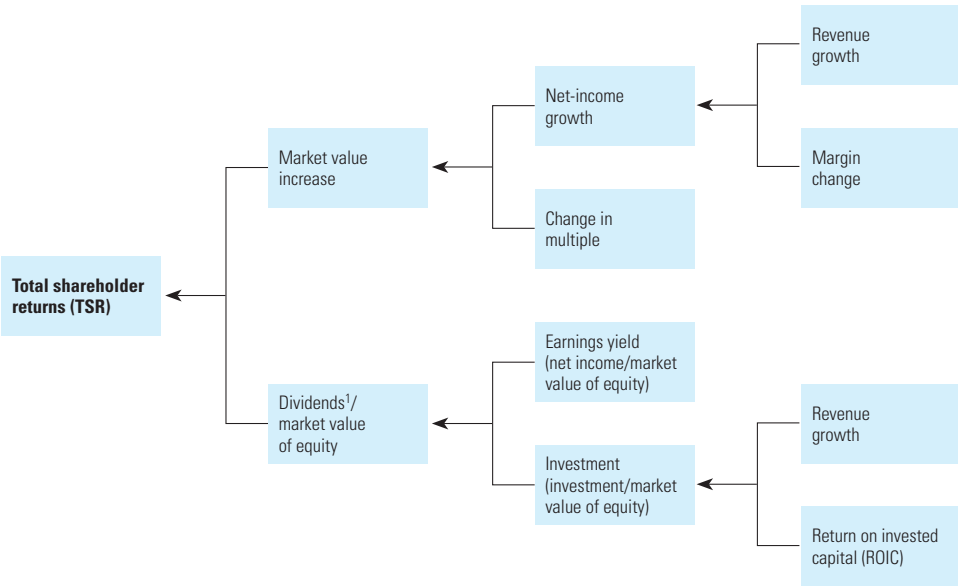
The change in market value is the change in net income plus the change in a company's price-to-earnings ratio (P/E).⁴ Adding the dividend yield gives the following equation for TSR:

$$\begin{aligned} \text{TSR} = & \text{Percent Change in Net Income} + \text{Percent Change in P/E} \\ & + \text{Dividend Yield} \end{aligned}$$

This equation expresses what we refer to as the "traditional" approach to analyzing TSR. While technically correct, however, this expression of TSR misses some important factors. For example, a manager might assume that all forms of net-income growth create an equal amount of value. Yet we know from Chapter 3 that different sources of earnings growth may create different amounts of value, because they are associated with different returns on capital and therefore generate different cash flows. For example, growth from acquisitions may reduce future dividend growth because of the large investments required.

⁴ Technically, there is an additional cross-term, which reflects the interaction of the share price change and the P/E change, but it is generally small, so we ignore it here.

EXHIBIT 5.2 **TSR Driven by Revenue Growth, Margin, ROIC, and Changes in Expectations**



Note: Assumes company has no debt and no share repurchases.

¹Dividends = Net Income – Investment

A second problem is that this approach assumes that the dividend yield can be increased without affecting future earnings and dividends, as if dividends themselves create value. But dividends are merely a residual. For example, if a company pays a higher dividend today by taking on more debt, that simply means future dividends must be lower because future interest expense and debt repayments will be higher. Similarly, if a company manages to pay a higher dividend by forgoing attractive investment opportunities, then future dividends will suffer, as future cash flows from operations will be lower.

Finally, the traditional expression of TSR fails to account for the impact of financial leverage: two companies that create underlying value equally well could generate very different TSR, simply because of the differences in their debt-to-equity ratios and the resulting differences in the risk to their investors.

To avoid these problems, we can decompose the traditional TSR components into ones that provide better insight into understanding the underlying sources of value creation. Exhibit 5.2 shows this graphically.

The derivation works as follows. Assume a company with no debt pays out all its cash flow as dividends. Start with the traditional definition:

$$\begin{aligned} \text{TSR} = & \text{Percent Change in Net Income} + \text{Percent Change in P/E} \\ & + \text{Dividend Yield} \end{aligned}$$

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The percent increase in earnings can be decomposed into the increase in revenues and the change in profit margin:⁵

$$\begin{aligned}\text{Percent Change in Net Income} = & \text{Percent Increase in Revenues} \\ & + \text{Impact of Increase in Profit} \\ & \text{Margin on Net Income}\end{aligned}$$

The dividend yield also can be decomposed:

$$\text{Dividend Yield} = \frac{\text{Dividends}}{\text{Market Value}}$$

In this simplified example, where the company pays out all its cash flow as dividends, dividends will equal net income less investment. Therefore, the dividend yield can be expressed as the earnings yield (net income divided by market value) less the percent of market value invested back into the business:

$$\text{Dividend Yield} = \frac{\text{Net Income}}{\text{Market Value}} - \frac{\text{Investment}}{\text{Market Value}}$$

Putting these components together gives the following expression for TSR:

$$\begin{aligned}\text{TSR} = & \text{Percent Change in Revenue} - \frac{\text{Investment}}{\text{Market Value}} \\ & + \text{Impact of Change in Profit Margin} \\ & + \frac{\text{Net Income}}{\text{Market Value}} + \text{Percent Change in P/E}\end{aligned}$$

To summarize, TSR is driven by these five factors:

1. Revenue growth
2. Investment required to achieve that revenue growth
3. Impact of a change in margin on net income growth
4. Starting ratio of net income to market value (which is the inverse of the P/E ratio)
5. Change in P/E ratio

The investment required for growth is a function of growth and ROIC, as described in Chapter 3:

$$\text{Investment} = \text{Net Income} \times \text{Growth} / \text{ROIC}$$

⁵ To be precise, there is an additional cross-term that reflects the interaction of these two effects, which we have omitted to focus on the key points.

EXHIBIT 5.3 Traditional vs. Enhanced TSR Decomposition

Company A financials			Decomposition of TSR			
\$ million	Base year	1 year later	%	Traditional	Enhanced	
Invested capital	100.0	107.0	Growth	7.0	7.0	
Earnings	12.5	13.4	Required investment	—	(5.6)	
			TSR from performance	7.0	1.4	
P/E (multiple)	10.0	10.3	→			
Equity value	125.0	137.5		Earnings yield	—	10.0
Dividends	5.0	5.5		Change in P/E	3.0	3.0
				Dividend yield	4.4	—
TSR, %		14.4	TSR, %	14.4	14.4	

The percent of market value invested therefore equals

$$\begin{aligned}\frac{\text{Investment}}{\text{Market Value}} &= \frac{\text{Net Income} \times \text{Growth} / \text{ROIC}}{\text{Market Value}} \\ &= \frac{\text{Net Income}}{\text{Market Value}} \times \frac{\text{Growth}}{\text{ROIC}}\end{aligned}$$

The ratio of net income to market value is just the inverse of the P/E ratio; therefore,

$$\frac{\text{Investment}}{\text{Market Value}} = \frac{1}{\text{P/E}} \times \text{Growth} / \text{ROIC}$$

Now we can see how TSR is driven by growth and ROIC, adjusted by the beginning P/E.

Exhibit 5.3 uses the financials of hypothetical Company A to compare the traditional method of TSR decomposition with our enhanced approach.⁶ Looking at the two decomposition approaches on the right side of the exhibit, the traditional approach indicates that Company A has a 14.4 percent TSR, based on 7 percent earnings growth, a 3 percent change in the company's P/E (a proxy for changed expectations), and a 4.4 percent dividend yield. The enhanced approach breaks down the TSR of Company A into three of the four parts of our full process of decomposition (for simplicity, in this example, Company A does not increase its margins). This shows that not much of the 14.4 percent TSR reflects the creation of new value. First, the reinvestment required to achieve 7 percent growth in earnings consumed most of the earnings growth itself, leaving TSR arising from performance at only 1.4 percent. Another 3 percent of TSR comes from a change in shareholder expectations (reflected in the P/E multiple increase), rather than performance, and the

⁶ The example assumes no changes in profit margins for both companies, so that earnings growth can arise only from investments.

EXHIBIT 5.4 Earnings Yield: TSR with Zero Growth

	Company H		Company L	
	Year 0	Year 1	Year 0	Year 1
Earnings, \$	100	100	100	100
P/E	20	20	15	15
Value, \$	2,000	2,000	1,500	1,500
Dividends (equals earnings), \$		100		100
Value plus dividends, \$		2,100		1,600
TSR, %		5.0		6.7
Inverse of P/E, %		5.0		6.7

remaining 10 percent is simply the earnings yield, reflecting what the TSR would have been with zero growth and if investors had not changed their expectations.

We have found that many people struggle with the earnings yield (zero-growth return) part of this decomposition. Here's a simple example of how this works. Suppose you have two companies, H and L, each with \$100 of earnings and zero growth. Since the companies aren't growing, they don't need to invest, so dividends to shareholders would equal earnings. Company H has a P/E of 20, and Company L has a P/E of 15. Exhibit 5.4 shows why the inverse of the P/E, the earnings yield, is the return the companies would earn if they didn't grow and their P/Es didn't change.

In the example, you can see that the TSR of Company H is 5.0 percent, exactly equal to the inverse of the P/E, the earnings yield. Similarly, Company L's TSR of 6.7 percent equals the inverse of its P/E. Note also that Company H, with the higher P/E, has the lower earnings yield (or zero-growth TSR). This demonstrates that companies with higher P/Es must achieve greater growth or improvements in ROIC to outperform the TSR of companies with lower P/Es.

The next example shows the impact of debt financing on the TSR decomposition. Suppose you own a house worth \$500,000 and you've borrowed \$200,000 against the house. If the house increases in value to \$550,000, your equity value would increase from \$300,000 to \$350,000. A 10 percent increase in the value of the house leads to a 17 percent return on your equity.

The same concept applies to companies. Consider Company B, which is identical to Company A (our simpler example in Exhibit 5.3) except for its debt financing. As detailed in Exhibit 5.5, the difference in financing means Company B generated a higher TSR of 18 percent. The traditional approach to decomposing TSR suggests that Company B's shareholders benefited from a higher dividend yield and a stronger increase in expectations. However, our more fundamental decomposition of Company B, based on earnings yield (zero-growth TSR) and changed expectations measured by the unlevered P/E

EXHIBIT 5.5 **Enhancing TSR Decomposition to Uncover Effects of Leverage**

Company B financials			Decomposition of TSR		
\$ million	Base year	1 year later	%	Traditional	Enhanced
Enterprise value	125.0	137.5	Growth	7.0	7.0
Debt ¹	(25.0)	(25.0)	Required investment	—	(5.6)
Equity value	100.0	112.5	TSR from performance	7.0	1.4
Dividends		5.5	Earnings yield	—	10.0
P/E (multiple)	8.0	8.4	Change in P/E ²	5.5	3.0
TSR, %		18.0	Impact of financial leverage	—	3.6
			Dividend yield	5.5	—
			TSR, %	18.0	18.0

¹ Assumes, for illustrative purposes, that debt carries no interest.

² Change in P/E multiple for traditional approach vs. change in unlevered P/E multiple in enhanced approach (enterprise value/earnings).

(ratio of enterprise value to earnings), shows that the first three parts of the company's decomposed TSR are in fact identical to those of Company A. The additional 3.6 percent TSR for Company B arises from the higher proportion of debt in its capital, rather than any newly created value. Adjusting for the higher financial risk associated with higher debt shows that Company B did not in fact create more value than Company A—an important fact for investors and the companies' executives.

We can apply this approach to our earlier comparison of Tyson and J&J Snack Foods. Exhibit 5.6 shows the TSR decomposition for the two companies. While Tyson's 27 percent annual TSR for 2013–2017 was higher than J&J's 14 percent, J&J outperformed Tyson on the TSR derived from growth: growth, net of investments, contributed 3 percent to TSR, versus a negative amount for Tyson, which made significant acquisitions that outweighed its modest revenue growth.

Tyson benefited from a much larger increase in operating profit margin: a TSR effect of 22 percent by increasing its margin from 4 percent to 9 percent, while J&J's margin was flat, at about 12 percent. Note that even though Tyson's

EXHIBIT 5.6 **Tyson Foods vs. J&J Snack Foods: TSR Decomposition, 2013–2017**

% Annualized

	Tyson Foods	J&J Snack Foods	Difference
Revenue growth	3	6	(3)
Investment for growth	(22)	(3)	(19)
Net impact of growth	(19)	3	(22)
Change in margin	22	0	22
TSR from performance	3	3	0
Earnings yield (zero growth return)	7	4	3
Change in earnings multiple	8	7	1
Impact of financial leverage	6	(1)	7
Nonoperating cash flows	3	1	2
TSR	27	14	13

margin increased more, J&J still earned a higher margin. Interestingly, both companies earned similar ROIC in 2017, about 22 percent, because Tyson had higher capital productivity.

While the impact of increasing expectations (the change in multiple) was similar at the two companies, J&J's multiple remained at a much higher level. Tyson's EV/NOPAT multiple increased from 13 times to 17 times, while J&J's increased from 23 times to 29 times.

Tyson had a further seven-percentage-point advantage in TSR due to higher financial leverage. The impact of leverage on J&J's TSR was actually negative, because it had more cash than debt. In contrast, Tyson's debt added six percentage points to its TSR.

UNDERSTANDING EXPECTATIONS

As the examples in this chapter have shown, investors' expectations at the beginning and end of the measurement period have a big effect on TSR. A crucial issue for investors and executives to understand, however, is that a company whose TSR has consistently outperformed the market will reach a point where the company will no longer be able to satisfy expectations reflected in its share price. From that point, TSR will be lower than it was in the past, even though the company may still be creating huge amounts of value. Managers need to realize and communicate to their boards and to investors that a small decline in TSR is better for shareholders in the long run at this juncture than a desperate attempt to maintain TSR through ill-advised acquisitions or new ventures.

This was arguably the point that Home Depot had reached in 1999. Earlier, we used earnings multiples to express expectations, but you can also translate those multiples into the revenue growth rate and ROIC required to satisfy current shareholder expectations by reverse engineering the share price. Such an exercise can also help managers assess their performance plans and spot any gaps between their likely outcome and the market's expectations. At the end of 1999, Home Depot had a market value of \$132 billion, with an earnings multiple of 47. Using a discounted-cash-flow model that assumes constant margins and return on capital, Home Depot would have had to increase revenues by 26 percent per year over the next 15 years to maintain its 1999 share price. Home Depot's actual revenue growth through 2006 averaged a very healthy 13 percent per year, an impressive number for such a large company but far below the growth required to justify its share price in 1999. It's no surprise, therefore, that Home Depot's shares underperformed the S&P 500 by 8 percent per year over the period. Since then, Home Depot's revenues increased from \$90 billion in 2006 to \$108 billion in 2018, an annualized increase of 2 percent per year. A large part of the slow growth was due to the weakness in the housing market, with revenue dropping to \$66 billion in 2010 before recovering to the current level.

What should Home Depot's board of directors have done immediately after 1999, given the company's high market value? Celebrating is definitely not the answer. Some companies would try to justify their high share prices by considering all sorts of risky strategies. But given Home Depot's size, the chances of finding enough high-ROIC growth opportunities to justify its 1999 share price were virtually nil.

Realistically, there wasn't much Home Depot could have done except prepare for an inevitable decline in share price: Home Depot's market value dropped from \$130 billion in December 1999 to \$80 billion in December 2006 (it increased to over \$200 billion by mid-2019). Some companies can take advantage of their high share prices to make acquisitions. But that probably wasn't a good idea for Home Depot because of its high growth—a large-enough management challenge to maintain—even without considering that the retail industry doesn't have a track record of making large acquisitions successfully.

Home Depot's situation in 1999 was unusual. Most companies, most of the time, will not have much trouble satisfying the shareholder expectations expressed in their current share price simply by performing as well as the rest of their industry. We have reverse engineered hundreds of companies' share prices over the years using discounted cash flows. With the exception of the Internet bubble era (1999–2000), at least 80 percent of the companies have had performance expectations built into their share prices that are in line with industry growth expectations and returns on capital. TSR for a company among these 80 percent is unlikely to be much different from the industry average unless the company performs significantly better or worse than expected, relative to its industry peers. The other 20 percent, however, should brace themselves for a significantly faster or slower ride on the treadmill. Managers who reverse engineer their share prices to understand expectations of their ROIC and growth can benefit from seeing on which side of this 80/20 divide they fall.

IMPLICATIONS FOR MANAGERS

The expectations treadmill makes it difficult to use TSR as a performance measurement tool. As we saw in the example of Tyson and J&J Snack Foods, the sizable differences in TSR for the two companies from 2013 to 2017 masked the big difference in expectations at the beginning of the measurement period. In Home Depot's case, living up to the expectations was virtually impossible, as no company can run that fast for very long.

As a result of the expectations treadmill, many executive compensation systems tied to TSR do not reward managers for their performance as managers, since the majority of a company's short-term TSR is driven by movements in its industry and the broader market. That was the case for the many executives who became wealthy from stock options in the 1980s and 1990s, a time

when share prices increased primarily because of falling inflation and interest rates, rather than anything those managers did. Conversely, many stock option gains were wiped out during the 2008 financial crisis. Again, the causes of these gains and losses were largely disconnected from anything managers did or didn't do (except for managers in financial institutions).

Instead of focusing primarily on a company's TSR over a given period, effective compensation systems should focus on growth, ROIC, and TSR performance relative to peers. That would eliminate much of the TSR that is not driven by company-specific performance.

In addition to fixing compensation systems, executives need to become much more sophisticated in their interpretation of TSR, especially short-term TSR. If executives and boards understand what expectations are built into their own and their peers' share prices, they can better anticipate how their actions might affect their own share prices when the market finds out about them. For example, if you're executing a great strategy that will create significant value, but the market already expects you to succeed, you can't expect to outperform on TSR. The management team and board need to know this, so the board will take a long-term view and continue to support management's value-creating priorities, even if these do not immediately strengthen the share price.

Executives also need to give up incessantly monitoring their stock prices. It's a bad habit. TSR is largely meaningless over short periods. In a typical three-month time frame, more than 40 percent of companies experience a share price increase or decrease of over 10 percent,⁷ movements that are nothing more than random. Therefore, executives shouldn't even try to understand daily share price changes unless prices move over 2 percent more than the peer average in a single day or 10 percent more in a quarter.

Finally, be careful what you wish for. All executives and investors like to see their company's share price increase. But once your share price rises, it's hard to keep it rising faster than the market average. The expectations treadmill is virtually impossible to escape, and we don't know any easy way to manage expectations down.

⁷ Share price movement relative to the S&P 500 index for a sample of nonfinancial companies with greater than \$1 billion market capitalization, measured during 2004–2007.

REVIEW QUESTIONS

1. What are total returns to shareholders (TRS), and why is this measure important?
2. What is the expectations treadmill, and how does it affect managers' ability to deliver above-average TRS over long periods of time?
3. What are the potential reasons why TRS over short periods of time may not reflect the actual performance of a company and its management?
4. What actions (good and bad) might managers take when investors have already-high expectations and managers want to outperform peers on TRS?
5. Suppose that you bought 100 shares of SodaCo one year ago for \$10 per share with the expectation of receiving a perpetual dividend of \$1 per share. What was your expected annual percentage return on this investment? Today, SodaCo announces that it will increase its annual dividend to \$2 per share. Upon this announcement, the stock price rises to \$20. If you then sell the stock, what percentage return would you realize on your investment? What annual return would the buyer of your stock expect in the future? Why is there such a difference in returns?
6. How can Company A outperform Company B on all key value drivers (e.g., growth and ROIC) but still deliver lower TRS?
7. Why is the old way of decomposing TRS (into changes in earnings, changes in P/E, and dividend yield) not the best way to understand a company's performance?
8. In the recommended approach to decomposing TRS, explain the theory behind the zero-growth return. What is it? What drives it?
9. Given that TRS is not a clean measure of management performance and is therefore a flawed basis for management compensation, how should a company gauge management performance? What measures should it use?

Valuation of ESG and Digital Initiatives

As we write this book at the beginning of 2020, two items on any executive's agenda are noteworthy for their emerging importance in creating value and their slipperiness when it comes to valuing them. One is managing the intertwined elements of environmental, social, and governance (ESG) concerns. The other is grappling with the myriad manifestations of technological improvement or transformation commonly referred to as "digital."

The principles of corporate valuation do not include simple prescriptions for assigning values to various approaches to ESG or individual digital assets or strategies. Today, for even the most proficient analyst seeking a corporate valuation, there is only so much that can be done with these elusive elements. Many services publish various ESG ratings, for example, but researchers have found that the ratings are uncorrelated across different services, because work is still in progress on how to identify robust metrics of their success.

This chapter offers instead a way to think about how to value strategies and decisions related to ESG and digital initiatives. Our view is that companies should focus on the few areas that make a difference in their industry—for example, water consumption for beverage makers, supply chain management for apparel companies, or carbon emissions for many industries. It's also important, particularly at times of rapid technological change, to fix a gimlet eye on the risks of embracing—or ignoring—trends big and small. In each case, we recommend trying out the basic principles of valuation to establish a foundation for measuring outcomes, combined with gathering data to improve their application in the future.

A COMMON FRAMEWORK

Before we dive into the details of ESG and digital valuation, it's worth pointing out that valuing these strategies or projects follows the same principles that apply to all investment decisions: use discounted cash flows, and compare scenario cash flows with a base case. Often, what is most critical for this analysis is the definition of the base case.

Sometimes executives argue that hard-to-quantify investments are necessary because they are “strategic,” or that their benefits can't be measured. This is rarely the case. The logic error is often in defining the base case. Take the decision by a bank to invest in a mobile-banking app. How would you quantify the value of this investment? The key is the base case. If all of a bank's competitors have mobile apps and the bank doesn't invest in one, its market share will likely fall over time as it loses customers (or fails to attract new ones). Therefore, the base case would be a decline in profits and cash flows, not stable profits and cash flows.

Companies are often reluctant to create business-as-usual projections that show declines in profits and cash flows. Yet such declines are what will most often happen when companies avoid change. Companies must become comfortable with declining-base cases; if they don't, they will have difficulty quantifying the value of many investments in ESG and digital. Quantifying the value is essential to making smart choices. It allows you to compare these initiatives against other investments that may be competing for scarce resources. And as in the example of the mobile-banking app, it may cause you to think about how much to invest in particular initiatives. It's not good enough to look at advancing technology or increasing demand for sustainability and act blindly, based on an uninformed sense of obligation to keep up with outside forces.

ENVIRONMENTAL, SOCIAL, AND GOVERNANCE (ESG) CONCERNS

Every business is deeply intertwined with environmental, social, and governance (ESG) concerns:¹

- Environmental criteria include the energy a company takes in and the waste it discharges, the resources it needs, and the consequences for living beings as a result. Some of the most significant measures are carbon emissions and climate change.

¹ This section on ESG is an adaptation of an article coauthored by one of this book's authors: W. Henisz, T. Koller, and R. Nuttall, “Five Ways That ESG Creates Value,” *McKinsey Quarterly* (November 2019), www.mckinsey.com.

- Social criteria address the relationships a company has and the reputation it fosters with people and institutions in the communities in which it does business. Important criteria include labor relations, diversity, and inclusion.
- Governance is the internal system of practices, controls, and procedures a company adopts in order to govern itself, make effective decisions, comply with the law, and meet the needs of external stakeholders.

These individual elements are themselves intertwined. For example, social criteria overlap with environmental criteria and governance when companies seek to comply with environmental laws and broader societal concerns about sustainability.

The combining of these reputation and business risks and benefits has more executives thinking and acting on ESG in a proactive way. As we discussed in Chapter 1, the U.S. Business Roundtable in August 2019 issued a statement strongly affirming businesses' connection with a broad range of stakeholders, including customers, employees, suppliers, communities, and shareholders.² Investors are becoming more interested in a company's ESG performance, and ESG-oriented investing is on the rise. ESG-related investment funds now top \$30 trillion—up 68 percent since 2014 and tenfold since 2004.³ The acceleration has been driven by heightened social, governmental, and consumer attention to the broader impact of corporations, as well as by the investors and executives who realize that a strong ESG proposition can safeguard a company's long-term success.

The weight of accumulated research finds that companies that pay attention to environmental, social, and governance concerns do not experience a drag on value creation.⁴ Better performance in ESG also corresponds with a

² The stakeholder-minded approach is elaborated upon in Witold J. Henisz, *Corporate Diplomacy: Why Firms Need to Build Ties with External Stakeholders* (New York: Routledge, 2016); J. Browne, R. Nuttall, and T. Stadlen, *Connect: How Companies Succeed by Engaging Radically with Society* (New York: PublicAffairs, 2016); and C. Mayer, *Prosperity: Better Business Makes the Greater Good* (Oxford: Oxford University Press, 2019).

³ *Global Sustainable Investment Review 2018*, Global Sustainable Investment Alliance, 2018, www.gsi-alliance.org.

⁴ W. J. Henisz and J. McGlinch, "ESG, Material Credit Events, and Credit Risk," *Journal of Applied Corporate Finance* 31 (July 2019): 105–117; M. Khan, G. Serafeim, and A. Yoon, "Corporate Sustainability: First Evidence on Materiality," *Accounting Review* 91, no. 6 (November 2016): 1697–1724; and Z. Nagy, A. Kassam, and L.-E. Lee, "Can ESG Add Alpha? An Analysis of ESG Tilt and Momentum Strategies," white paper, MSCI, June 2015, msci.com.

reduction in downside risk, as evidenced, among other ways, by lower loan and credit default swap spreads and higher credit ratings.⁵

In a 2019 McKinsey survey of 558 executives from around the globe and in different industries, 57 percent said they believe that ESG programs create shareholder value.⁶ While nearly all of the 57 percent said these programs create long-term value, two-thirds of them also reported that such programs create value in the short term. Among the major benefits driving value creation, according to respondents, are maintaining a good reputation and brand equity, attracting and maintaining talented employees, and strengthening the company's competitive position. Respondents across the spectrum also said they would be willing to pay a 10 percent premium for a company with a positive ESG record versus one with a negative record.

These favorable opinions do not mean that a company should undertake every ESG idea that comes along. Consistent with valuation principles, our point is that companies should take ESG considerations into account when they make important decisions and that companies should actively look for opportunities to invest in projects that have ESG benefits. Those who look actively may find more positive present value opportunities than they had expected. Where can they look for a strong ESG proposition that makes financial sense? ESG may link to cash flow in five important ways: (1) facilitating revenue growth, (2) reducing costs, (3) minimizing regulatory and legal interventions, (4) increasing employee productivity, and (5) optimizing investment and capital expenditures.

Revenue Growth

A strong ESG proposition helps companies tap new markets and expand in existing ones. When governing authorities trust corporate actors, they are more likely to award them the access, approvals, and licenses that afford fresh

⁵ See, for example, S. A. Lundqvist and A. Vilhelmsson, "Enterprise Risk Management and Default Risk: Evidence from the Banking Industry," *Journal of Risk and Insurance* 85, no. 1 (March 2018), <https://onlinelibrary.wiley.com>; E. Landry, M. Lazaro, and A. Lee, "Connecting ESG and Corporate Bond Performance," MIT Management Sloan School and Breckinridge Capital Advisors, 2017, mitsloan.mit.edu; and M. Reznick and M. Viehs, "Pricing ESG Risk in Credit Markets," Hermes Credit and Hermes EOS, 2017, hermes-investment.com. Similar benefits are found in yield spreads attached to loans; see A. Goss and G. S. Roberts, "The Impact of Corporate Social Responsibility on the Cost of Bank Loans," *Journal of Banking and Finance* 35, no. 7 (2011): 1794–1810, [sciencedirect.com](https://www.sciencedirect.com); S. Chava, "Environmental Externalities and Cost of Capital," *Management Science* 60, no. 9 (September 2014): 2111–2380; S. C. Bae, K. Chang, and H.-C. Yi, "The Impact of Corporate Social Responsibility Activities on Corporate Financing: A Case of Bank Loan Covenants," *Applied Economics Letters* 23, no. 17 (2016): 1234–1237; and S. C. Bae, K. Chang, H.-C. Yi, "Corporate Social Responsibility, Credit Rating, and Private Debt Contracting: New Evidence from Syndicated Loan Market," *Review of Quantitative Finance and Accounting* 50, no. 1 (2018): 261–299.

⁶ See L. Delevingne, A. Gründler, S. Kane, and T. Koller, "The ESG Premium: New Perspectives on Value and Performance," *McKinsey on Finance* 73 (January 2020), www.mckinsey.com.

opportunities for growth. For example, in a recent, massive public–private infrastructure project in Long Beach, California, the for-profit companies selected to participate were screened based on their prior performance in sustainability. Superior ESG execution has demonstrably paid off in mining as well. Consider gold, a commodity (albeit an expensive one) that should, all else being equal, generate the same returns for the companies that mine it regardless of their ESG propositions. Yet one major study found that companies with social engagement activities perceived to be beneficial by public and social stakeholders had an easier go at extracting those resources, without extensive planning or operational delays. These companies achieved demonstrably higher valuations than competitors with lower social capital.⁷

ESG can also drive consumer preference. McKinsey research has shown that customers say they are willing to pay to “go green.” Although there can be wide discrepancies in practice, including customers who refuse to pay even 1 percent more, the researchers found that when consumers were surveyed on purchases in multiple industries, including the automotive, building, electronics, and packaging categories, upward of 70 percent said they would pay an additional 5 percent for a green product if it met the same performance standards as a nongreen alternative. In another study, nearly half (44 percent) of respondents identified business and growth opportunities as the impetus for their companies to start sustainability programs.

The payoffs are real. When Unilever developed Sunlight, a brand of dishwashing liquid that uses much less water than its other brands, sales of Sunlight and Unilever’s other water-saving products proceeded to outpace category growth by more than 20 percent in a number of water-scarce markets. Procter & Gamble, too, is taking aim at developing an estimated \$20 billion product line of detergents that are effective in cold water.⁸ And Finland’s Neste, founded as a traditional petroleum-refining company more than 70 years ago, now generates more than two-thirds of its profits from renewable fuels and sustainability-related products.

Cost Reductions

ESG can also reduce costs substantially. Among other advantages, executing ESG effectively can help combat rising operating expenses (such as raw materials costs and the true cost of water or carbon), which McKinsey research found can boost operating profits by as much as 60 percent. The researchers created a metric—the amount of energy use, water use, and waste created in relation to revenue—to analyze the relative resource efficiency of companies

⁷ W. J. Henisz, S. Dorobantu, and L. J. Nartey, “Spinning Gold: The Financial Returns to Stakeholder Engagement,” *Strategic Management Journal* 35, no. 12 (December 2014): 1727–1748.

⁸ Henisz, *Corporate Diplomacy*.

within various sectors. They found a significant correlation between resource efficiency and financial performance. The study also identified companies across sectors that did particularly well in terms of resource efficiency and financial performance—precisely the companies that had taken their sustainability strategies the furthest.

As with each of the five links to ESG value creation, the first step to realizing value begins with recognizing the opportunity. Consider 3M, which has long understood that being proactive about environmental risk can be a source of competitive advantage. The company has a program called “pollution prevention pays,” which aims to prevent rather than clean up pollution; efforts have included reformulating products, improving manufacturing processes, redesigning equipment, and recycling and reusing waste from production. Since introducing the program in 1975, 3M has saved \$2.2 billion. Another enterprise, a major water utility, achieved cost savings of almost \$180 million per year through lean initiatives aimed at improving preventive maintenance, refining spare-parts inventory management, and tackling energy consumption and recovery from sludge. FedEx, for its part, aims to convert its entire 35,000-vehicle fleet to electric or hybrid engines. To date, 20 percent have been converted, which has already reduced fuel consumption by more than 50 million gallons.⁹

Reduced Regulatory and Legal Interventions

A stronger external-value proposition can enable companies to achieve greater strategic freedom, easing regulatory pressure. In case after case, across sectors and geographies, we’ve seen that strength in ESG helps reduce companies’ risk of adverse government action. It can also engender government support.

The value at stake may be higher than you think. Typically one-third of corporate profits are at risk from state intervention.¹⁰ Regulation’s impact, of course, varies by industry. For pharmaceuticals and health care, the profits at stake are about 25 to 30 percent. In banking, where provisions on capital requirements, “too big to fail” regulations, and consumer protection are so critical, the value at stake is typically 50 to 60 percent. For the automotive, aerospace and defense, and tech sectors, where government subsidies (among other forms of intervention) are prevalent, the value at stake can reach 60 percent as well.

⁹ W. J. Henisz, “The Costs and Benefits of Calculating the Net Present Value of Corporate Diplomacy,” *Field Actions Science Reports*, special issue 14 (2016), <https://journals.openedition.org/factsreports/4109>.

¹⁰ See Henisz et al., “Five Ways That ESG Creates Value.”

Employee Productivity Uplift

A strong ESG proposition can help companies attract and retain quality employees, enhance employee motivation by instilling a sense of purpose, and increase productivity overall. Employee satisfaction is positively correlated with shareholder returns.¹¹ For example, the London Business School's Alex Edmans found that the companies that made *Fortune's* list of the 100 Best Companies to Work For in America generated 2.3 to 3.8 percent higher stock returns per year than their peers, measured over a period of more than 25 years.¹² Moreover, it has long been observed that employees who report feeling not just satisfied but also connected perform better. The stronger an employee's perception of impact on the beneficiaries of their work, the greater the employee's motivation to act in a "prosocial" way.¹³

Recent studies have also shown that positive social impact correlates with higher job satisfaction, and field experiments suggest that when companies "give back," employees react with enthusiasm. For instance, randomly selected employees at one Australian bank who received bonuses in the form of company payments to local charities reported greater and more immediate job satisfaction than their colleagues who were not selected for the donation program.¹⁴

Just as a sense of higher purpose can inspire employees to perform better, a weaker ESG proposition can drag productivity down. The most glaring examples are strikes, worker slowdowns, and other labor actions within an organization. But it's worth remembering that productivity constraints can also manifest outside of a company's four walls, across the supply chain.¹⁵ Primary suppliers often subcontract portions of large orders to other firms or rely on purchasing agents, and subcontractors are typically managed loosely, sometimes with little oversight regarding workers' health and safety.

¹¹ A. Edmans, "Does the Stock Market Fully Value Intangibles? Employee Satisfaction and Equity Prices," *Journal of Financial Economics* 101, no. 3 (September 2011): 621–640.

¹² A. Edmans, "The Link between Job Satisfaction and Firm Value, with Implications for Corporate Social Responsibility," *Academy of Management Perspectives* 26, no. 4 (November 2012): 1–19.

¹³ A. M. Grant, "Does Intrinsic Motivation Fuel the Prosocial Fire? Motivational Synergy in Predicting Persistence, Performance, and Productivity," *Journal of Applied Psychology* 93, no. 1 (2008): 48–58; A. M. Grant, "Relational Job Design and the Motivation to Make a Prosocial Difference," *Academy of Management Review* 32, no. 2 (April 2007): 393–417; and J. S. Bunderson and J. A. Thompson, "Violations of Principle: Ideological Currency in the Psychological Contract," *Academy of Management Review* 28, no. 4 (2003): 571–586.

¹⁴ J.-E. De Neve et al., "Work and Well-Being: A Global Perspective," in *Global Happiness Policy Report*, ed. Global Council for Happiness and Wellbeing (New York: Sustainable Development Solutions Network, 2018).

¹⁵ A.-T. Bové and S. Swartz, "Starting at the Source: Sustainability in Supply Chains," McKinsey & Co., November 2016, www.mckinsey.com.

Farsighted companies pay heed. Consider General Mills, which works to ensure that its ESG principles apply “from farm to fork to landfill.” Walmart, for its part, tracks the work conditions of its suppliers, including those with extensive factory floors in China, according to a proprietary company scorecard. And Mars seeks opportunities where it can deliver what it calls “win-win-wins” for the company, its suppliers, and the environment. Mars has developed model farms that not only introduce new technological initiatives to farmers in its supply chains, but also increase farmers’ access to capital so they are able to obtain a financial stake in those initiatives.¹⁶

Investment and Asset Optimization

A strong ESG proposition can enhance investment returns by allocating capital to more promising and more sustainable opportunities (for example, renewables, waste reduction, and scrubbers). It can also help companies avoid stranded investments that may not pay off because of longer-term environmental issues (such as massive write-downs in the value of oil tankers). Remember, taking proper account of investment returns requires that you start from the proper baseline. When it comes to ESG, it’s important to bear in mind that a do-nothing approach is usually an eroding line, not a straight one. Continuing to rely on energy-hungry plants and equipment, for example, can drain cash going forward. While the investments required to update operations may be substantial, choosing to wait it out can be the most expensive option of all.

The rules of the game are shifting: regulatory responses to emissions will likely add to energy costs and could especially affect balance sheets in carbon-intensive industries. And bans or limitations on such things as single-use plastics or diesel-fueled cars in city centers will introduce new constraints on an immense number of businesses, many of which could find themselves having to play catch-up. One way to get ahead of the future curve is to consider repurposing assets right now—for instance, converting failing parking garages into uses with higher demand, such as residences or day-care facilities, a trend we’re beginning to see in reviving cities.

Foresight flows to the bottom line, and riding sustainability’s tailwinds presents new opportunities to enhance investment returns. “Consider China, for example. The country’s imperative to combat air pollution is forecast to create more than \$3 trillion in investment opportunities through 2030, ranging across industries from air-quality monitoring to indoor air purification and even cement mixing.

¹⁶ K. Askew, “‘Extended Supply Chains Are Broken’: Why Mars Thinks the Commodities Era Is Over,” *Food Navigator*, June 6, 2018, www.foodnavigator.com.

DIGITAL INITIATIVES

The definition of *digital* is fuzzy. Some view it as simply the upgraded term for what their IT function does. Others focus on digital marketing and sales, providing digital services to customers, or connecting devices. The applications of digital technology in organizations involve all of these and probably ideas that haven't been thought of yet. Several of our colleagues examined a typical consumer packaged-goods company to see how many ways digitization and digital applications could be used to improve performance. They identified at least 33 possibilities, including digital marketing, optimizing trade spending, improving sales force coverage, predictive maintenance, supply chain planning, and robotic process automation in the back office.

Given the wide scope of potential digital initiatives, it is no surprise that most companies are launching them. In a 2018 survey of 1,733 managers, about eight in ten said their organizations had begun a digital transformation. However, just 14 percent said their efforts had made and sustained performance improvements. What's more, only 3 percent reported complete success at sustaining their change.¹⁷ Evidently, digital is an area where management discipline is much needed.

Measuring the Value of Digitization

It's not surprising that companies struggle with how to evaluate the myriad "digital" initiatives being proposed. Yet the fundamental principle still applies: evaluate digital projects based on the cash flow they are expected to generate. While it sounds simple, getting it right requires some thoughtful strategic analysis.

Ideally, all investment decisions should be analyzed against an alternative course of action. For digital projects, the alternative may be to do nothing. But the do-nothing case doesn't mean zero cash flows. In fact, the do-nothing or business-as-usual case is often the key to evaluating digital projects.

Banks have faced this challenge several times over the past 40 years. In the 1970s and 1980s, banks introduced automated teller machines. In the 2000s, banks set up online banking. In the 2010s, banks developed mobile-banking apps. It seems obvious that banks needed to introduce all these innovations. But these innovations probably didn't generate new revenues, because customers expected them. Thus, although a mobile-banking app makes strategic sense, it appears to create a negative present value due to its additional costs with no added revenues.

¹⁷ J. Deakin, L. LaBerge, and B. O'Beirne, "Five Moves to Make During a Digital Transformation," McKinsey & Company, April 2019.

Here's where the importance of the base case comes in. If the bank doesn't build a mobile app, it will likely lose market share and revenues over time. In this case, the cash inflows are the avoidance of lost revenues, which could be substantial. So this project likely does have a positive present value.

Ideally, the bank would estimate the timing of market-share loss to decide on the best time to build the app. Perhaps delaying a year or two might maximize value if the bank's customer base isn't clamoring for it yet. The bank should also consider alternative features for the app and ways to build it. Should it start with something simple and low cost to roll out and then improve it over time? Or should it spend more up front on a more feature-laden product? As you can see, there are many different cash flow scenarios to analyze when making this decision.

Paths to Improved Performance

Digital initiatives can improve a company's performance in numerous ways. To analyze the potential impact of digital, it helps to frame the discussion as two opportunities or threats. The first—and the highest-profile manifestation of digital in the business press—is an application of digital tools that fundamentally disrupts an industry, requiring a major revamp of a company's business model.

The second kind of impact, less dramatic but also important, occurs when companies use digital to simply do the things they already do, only better. Digital strategies can be applied in more mundane but also important ways including cost reduction, improved customer experience, new revenue sources, and better decision making. The line between the two applications can blur, such as when clothing retailers integrate their physical and online sales. The retailer is still selling clothes, but the customer's experience has changed, and the retailer must substantially retool its business.

New Business Models In some cases, digital disruption upends entire business models or creates entirely new businesses. The Internet changed the way consumers research and purchase airline tickets and hotel rooms, disintermediating many traditional travel agents. The introduction of video-streaming services has disrupted the economics of traditional broadcast and cable TV channels. In some cases, digital has created enormous new businesses. Cloud computing services generated between \$80 billion and \$100 billion of revenues in 2019, up from less than \$10 billion ten years earlier. The rise of cloud computing disrupted two other industries. First, the standardization of servers by leading players disrupted the manufacturers of mainframe and server computers. Second, it disrupted the IT services business that ran companies' data centers.

To value these new businesses, use the standard DCF approach. The fact that these businesses are often growing fast and don't earn profits early on does not affect the valuation approach. Eventually, they will need to generate profits and cash flow and earn an attractive ROIC. In Chapter 36, we describe how to value high-growth companies. The key point is that with high-growth companies, you must start in the future to estimate revenues when the market begins to stabilize, based on the market's potential size. You should estimate ROIC based on an assessment of the fundamental economics of the business.

An important consideration in estimating the potential size and ROIC of a new digital business is whether or not it will have *network effects*, also called increasing returns to scale. The basic idea is this: in certain situations, as companies grow, they can earn higher margins and return on capital because their product becomes more valuable with each new customer. In most industries, competition forces returns back to reasonable levels. But in industries with network effects, competition is kept at bay by the low and decreasing unit costs of the market leader (hence the tag "winner take all" for this kind of industry).

Take Microsoft's Office software, a product that provides word processing, spreadsheets, and graphics. Office has long been the standard used by most companies and other users. Early on, as the installed base of Office users expanded, it became ever more attractive for new customers to use Office for these tasks, because they could share documents, calculations, and images with so many others. As the customer base grew, margins were very high, because the incremental cost of providing software through DVD or download was so low. Office is one of the most profitable products of all time. That said, even such a successful product may be threatened by competition as more computing moves to the cloud.

Such network effects are not the usual case. The history of innovation shows how difficult it is to earn monopoly-size returns on capital for any length of time except in very special circumstances. Many companies and investors didn't realize how rare this was during the dot-com bubble of 1999–2000. More recently, investors again may have gone overboard with the number of "unicorns," typically defined as start-up companies with values above \$1 billion (usually still private) and negative profits. In 2019, as some unicorns went public or tried to, there was a renewed realization that not all these companies could earn extraordinary returns from network effects, and values fell considerably. It's unlikely that companies offering analytics services, selling e-cigarettes, or renting out short-term office space will achieve long-term network effects.

Cost Reduction Many digital initiatives can help companies reduce their operating costs. Predictive maintenance on factory equipment reduces both maintenance costs and lost production from downtime. Another example is

the grandly named robotic process automation (RPA). This doesn't refer to physical robots, but rather to software that automates processes like accounts-payable processing. As these robots become more sophisticated, they can take on even more difficult tasks, handling exceptions in addition to plain-vanilla accounts payable.

Some examples show great progress for this kind of cost reduction. One mining company saved over \$360 million per year from process automation in the field that gave managers more insight into what exactly was happening, enabling managers to make adjustments and anticipate needed ones. Fossil-fuel power generators have improved a plant's heat rate (how efficiently the plant uses fuel) by up to 3 percent by using sensors and actuators for remote monitoring and automated operations, as well as employing smart valves that self-report and repair leakages. They've also used automated work-order generation, remote expert support using virtual-reality devices, and automated warehouses to reduce operating costs by 5 to 20 percent. At the same time, they have improved safety by using robots for tasks in confined spaces, as well as advanced analytics to prevent accidents due to fatigue or distraction.¹⁸

Understanding the economics of cost reduction is not as straightforward as it may seem. You might be tempted to estimate the present value by simply discounting the expected savings and subtracting the investments required. But you also must examine the second-order effects. Are your competitors pursuing the same initiatives? In a competitive industry like the chemicals business, those cost reductions might simply be passed through to customers as price reductions. Chemical companies typically find ways to reduce costs by around 2 percent per year, but their margins don't increase, because industry players pass the savings on to customers.

In a situation like this, where the present value of cost reduction efforts is zero because the savings are passed on to customers, the alternative case becomes important. If your competitors are pursuing digital initiatives to reduce costs and you are not, you'll still have to reduce your prices in line with your competitors'. The alternative to the digital initiative would be a decline in cash flows due to lower prices without reduced costs. So the present value of the initiative may turn positive again, once you compare your initiative to the right base case. In practice, whether the savings are passed on to customers will vary by industry, but it's critical to think carefully through the alternative case.

Improved Customer Experience Consumers have benefited tremendously from the digital actions of companies serving them. Many retailers have become "omnichannel," giving consumers a high degree of flexibility. Consumers can

¹⁸ G. Guzman, A. Prasanna, P. Safarik, and P. Tanwar, "Unlocking the Value of Digital Operations in Electric-Power Generation," McKinsey & Company, October 2019, www.mckinsey.com.

purchase an item of clothing in a store or online, to be shipped to the buyer's home or to a local store. If the local store doesn't have the right size for an in-store shopper, the customer can order it on the spot and have it delivered to the customer's home. A customer who decides to return an item can return it to any store or mail it back, regardless of how it was purchased. Consumers can also track in real time the progress of shipments heading their way.

Using digitization to improve customer experience can add value to the business in a variety of ways. One leading manufacturer of agricultural products was struggling with low customer satisfaction scores and an erosion of its customer base. Using digital solutions, the company created a seamless online process for ordering, tracking, and query management. This increased the company's customer satisfaction score by 24 percentage points and improved throughput by 20 percent.¹⁹ In some cases, improved customer service also reduces costs. An electricity distribution company fully redesigned its customer interfaces in a "digital-first" way that made a priority of the customer's online interaction. Customer satisfaction rose 25 percentage points, employee satisfaction increased by 10 percentage points, and customer service costs fell 40 percent.

As is the case with applying digital solutions to reduce costs, it's critical to think through the competitive effects of investing in digital to gain a superior customer experience. Recall our earlier example of the mobile-banking app. The value proposition boils down to cash flow, but special considerations emerge. Does the improved customer service lead to higher market share because your customer service is better than that of your competitors? Or does it maintain your market share or avoid losing market share because your competitors are doing the same thing?

In many situations, customers have come to expect an improved customer experience and are unwilling to pay extra for it. In the case of omnichannel retailers, today's customers routinely expect seamless transactions across channels from many retailers, but for the retailers, providing omnichannel services is expensive. The cost to ship online orders often makes these sales unprofitable, while in-store sales may be declining, leading to lower margins, as some costs are fixed. Even so, retailers have no choice but to provide the omnichannel services despite lower profitability. If they don't, they'll lose even more revenues and profits.

New Revenue Sources Some companies have been able to create new revenue sources through digital initiatives. In these cases, the economic analysis versus the base case is more straightforward, because at least for a while, you (and maybe your competitors) are making the pie bigger for the whole

¹⁹ J. Boringer, B. Grehan, D. Kiewell, S. Lehmitz, and P. Moser, "Four Pathways to Digital Growth That Work for B2B Companies," McKinsey & Company, October 2019.

industry. However, genuinely new revenue sources can be hard to find and difficult to convince customers to pay for.

Imagine you are sitting at home with an urge for some ice cream but don't want to go out to the local convenience store. Ben & Jerry's in the United Kingdom has set up centralized ice-cream freezers where a delivery company picks up the ice cream and delivers it to the customer within a short time period. These centralized freezers generate ten times the volume of convenience store freezers—mostly additional sales, because without the convenient delivery, many customers would simply skip the ice cream.

Or consider farm equipment manufacturer John Deere's introduction of precision farming services. The company has created a data-driven service business that collects soil samples and analyzes weather patterns to help farmers optimize crop yields. Sensors in tractors and other machinery provide data for predictive maintenance, automated sprinkler systems synchronize with weather data, and an open-software platform lets third parties build new service apps.²⁰

Then there's one transportation company's digital solution to help its customers improve fleet maintenance. That solution helped generate more than \$10 million of additional revenue through software subscriptions and after-market parts sales.²¹

These new revenue sources can create value because they don't involve just keeping up with the competition. In two of the examples, digital innovations created an overall increase in the revenue pool for the industry. In Ben & Jerry's case, the overall consumption of ice cream increased. In John Deere's case, a new product offering also increased overall demand.

Better Decision Making Finally, some executives are pairing the trove of data being generated and new advanced analytics techniques to enable managers to make better decisions about a broad range of activities, including how they fund marketing, utilize assets, and retain customers.

Consider two examples. A maker of high-tech hardware implemented a partially automated solution to improve pricing for thousands of product configurations. Key features included configuration-based price benchmarking, analysis of price trends, and automated pricing recommendations with weekly updates of up to 200,000 price points for up to 20,000 products. A consumer products company used advanced analytics to improve the design of its planograms. A planogram is a model of how a consumer packaged-goods company allocates its limited space on retail shelves. It describes which products will be included and how to display them. Analytics showed decision

²⁰ J. Bughin, T. Catlin, M. Hirt, and P. Willmott, "Why Digital Strategies Fail," *McKinsey Quarterly* (January 2018), www.mckinsey.com.

²¹ M. Banholzer, M. Berger-de Leon, S. Narayanan, and M. Patel, "How Industrial Incumbents Create New Businesses," McKinsey & Company (September 2019), www.mckinsey.com.

makers at the company that they could dramatically improve effectiveness. At the same time, they reduced the number of people required to design planograms from ten to just two.

Advanced analytics to improve decision making can generate additional revenues, reduce costs, or both. In the planogram example, the improvement can increase total customer spending by getting customers to upgrade to more profitable products. In this case, because the change involves only choices within the company's product mix, the improvement can create value without necessarily inviting a competitive response. In other cases, the benefits may be diluted because competitors take similar actions, but the investment in analytics still may create value by maintaining competitive parity.

CLOSING THOUGHTS

As executives and investors alike grapple with understanding the impact on business value of emerging environmental, social, and governance issues, as well as the competitive implications of digital technologies in all their forms, it bears remembering that these topics and the management responses to them will likely be fluid for some time to come. Still, as new as these topics may seem, existing valuation principles can help with framing initial responses and surfacing techniques for dealing with such challenges as they evolve. Companies will also do well if they concentrate on the specific areas that will have the greatest impact on the factors that create value.

REVIEW QUESTIONS

1. Suppose that you manage CoffeeCo, a coffee store chain with thousands of stores globally. You are considering investing in an app that allows customers to pre-order coffee and food to pick up at their local stores. How should you model this investment decision?
2. What do managers mean when they refer to ESG concerns? What does current research say on how concerned managers are about ESG issues and the relationship between ESG performance and stock performance?
3. In what ways may ESG link to firm cash flows? What are examples of each way?
4. What are two main ways that a firm can improve its performance using digital initiatives?
5. What are network effects? How might digital initiatives create them and why are they significant for a firm's value?
6. In addition to potential industry disruption, what are potential ways that digital initiatives may enhance firm value? What are examples of each way?

The Stock Market Is Smarter Than You Think

The stock market's volatility and the sometimes-erratic pricing of companies' shares have always raised questions about the link between stock prices and economic fundamentals. Some experts have at times even posited that stock markets seem to lead lives of their own. In 2017 the level of market valuations led Nobel laureate Richard Thaler to comment, "We seem to be living in the riskiest moment of our lives, and yet the stock market seems to be napping. . . . I admit to not understanding it."¹ Several years earlier, another Nobel Prize-winning economist, Robert Shiller, wrote, "Fundamentally, stock markets are driven by popular narratives, which don't need basis in solid facts."² American investor Bill Gross claimed in 2012 that the last 100 years of U.S. stock returns "belied a commonsensical flaw much like that of a chain letter or yes—a Ponzi scheme."³

Does it make sense to view the stock market as an arena where emotions rule supreme? We think not. Certainly, irrational behavior can drive prices for some stocks in some sectors in the short term. And for shorter periods of time, even the market overall can lose touch with economic fundamentals. But in the long term, the facts clearly show that individual stocks and the market as a whole track return on invested capital (ROIC) and growth. For this reason, managers should continue to make decisions based on these fundamental drivers of value. By doing so, managers can also detect and perhaps exploit any irrational market deviations if and when they occur.

In this chapter, we'll explain how a market with different types of investors can lead to rational prices most of the time, even if some of the investors don't

¹ J. Smialek, "Nobel Economist Thaler Says He's Nervous about Stock Market," *Bloomberg News*, October 10, 2017, www.bloomberg.com.

² R. Shiller, "When a Stock Market Is Contagious," *New York Times*, October 18, 2014, www.nytimes.com.

³ W. H. Gross, "Cult Figures," *Investment Outlook* (PIMCO), August 2012, www.pimco.com.

make decisions based on economic fundamentals. Then we'll show the empirical evidence that growth and return on invested capital (ROIC) are, in fact, the key drivers of value. Finally, we'll explode the myths behind some commonly accepted beliefs that are at odds with the fundamental principles of valuation.

MARKETS AND FUNDAMENTALS: A MODEL

We use a straightforward model to illustrate how market trading by both fundamental, or informed, investors and nonfundamental investors (what we call "noise traders") will produce prices that are generally in line with intrinsic value but can still be volatile.⁴ These prices may even deviate significantly from intrinsic value under certain, albeit rare, conditions.

Assume a basic market where trading is limited to one company's stock and, for comparison, a risk-free asset. Two types of investors trade in this market. Informed investors develop a point of view about the intrinsic value of the company's shares based on its underlying fundamentals, such as return on capital and growth. They base their buy and sell decisions on this informed point of view. They may not all agree on the intrinsic value. Some may believe the company's shares are worth \$40, others \$50, and others \$60. Because of transaction costs and uncertainty about the intrinsic value, they will trade only if the stock price deviates by more than 10 percent from their value estimates.

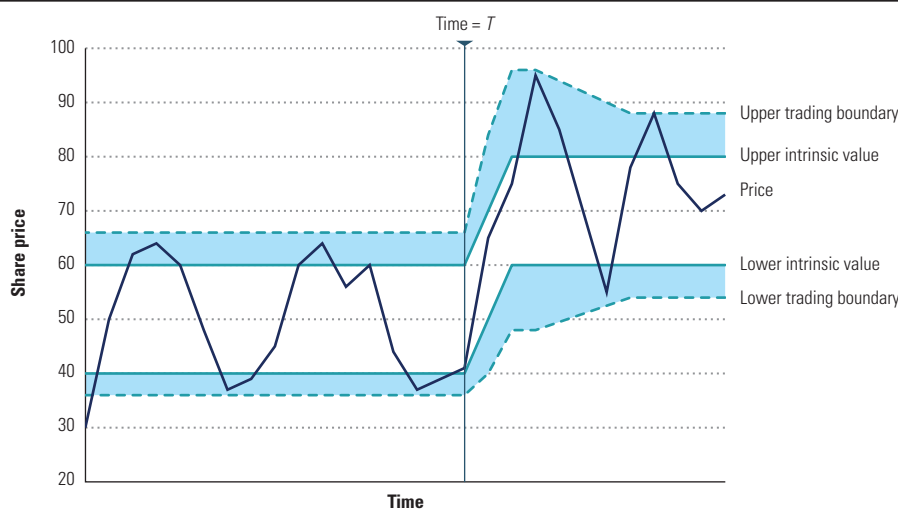
The other investors in this market are the noise traders. These traders may be news oriented, trading on any event they believe will move the share price in the near term, without having a point of view on the company's intrinsic value. Noise traders can also trade on momentum, basing their trades only on price trends: when shares are going up, they buy, assuming the price will continue to increase, and when prices are going down, they sell.⁵

Say trading starts when the price of a single share in the market is \$30. Informed investors start buying shares because they believe the shares should be worth \$40 to \$60; such buying drives up the share price. Some noise traders notice the rising share price and begin to purchase as well. This accelerates the share price increase, attracting more and more noise traders. As the share price increases, the informed investors gradually slow their purchases. At \$44, the most pessimistic begin to sell. Once the price passes \$66, all informed investors are selling. Momentum declines, which some of the noise traders sense, so they begin to sell as well. The selling pressure builds, and the stock

⁴Nonfundamental investors could be called "irrational" because they don't make decisions based on an economic analysis of a company. We call them nonfundamental, because their strategies might be rational and sophisticated even though not based on fundamentals.

⁵Our two investor groups are similar to feedback traders and smart-money investors, as in W. N. Goetzmann and M. Massa, "Daily Momentum and Contrarian Behavior of Index Fund Investors," *Journal of Financial and Quantitative Analysis* 37, no. 3 (September 2002): 375–389.

EXHIBIT 7.1 Model of Share Price Trading Boundaries



price begins to fall. The noise investors accelerate the fall, but this slows as more and more informed investors begin to buy until, at \$36, all informed investors are buying again, and the fall is reversed.

The pattern continues, with the share price oscillating within a band whose boundaries are set by the informed investors, as shown in Exhibit 7.1. If the noise traders act not only on price movements but also on random, insignificant events, there will also be price oscillations within the band. The band itself can change over time, depending on the uncertainty among informed investors about the company's intrinsic value. For example, product launches or successes in research and development can lead informed investors to increase their value estimates as well as their trading bandwidth. As a result, price volatility will be temporarily higher while investors are absorbing the new information, as shown in the period after time T in Exhibit 7.1.

In this model, prices will move within the bandwidth if there is enough informed capital. This mechanism can break down, but only in rare situations. For example, when fundamental investors are vastly outnumbered by noise traders, their sales of stocks might not be able to stop a price rally. Such circumstances are unlikely, given the amounts of capital managed by sophisticated, professional—that is to say, fundamental—investors today.⁶ Nevertheless, once they have sold all the overvalued stock, some fundamental investors can be reluctant to engage in short sales for fear of losing significant amounts before prices revert to lower levels. Others can face institutional or regulatory

⁶ This is also what the academic literature predicts: informed investors outweigh and ultimately survive noise traders. See, for example, L. Blume and D. Easley, "Market Selection and Asset Pricing," in *Handbook of Financial Markets: Dynamics and Evolution*, ed. T. Hens and K. Hoppe (Amsterdam: Elsevier, 2009); and J. De Long, A. Shleifer, L. Summers, and R. Waldman, "The Survival of Noise Traders in Financial Markets," *Journal of Business* 64, no. 1 (1991): 1–19.

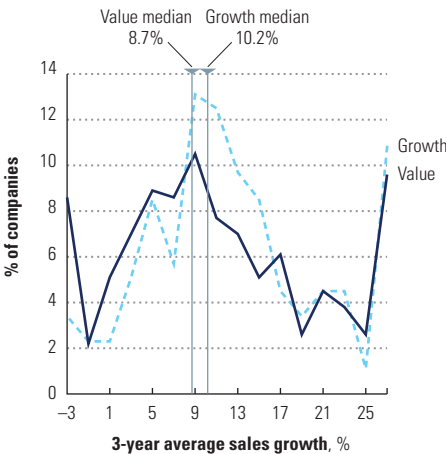
restrictions. As a result, the price rally might continue. But noise traders cannot push share prices above their intrinsic levels for prolonged periods; at some point, fundamentals prevail in setting prices in the stock market. In extreme cases, such as the technology bubble of the 1990s, this could take a few years, but the stock market always corrects itself to align with the underlying fundamental economics.

MARKETS AND FUNDAMENTALS: THE EVIDENCE

In general, the empirical evidence supports the idea that growth and ROIC are the key drivers of value. That means the evidence tends not to support beliefs that value is shaped as much by other measures. Even some of the most conventional beliefs about the stock market are not supported by the facts. For example, most growth and value indexes, like those of Standard & Poor's, categorize companies as either "value" or "growth" based on a combination of factors, including market-to-book ratios and price-to-earnings ratios (P/Es). Typically, companies with high market-to-book ratios and high P/Es end up in the growth category, while the others fall in the value category. However, growth is only one factor driving differences in market-to-book ratios and P/Es. ROIC also is important. In fact, we have found no difference in the distribution of growth rates between so-called value and growth stocks (see Exhibit 7.2). We did, however, find that so-called growth stocks tend to have high ROIC, and value stocks have lower ROIC. The median return on capital for so-called value companies was 15 percent, compared with 35 percent for the growth companies. So the companies classified as growth did not grow faster on average, but they did have higher returns on capital. That's

EXHIBIT 7.2 Distribution of Growth Rates for Growth and Value Stocks

Growth stocks do not grow materially faster . . .



. . . but do have higher ROICs

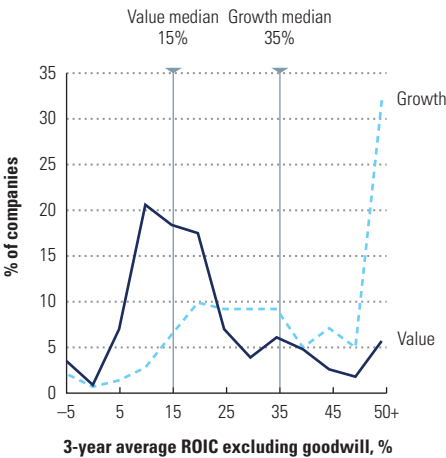
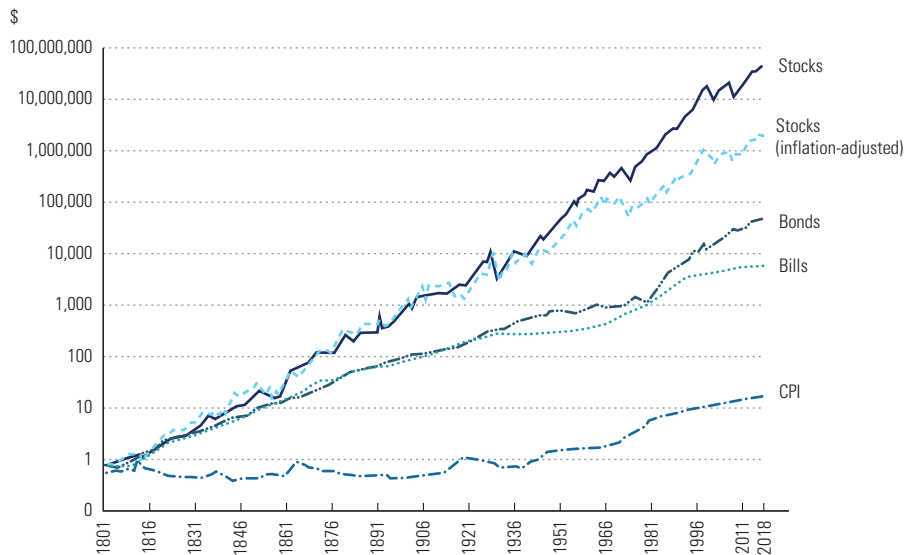


EXHIBIT 7.3 **Stock Performance against Bonds in the Long Run, 1801–2018**

Source: J. J. Siegel, *Stocks for the Long Run: The Definitive Guide to Financial Market Returns and Long-Term Investment Strategies* (New York: McGraw-Hill, 2014); R. G. Ibbotson, 2019 SBBI Yearbook (Duff & Phelps).

why a modestly growing company, like the high-ROIC consumer packaged goods company Clorox, ends up on the growth-stock list.

Decades of Consistent Returns

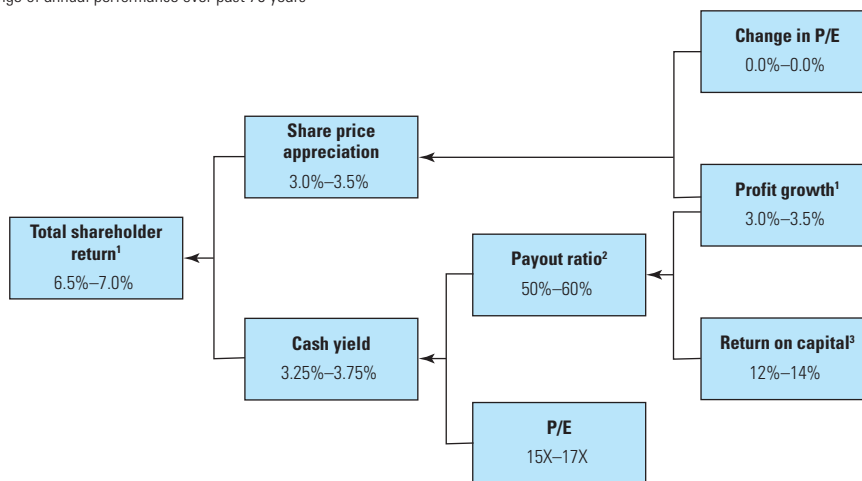
Similarly, market bubbles and crises have always captured public attention, fueling the belief that the stock market moves in chaotic ways, detached from economic fundamentals. The 2008 financial crisis, the technology bubble of the 1990s, the Black Monday crash of October 1987, the leveraged-buyout (LBO) craze of the 1980s, and, of course, the Wall Street crash of 1929 appear to confirm such ideas. But the facts tell a different story. Despite these occurrences, U.S. equities over the past 200 years have delivered decade after decade of consistent returns to shareholders of about 6.75 percent annually, adjusted for inflation. Over the long term, the stock market has been far from chaotic (Exhibit 7.3).

The origins of this 6.75 percent total shareholder return (TSR) lie in the fundamental performance of companies and the long-term cost of equity. TSR is simply the sum of the relative share price appreciation plus the cash yield (see Exhibit 7.4). Over the past 70 years, corporate profits in the United States have grown about 3 to 3.5 percent per year in real terms, and the median P/E has hovered around a level of about 15 to 17.⁷ If P/Es revert to a normal level over time, share price appreciation should therefore amount to around 3 to 3.5 percent per year. Moreover, corporate America typically reinvests about

⁷ Note that the P/E is stable if long-term growth rates, returns on capital, and costs of equity are stable.

EXHIBIT 7.4 Economic Fundamentals Explain Long-Term Total Shareholder Returns

Range of annual performance over past 70 years

¹ Measured in real terms.² Estimated as $(1 - \text{growth}/\text{return on capital})$, where growth is real-terms profit growth plus inflation at 2.0%–2.5%.³ Long-term average ROIC—recent years have been above average levels.

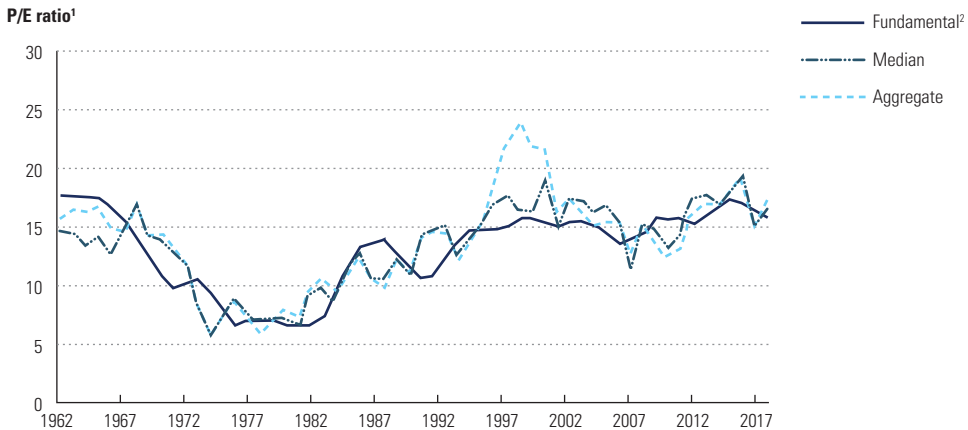
40 to 50 percent of profits every year to achieve this profit growth, leaving the remainder to be paid to shareholders as dividends and share repurchases. The resulting 50 to 60 percent payout ratio is not a coincidence: it follows from a typical 12 to 14 percent return on equity for U.S. companies, combined with 3 to 3.5 percent growth in real terms, or 5 to 6 percent including inflation. It translates to a cash yield to shareholders (that is, the inverse of the P/E times the payout ratio) of around 3.5 percent at the long-term average P/E of 15 to 17. Adding the cash yield to the annual 3 to 3.5 percent share price appreciation results in total real shareholder returns of about 6.5 to 7 percent per year.

P/E Fundamentals

Some analysts miss an important element of stock returns: the gains are driven by both share price appreciation and cash yields. In the view of these analysts share prices cannot increase faster than corporate profits. But this perspective erroneously misses the cash distributions entirely. Other experts have been too pessimistic about the share price appreciation component when they've predicted convergence of the P/E toward long-term average levels. Their estimates for the long-term average level are too low because they incorporate the 1970s and 1980s, when P/Es were severely depressed because of exceptionally high inflation levels.⁸

⁸ In addition, Robert Shiller's measure of the current P/E is overestimated because it does not exclude extraordinary losses such as goodwill impairments. See J. Siegel, "Don't Put Faith in Cape Crusaders," *Financial Times*, August 20, 2013; "Siegel vs. Shiller: Is the Stock Market Overvalued?," Knowledge@Wharton, September 18, 2018, knowledge.wharton.upenn.edu/article/siegel-shiller-stock-market/.

EXHIBIT 7.5 Estimating Fundamental Market Valuation Levels



¹ Price-to-earnings ratio on 12-month forward-looking earnings for S&P 500.

² Moving average over three years.

The fundamental performance of companies and of the economy also explains the level of the stock market over shorter periods of time. We estimated a fundamental P/E for the U.S. stock market for each year from 1962 to 2019, using the simplest equity discounted-cash-flow (DCF) valuation model, following the value driver formula first presented in Chapter 2. We estimated what the price-to-earnings ratios would have been for the U.S. stock market for each year, had they been based on these fundamental economic factors. Exhibit 7.5 shows how well even a simple fundamental valuation model fits the stock market's actual P/E levels over the past decades, despite periods of extremely high economic growth in the 1960s and 1990s, as well as periods of low growth and high inflation in the 1970s and 1980s. By and large, the U.S. stock market has been fairly priced and in general has oscillated around its fundamental P/Es. We conducted a similar analysis of the European stock markets and obtained similar results.

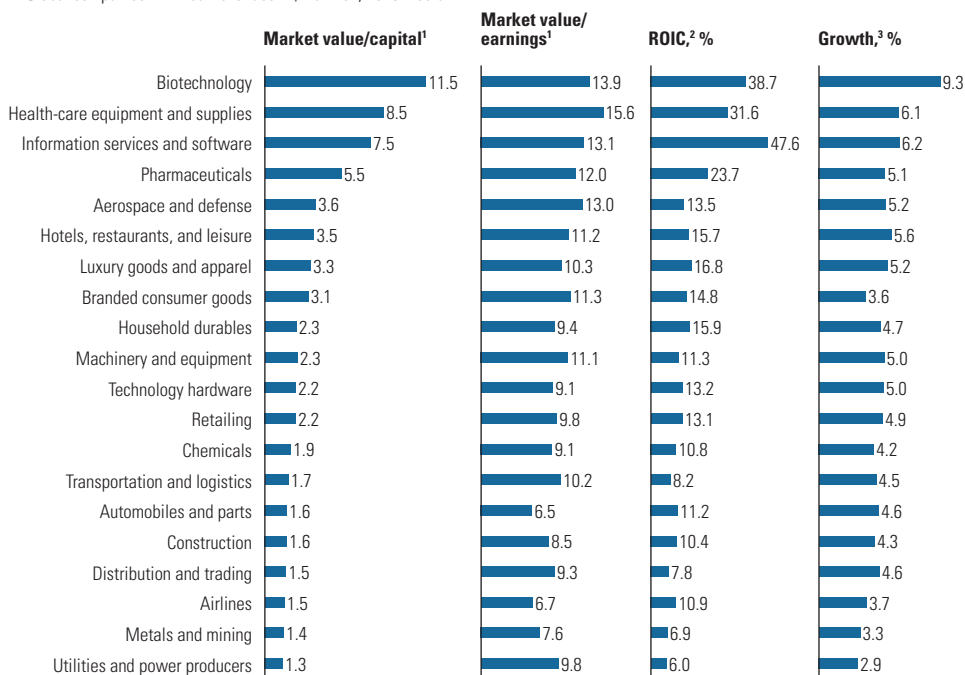
Note that both the fundamental and actual P/Es have shown an upward trend over the past 35 years, rising toward 17 in 2019. To a large extent, this pattern is driven by steadily increasing margins and returns on capital.⁹ Excess cash balances held by large companies form another factor. Cash has a high implied P/E because it carries little after-tax interest. Correcting for the excess cash balance in corporate P/Es lowers the 2017 ratio for the market as a whole by a full point, from 19 to 18.¹⁰

⁹ See also Chapter 8 and R. Jain, B. Jiang, and T. Koller, "What's behind This Year's Buoyant Market," *McKinsey on Finance*, no. 52 (Autumn 2014): 27–31.

¹⁰ See R. Gupta, B. Jiang, and T. Koller, "Looking behind the Numbers for US Stock Indexes," *McKinsey on Finance*, no. 65 (January 2018): 11–15.

EXHIBIT 7.6 **Market Value vs. ROIC and Growth across Selected Industry Sectors**

Global companies with real revenues > \$1 billion, 2018 median



¹ Market value is enterprise value, capital is invested capital excluding goodwill, and earnings is earnings before interest, taxes, depreciation, and amortization (EBITDA).

² Average return on invested capital excluding goodwill over 2015–2017.

³ Analyst consensus forecast of annual revenue growth from 2018 to 2020.

Source: Corporate Performance Analytics by McKinsey.

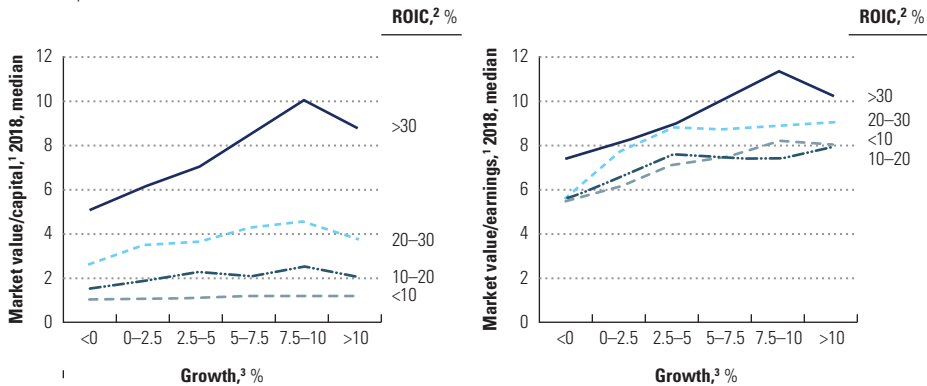
Higher Returns, Higher Value

What holds for the stock market as a whole also holds across industries. For the largest listed companies in the world grouped by industry in 2018,¹¹ we took their average ROIC for the previous three years as a proxy for expected future returns and used the analysts' consensus estimate of their three-year growth outlook as the proxy for long-term expected growth (see Exhibit 7.6). Industries with higher ratios of market value to capital or market value to earnings also have higher growth and/or higher ROIC driven by better sales margins and capital turnover. Life science and technology companies had the highest valuation levels, thanks to having the highest ROIC combined with superior growth. Other companies, like those in the hotels and restaurants or luxury-goods sectors, receive high valuations from strong growth at average

¹¹ This sample comprises all listed companies (excluding financial institutions) with revenues exceeding \$1 billion from the United States, Europe, Australia, New Zealand, and Japan.

EXHIBIT 7.7 **Market Value, ROIC, and Growth: Empirical Relationship**

Global companies with real revenues > \$1 billion



¹ Market value is enterprise value, capital is invested capital excluding goodwill, and earnings is earnings before interest, taxes, depreciation, and amortization (EBITDA).

² Average return on invested capital excluding goodwill over 2016–2017.

³ Analyst consensus forecast of annual earnings growth from 2018 to 2020.

Source: Corporate Performance Analytics by McKinsey.

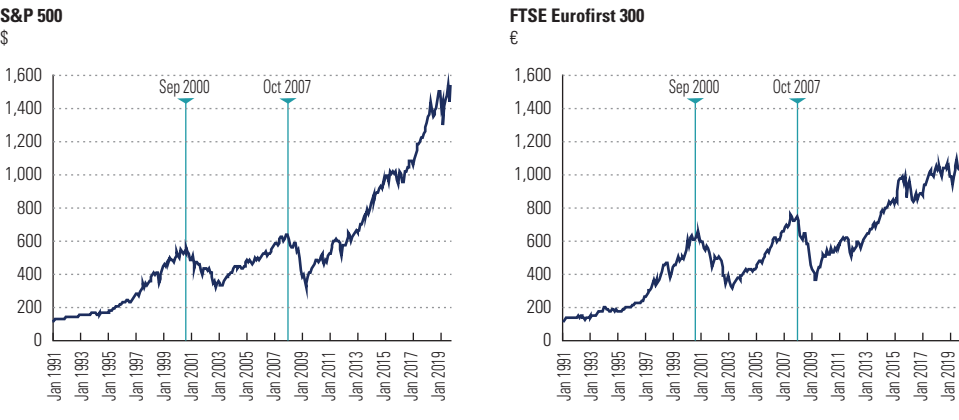
levels of ROIC. Utilities and companies in metals and mining were valued at low market-value-to-capital multiples because of their low returns on capital and low expected growth. Note that the ratios of market value to earnings show less variation across sectors, reflecting investor expectations of converging earnings growth in the long term.

The same principles apply to individual companies. We compared the ratios of market value to capital of all the companies in the same sample versus their expected ROIC and growth. Exhibit 7.7 shows that, for a given level of growth, higher rates of ROIC generally lead to higher market values, and above a given level of ROIC, higher growth also leads to higher value. Although the empirical results do not fit the theoretical model perfectly, they still clearly demonstrate that the market values companies based on growth and ROIC.

For example, consider the fact that valuation multiples in the United States tend to be higher than in most other countries. That fact has even made some European companies consider relisting their stocks in the U.S. stock market in the hope of obtaining a higher valuation. As we discuss later in this chapter, however, such hope is false. U.S. investors do not pay more than European investors for the same stock. The difference in valuation multiples can be explained by underlying fundamentals. First, there is a marked difference in sector composition between the U.S. and European economies. The technology and life science sectors, which have high valuation multiples, carry far more weight in the U.S. economy. Second, we find that U.S. companies typically generate higher returns on capital than European companies in the same sector.

EXHIBIT 7.8 U.S. and European Equity Markets in High-Tech and Credit Bubbles

U.S. and European total return index (January 1991 = 100)



Source: S&P Capital IQ.

Deviations from Fundamentals

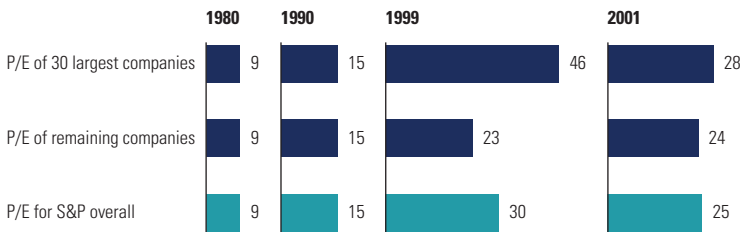
Nevertheless, there have been periods when deviations from economic fundamentals were so significant and widespread that they affected the stock market as a whole. Two examples are the technology bubble that burst in 2000 and the credit bubble that collapsed in 2007 (see Exhibit 7.8).

The technology market boom is a classic example of a valuation bubble, in which stocks are priced at earnings multiples that underlying fundamentals cannot justify. When Netscape Communications became a public company in 1995, it saw its market capitalization soar to \$6 billion on an annual revenue base of just \$85 million. As investors quickly became convinced that the Internet would change the world, they sent the Standard & Poor's (S&P) 500 index to a new peak in 2000. By 2003, the index had tumbled back to half that level.

Although the valuation of the market overall was affected, the technology bubble was concentrated in technology stocks and certain very large stocks (so-called megacaps) in other sectors. Before and after the bubble, the P/E's of the 30 largest companies were about the same on average as those of the other 470 companies in the index (see Exhibit 7.9). However, in 1999, the average

EXHIBIT 7.9 Impact of Largest Stocks on Overall Market Valuation

12-month trailing price-to-earnings ratios



Source: Compustat.

top-30 company had a P/E of 46 times, compared with an average of 23 times for the other 470 companies. As a result, the weighted average P/E for the market overall reached 30 times.

Most of the large-capitalization companies with high P/Es were clustered in just three sectors: technology, media, and telecommunications (TMT). Of course, some of the companies born in this era (including Amazon and eBay) have created substantial economic value. But for every solid, innovative new business idea, there were dozens of companies that forgot or purposely threw out fundamental rules of economics.

By 2007, stock markets around the world had more than recovered from the technology bubble fallout, and the S&P 500 reached a new peak value (see Exhibit 7.8). The largest property boom and credit expansion in U.S. and European history drove corporate earnings to exceptional levels that ultimately proved unsustainable. Although all companies were affected, this bubble too was mainly driven by a few sectors. The financial, energy, utilities, and materials sectors showed sharply inflated earnings, from 41 percent of total S&P earnings in 1997 to 51 percent in 2006. But in 2007, a chain reaction of collapsing funding structures for mortgages and other forms of credit brought financial institutions across the world into distress. U.S. and European stock markets lost more than half of their value as the world's economy experienced the steepest downturn since the 1930s.

After 2009, U.S. stock markets quickly regained momentum and reached new record levels, this time fueled by strong increases in returns on capital and revenue growth, especially in the life science and technology sectors (see also Chapter 8). The megacap phenomenon emerged again, although on a far more modest scale than in the high-tech bubble. As of 2018, just four megacap companies—Alphabet (Google), Amazon, Facebook, and Microsoft—accounted for 10 percent of the S&P 500 index.¹² European stock markets took much longer to regain pre-crisis levels, due to weaker underlying return on capital and growth. The 2010 sovereign debt crisis caused a prolonged slowdown of economic activity across the largest European countries. In addition, European countries did not experience the emergence and ongoing growth of a strong technology sector, as the United States did.

Paradoxically, the fact that market deviations do occur from time to time makes it even more important for corporate managers and investors to understand the true, intrinsic value of their companies; otherwise, they will be unsure how to exploit any market deviations, if and when they occur. For instance, they might use shares to pay for acquisitions when those shares are overvalued by the market, or they might divest particular businesses at times when trading and transaction multiples in those sectors are higher than underlying fundamentals can justify.

¹² See Gupta et al., "Looking behind the Numbers for US Stock Indexes."

MYTHS ABOUT EARNINGS

So far, we've made the positive case for managers to focus their energy on growth at an attractive ROIC. Yet some companies go to great lengths to achieve a certain earnings per share (EPS) number or to smooth out their earnings. This is wasted energy. The evidence shows that these efforts aren't worth it, and they may actually hurt the company.

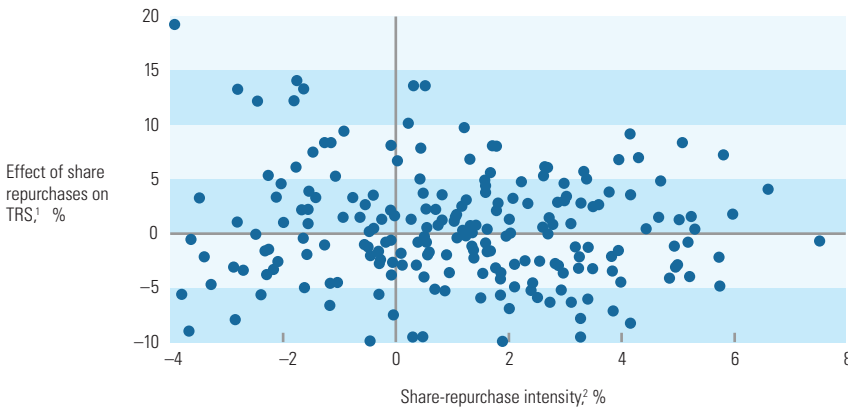
We're not saying that EPS doesn't matter. Companies that create value often have attractive earnings growth, and earnings will equal cash flow over the life span of the company. But not all earnings growth creates value. Consider the three most important drivers of EPS growth: revenue growth, margin improvement, and share repurchases. As we've pointed out, revenue growth (especially organic growth) is a powerful driver of value if it generates a return on invested capital exceeding the cost of capital. Margin improvements that are coming purely from cost cutting are not sustainable in the long term and might even hurt a company's future growth and value creation if investments in research or marketing are cut back. Share repurchases typically increase EPS but also increase a company's debt or reduce its cash. In either case, this leads to a decline in a company's P/E, which affects the increase in EPS so that value per share does not change. Consider Microsoft, with around \$130 billion in liquid assets in 2019. The liquid assets are low risk and low return, so they have a high P/E (higher than for Microsoft's operating assets). Paying out the liquid assets would reduce the proportion of high-P/E assets relative to lower-P/E assets, reducing the overall (weighted-average) P/E for Microsoft as a whole.

In this section, we'll show that the sophisticated investors who drive stock market values dig beneath a company's accounting information to understand the underlying economic fundamentals. A classic example is the share price reaction to changes in inventory accounting by U.S. companies in the 1960s and 1970s. Because of rising price levels in these years, changing from first-in-first-out (FIFO) to last-in-first-out (LIFO) accounting decreased reported profits as well as taxable income. But the investor reaction reflected by the share price was typically positive, because investors understood that free cash flows would be higher as a result of lower taxes.¹³

Sometimes investors have difficulty detecting the true economic situation behind accounting information. For example, investors found it hard to assess the true risks and returns on capital of many financial institutions prior to the 2008 credit crisis because the financial reports were so opaque. Some companies, including Enron and WorldCom, misled stock markets by purposely manipulating their financial statements. But all managers should understand that markets can be mistaken or fooled for only so long. Sooner or later, share prices need to be justified by cash flows rather than accounting earnings.

¹³ G. Biddle and F. Lindahl, "Stock Price Reactions to LIFO Adoptions: The Association between Excess Returns and LIFO Tax Savings," *Journal of Accounting Research* 20, no. 2 (1982): 551–588.

EXHIBIT 7.10 Relationship between Share Repurchases and Shareholder Returns



Based on a sample set of more than 250 nonfinancial S&P 500 companies.

¹ Effect of share repurchases on TRS is measured by residuals of multivariate regression. Variables are share-repurchase intensity and economic-profit growth. Economic-profit growth is a measure that combines earnings growth and return on capital (relative to cost of capital). This regression shows that the effect of share-repurchase intensity is not statistically significant.

² Difference between EPS growth and net income growth used as proxy for degree of share-repurchase intensity.

Source: Corporate Performance Analytics by McKinsey.

EPS Growth from Share Repurchases

Even though EPS is not a reliable indicator of value creation, many companies still use it as a key measure of financial performance and an important input for executive compensation. Not surprisingly, we find that executives pursue share repurchase programs mainly because they believe the resulting EPS growth creates value for shareholders. But savvy markets see through such moves with a gimlet eye. One company managed to create strong growth in EPS while its net income was falling, simply by retiring its shares even faster.¹⁴ When investors understood that business results were declining, the company's share price dropped by 40 percent relative to the overall market change.

The empirical evidence is clear. At face value, there appears to be a correlation between shareholder value creation and the intensity of a company's share repurchase program. But that is simply because companies with higher returns on capital and growth also tend to pay out more cash to shareholders. After we control for differences in growth and return on capital, no relationship is left between share repurchases and shareholder value creation (see Exhibit 7.10).¹⁵

¹⁴ See O. Ezekoye, T. Koller, and A. Mittal, "How Share Repurchases Boost Earnings without Improving Returns," *McKinsey on Finance*, no. 58 (2016): 15–24.

¹⁵ If companies could time share repurchases when share prices are truly low, they could create value for shareholders who do not sell. However, as we discuss in Chapter 33, most companies do not time repurchases effectively.

Earnings from Mergers and Acquisitions

There is yet another way for companies to increase their earnings: buying another company. Say a company has \$1 billion of excess cash. It uses the cash to buy another company earning \$50 million per year at a P/E multiple of 20 times. Its earnings will increase by \$50 million, less the forgone interest it was earning on the excess cash; assuming that equals \$5 million (at a 0.5 percent after-tax return on cash), the net increase is \$45 million. Though the company's earnings have increased, we can't tell whether it has created value. At a 20 P/E purchase price, it will be earning only 5 percent on its invested capital. If it has a 10 percent cost of capital, it will need to double the earnings of the acquired company to earn its cost of capital on the \$1 billion it just invested.

Investors see through the accounting earnings. Chapter 31 shows that whether an acquisition increases or decreases earnings in the first year or two after the acquisition has no correlation with the stock market's reaction to the transaction.

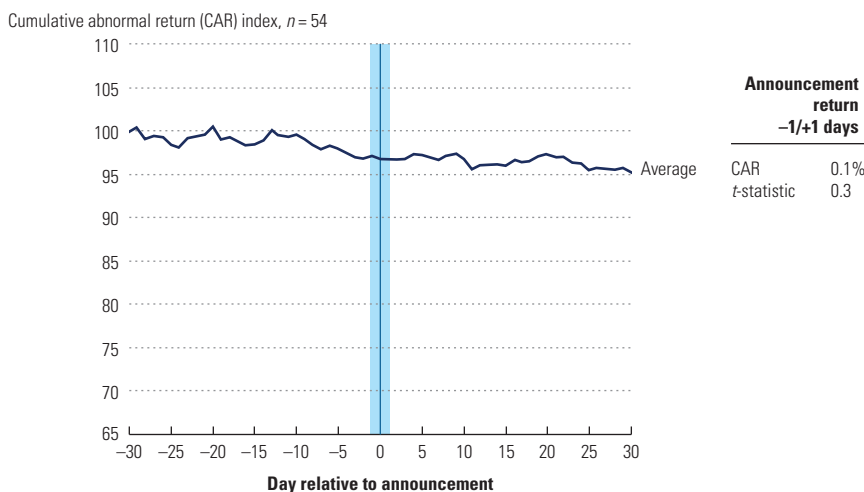
Investors also see through the illusion of "multiple expansion," as we discussed in Chapter 3. There is no empirical evidence or economic logic that the stock market will value an acquired business at the earnings multiple of the acquiring business. The earnings multiple of two combined businesses will simply equal the weighted average of the individual earnings multiples. Any value increase must come from additional cash flows over and above those of the individual businesses.

Write-Downs

Executives are often reluctant to take the earnings hit from writing down the value of assets, assuming that investors will react negatively. But investors don't respond mechanically to write-downs. Rather, they assess what information the write-down conveys about the future performance of the company.

We looked at 99 companies in the United States that had written off at least \$2 billion of impaired goodwill against their profits from 2007 to 2011.¹⁶ There was no statistically significant drop in share prices on the day a write-off was announced. The markets had already anticipated the lower benefits from past acquisitions and reduced the share prices long before the write-off announcements. For example, prices jumped nearly 10 percent when Boston Scientific announced a \$2.7 billion write-down associated with its 2006 acquisition of Guidant. Prices rose almost 8 percent when U.S. Steel announced a goodwill impairment charge of \$1.8 billion with its third-quarter earnings in 2013. We found a similar pattern for the 15 largest goodwill impairments by European companies from 2010 to 2012. The pattern is consistent over many

¹⁶ See B. Cao, M. Goedhart, and T. Koller, "Goodwill Shunting: How to Better Manage Write-Downs," *McKinsey on Finance*, no. 50 (Spring 2014): 13–15.

EXHIBIT 7.11 **No Market Reaction to Announcement of Goodwill Impairment**

Source: SEC filings, Datastream, Bloomberg.

years. Likewise, Exhibit 7.11 shows there was no statistically significant drop in share prices on the announcement of goodwill impairments in an earlier sample of 54 companies in the United States and Europe from 2002 to 2004.¹⁷

Stock markets clearly look at the underlying cash flows and business fundamentals rather than reported earnings and goodwill impairments. In the 2010 to 2012 sample of European write-offs, in fact, only one analyst report issued after one announcement even commented on the size of the impairment. Analysts did, however, comment strongly on indications of how the company would move forward. Changes in signals or explicit guidance about future operating earnings, the outlook for the market and business units, and any management actions or plans to address changing conditions are important.

Employee Stock Options

In the early 2000s, proposed new accounting rules requiring employee stock options to be expensed in the income statement caused much concern. Some executives and venture capitalists claimed that expensing stock options would reduce the earnings of small high-growth companies so much that they would not be able to take the companies public.

Of course, there was no need for concern, because stock prices are driven by cash flows, not reported earnings. Academic research has shown that the stock market already took account of employee options in its valuation of

¹⁷ The sample comprises selected U.S. and European companies with a market capitalization of at least \$500 million and an impairment charge of at least 2 percent of market capitalization.

companies that give full information about their options schemes—even when the option values are not explicitly expensed in the companies' income statements.¹⁸ In fact, companies that voluntarily expensed their employee options before doing so became mandatory experienced no decrease in share price, despite the negative implications for reported earnings.¹⁹

We came to a similar conclusion after examining 120 U.S. companies that began expensing their stock options between July 2002 and May 2004. Furthermore, we found no relationship between the size of the earnings decrease due to option expensing and any abnormal returns during the days surrounding the new policy's announcement. The market already had the relevant information on the option plans and was not confused by a change in reporting policy.

Different Accounting Standards

Share price data for companies that report different accounting results in different stock markets provide additional evidence that stock markets do not take reported earnings at face value. Prior to 2008, non-U.S. companies that had securities listed in the United States and did not report under U.S. Generally Accepted Accounting Principles (GAAP) or International Financial Reporting Standards (IFRS), for example, were required to report equity and net profit under U.S. GAAP.²⁰ These could have provided results that differed significantly from the equity and net profit reported under their domestic accounting standards. We analyzed a sample of 50 European companies that began reporting reconciliations of equity and profit to U.S. GAAP after obtaining U.S. listings between 1997 and 2004. The differences between net income and equity under U.S. and local accounting standards were often quite large; in more than half the cases, the gap exceeded 30 percent.

Many executives probably worried that lower earnings under U.S. GAAP would translate directly into a lower share price. But this was not the case. Even though two-thirds of the companies in our sample reported lower earnings following U.S. disclosure, the stock market reaction to their disclosure was positive, as shown in Exhibit 7.12. At that time, following U.S. GAAP standards also generally meant disclosing more information than required by local standards. Evidently, improved disclosure outweighed any artificial accounting effects.

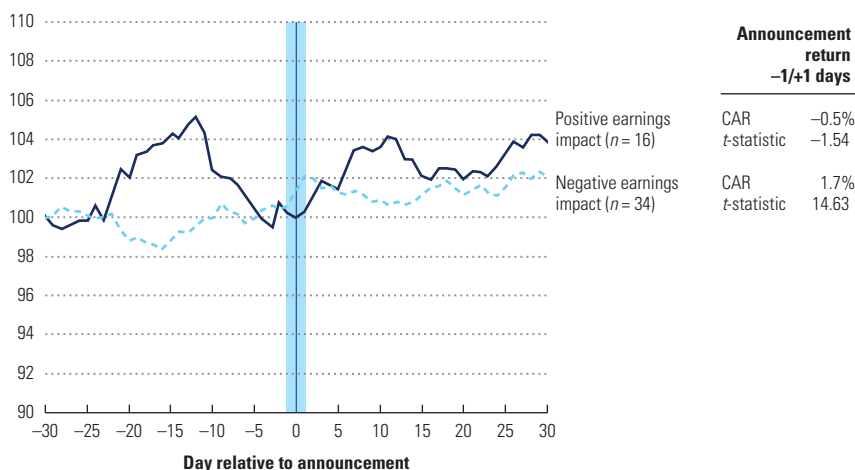
¹⁸ D. Aboody, M. Barth, and R. Kasznik, "SFAS No. 123 Stock-Based Compensation Expense and Equity Market Values," *Accounting Review* 79, no. 2 (2004): 251–275.

¹⁹ D. Aboody, M. Barth, and R. Kasznik, "Firms' Voluntary Recognition of Stock-Based Compensation Expense," *Journal of Accounting Research* 42, no. 2 (December 2004): 251–275.

²⁰ Since March 2008, non-U.S. companies reporting under IFRS are no longer required to reconcile financial statements to U.S. GAAP in their Securities and Exchange Commission (SEC) filings.

EXHIBIT 7.12 **No Clear Impact of U.S. GAAP Reconciliation**

Average cumulative abnormal return (CAR) index



Source: SEC filings, Datastream, Bloomberg.

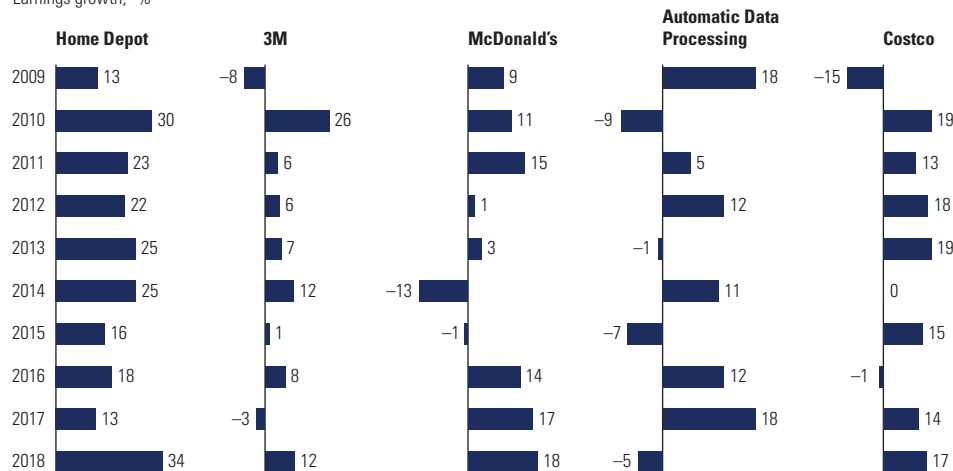
MYTHS ABOUT EARNINGS MANAGEMENT

On July 17, 2019, Internet entertainment service company Netflix reported second-quarter earnings of \$0.56 per share, just four cents short of the \$0.60 analyst consensus expectations. In addition, it had generated \$4.92 billion in revenues, 25 percent higher than for the same quarter the year before, but missed analyst revenue targets by \$10 million. On the same day, its share price dropped by more than 10 percent. The trigger for the price decline was not the company's missing its earnings or revenue targets. Rather, investors were concerned about the company's long-term outlook because of a decline in U.S. subscribers when an increase was expected, as well as significantly lower growth in international subscribers. Still, events such as this have led many managers to believe that stock markets are increasingly sensitive to short-term earnings that undershoot analysts' expectations or to volatility in earnings generally. As we'll show, events like these are not driven by the earnings announcement itself, but by other information that accompanies the earnings, such as the underlying subscriber base growth in the case of Netflix. Furthermore, investors are not much concerned by earnings volatility and offer no rewards for predictable earnings or earnings guidance.

Earnings Volatility

Some managers believe investors will pay a premium for steady earnings growth. Indeed, executives regularly cite stabilizing earnings growth as a reason for strategic actions. For example, the CEO of Conoco once justified a

EXHIBIT 7.13 Earnings Growth of Least Volatile Companies: Not So Smooth

Earnings growth,¹ %¹ Earnings is net income before extraordinary items, adjusted for goodwill impairment.

Source: S&P Capital IQ.

pending merger with Phillips Petroleum in part by asserting that the merger would offer greater earnings stability over the commodity price cycle.²¹

In contrast, academic research finds that earnings variability has either limited or no impact on market value and shareholder returns. Ratios of market value to capital are diminished by cash flow volatility, but not by earnings volatility. Investors see through earnings smoothing that is unconnected to cash flow.²² In 30 years of U.S. profit data, there is no correlation between variability in EPS and a company's market value.²³ Some researchers find a statistically significant, but practically negligible, relationship between the two: between the 1 percent of companies with the lowest earnings volatility and the 1 percent with the highest lies a difference in market-to-book ratios of less than 10 percent.²⁴

Part of the explanation for the results is that smooth earnings growth is a myth. Almost no companies demonstrate smooth earnings growth. Exhibit 7.13 shows the earnings growth of the five firms among the 10 percent of large listed U.S. companies that had the least volatile earnings growth from 2008 to 2018.²⁵ Of the companies examined, Home Depot was the only one with ten years of steady earnings growth. Only a handful had earnings growth that was steady for four or

²¹ Analyst teleconference, November 19, 2001.²² See B. Rountree, J. Weston, and G. Allayannis, "Do Investors Value Smooth Performance?" *Journal of Financial Economics* 90, no. 3 (December 2008): 237–251.²³ J. McNnis, "Earnings Smoothness, Average Returns, and Implied Cost of Equity Capital," *Accounting Review* (January 2010).²⁴ R. Barnes, "Earnings Volatility and Market Valuation: An Empirical Investigation" (LBS Accounting Subject Area Working Paper ACCT 019, 2003). The difference was 0.2, and the average market-to-book ratio for the entire sample was around 2.²⁵ These were all listed nonfinancial U.S. companies with revenues of more than \$1 billion in 2018.

more years. Most companies with relatively stable earnings growth follow a pattern similar to the four companies other than Home Depot in Exhibit 7.13: several years of steady growth interrupted by a sudden decline in earnings.

Meeting Consensus Earnings Estimates

When a high-profile company misses an earnings target, it certainly makes headlines, but the impact of short-term earnings on share prices should not be overstated. For example, empirical research has shown that earnings surprises explain less than 2 percent of share price volatility in the four weeks surrounding the announcements.²⁶ Investors place far more importance on a company's economic fundamentals than on reported earnings. Sometimes, however, short-term earnings are the only data investors have on which to base their judgment of fundamental corporate performance. In these cases, investors may interpret a missed EPS target as an omen of a decline in long-term performance and management credibility, so they lower the company's share price accordingly. As we describe in more detail in Chapter 34, the announcement of lower-than-expected earnings only drives share prices down in case of downward revisions of long-term fundamental prospects.

Similarly, share prices do not rise if the market believes a positive earnings surprise is simply the result of some imaginative accounting, such as deliberate timing of book gains from asset divestments or acceleration of sales from deep discounts to customers. For such accruals-dependent earnings increases, subsequent shareholder returns are poor, relative to peers.²⁷ And investors are wise to be wary when accruals contribute substantially to earnings, because this typically indicates that a company has reached a turning point and will post lower earnings in the future.

Earnings Guidance

Many companies believe that providing guidance on their expected earnings for the upcoming quarter or year can lead to higher valuations, lower share price volatility, and greater market liquidity for their shares at what they perceive to be limited costs. Unfortunately, there is no evidence that guidance delivers any of these benefits. As we discuss in Chapter 34, we find that whether companies issue guidance does not affect their earnings multiples, returns to shareholders, or share price volatility. The impact of guidance on a stock's liquidity, if any, typically disappears in the following year, making it practically irrelevant from a shareholder's perspective.²⁸

²⁶ W. Kinney, D. Burgstahler, and R. Martin, "Earnings Surprise 'Materiality' as Measured by Stock Returns," *Journal of Accounting Research* 40, no. 5 (December 2002): 1297–1329.

²⁷ K. Chan, L. Chan, N. Jegadeesh, and J. Lakonishok, "Earnings Quality and Stock Returns," *Journal of Business* 79, no. 3 (2006): 1041–1082.

²⁸ See T. Koller, B. Jiang, and R. Raj, "Three Common Misconceptions about Markets," *Journal of Applied Corporate Finance* 25, no. 3 (2006): 32–38.

However, earnings guidance could lead to significant but hidden costs. Companies at risk of missing their own forecasts could be tempted to artificially improve their short-term earnings. As described previously, that is not likely to convince the market and could come at the expense of long-term value creation. When providing guidance at all, companies are therefore better off if they present ranges rather than point estimates and if they present these for underlying operational performance (for example, targets for volume and revenue, operating margins, and initiatives to reduce costs) rather than for earnings per share.

MYTHS ABOUT DIVERSIFICATION

Diversification is intrinsically neither good nor bad; it all depends on whether the parent company is the best owner of the businesses in its portfolio. Some executives believe that diversification brings benefits, such as more stable aggregate cash flows, tax benefits from higher debt capacity, and better timing of investments across business cycles. However, as we discuss in Chapter 28, there is no evidence of such advantages in developed economies. Yet the evidence does point to *costs* of diversification: the business units of diversified companies often underperform their focused peers because of added complexity and bureaucracy.

Another misconception about diversification is that it leads to so-called conglomerate discounts to the fair value of the business. According to this viewpoint, spin-offs and other forms of divestment are effective instruments to unlock these conglomerate discounts. Those who hold this view note that share price reactions to divestment announcements are typically positive, which is taken as evidence that such transactions are an easy solution to low valuations.

Typically, this misunderstanding is based on a misleading sum-of-the-parts calculation, in which analysts estimate the value of each of a company's businesses based on the earnings multiples of each business's industry peers. If the value of the sum of the businesses exceeds the company's current market value, the analysts assume the market value includes a conglomerate discount. However, as we discuss in Chapter 19, the analyses are often based on industry peers that are not actually comparable in terms of performance or sector. When the analysis uses true industry peers, the conglomerate discount disappears.

Positive share price reactions to divestment announcements therefore do not represent any correction of undervaluation or oversight by investors. The reactions simply reflect investor expectations that performance will improve at both the parent company and the divested business once each has the freedom to change its strategies, people, and organization. As a large body of

empirical evidence shows, investors are right in anticipating performance step-ups.²⁹ For example, we found that for 85 major spin-offs since 1992, both the divested businesses and the parent companies delivered significant improvements in operating profit margins over five years following the transaction (see Chapter 32).

MYTHS ABOUT COMPANY SIZE

Many executives are tempted by the illusion that the absolute size or scale of a company brings benefits in the form of either higher share prices in the stock market or higher ROIC and growth in the businesses. Academics and practitioners have claimed that larger companies are in higher demand by investors because they get more coverage from equity analysts and media. Or they say the cost of capital is lower because large companies are less risky and their stocks more liquid. Higher demand and lower cost of capital should lead to higher valuation in the market.

However, there is no evidence that size matters once companies have reached a certain size. The cutoff point probably lies in the range of a market capitalization of \$250 million to \$500 million.³⁰ Only below that range is there some indication of higher cost of capital, for example. Whether a company has a market capitalization of \$1 billion, \$5 billion, or more does not matter for its relative valuation in the market.

The same holds for any positive effect of a company's size on its ROIC and growth. In most businesses, economies of scale make a difference only up to a certain size of the business. Large (and medium-size) companies have typically already extracted maximum benefits from such economies of scale. For example, it is tempting to believe that package-delivery companies such as FedEx or UPS can easily process more packages at limited additional costs (the planes and trucks are already in place). But the networks of these companies are finely tuned and optimized for minimum unused capacity. Increasing volume by 10 percent might in fact require 10 percent more planes and trucks.

For most companies, increases in size alone no longer automatically bring further improvements in performance but just generate more complexity.

²⁹ See, for example, J. Miles and J. Rosenfeld, "The Effect of Voluntary Spin-Off Announcements on Shareholder Wealth," *Journal of Finance* 38 (1983): 1597–1606; K. Schipper and A. Smith, "A Comparison of Equity Carve-Outs and Seasoned Equity Offerings: Share Price Effects and Corporate Restructuring," *Journal of Financial Economics* 15 (1986): 153–186; K. Schipper and A. Smith, "Effects of Recontracting on Shareholder Wealth: The Case of Voluntary Spin-Offs," *Journal of Financial Economics* 12 (1983): 437–468; J. Allen and J. McConnell, "Equity Carve-Outs and Managerial Discretion," *Journal of Finance* 53 (1998): 163–186; and R. Michaely and W. Shaw, "The Choice of Going Public: Spin-Offs vs. Carve-Outs," *Financial Management* 24 (1995): 5–21.

³⁰ See R. McNish and M. Palys, "Does Scale Matter to Capital Markets?" *McKinsey on Finance* (Summer 2005): 21–23.

Growth often means adding more business units and expanding geographically, which lengthen the chain of command and involve more people in every decision. Smaller, nimbler companies can well end up with lower costs. Whether size helps or hurts, whether it creates scale economies or diseconomies, depends on the unique circumstances of each company.

MYTHS ABOUT MARKET MECHANICS

Conventional wisdom has long held that companies can capture benefits for their shareholders without any improvements to underlying cash flows by having their stock included in a key market index, listing it in multiple markets, or splitting their stocks. True, a company from an emerging market in Asia securing a U.S. listing or a little-known European company joining a leading global stock index might secure some appreciable uplift. But well-functioning capital markets are entirely focused on the fundamentals of cash flow and revenue growth.

Index Membership

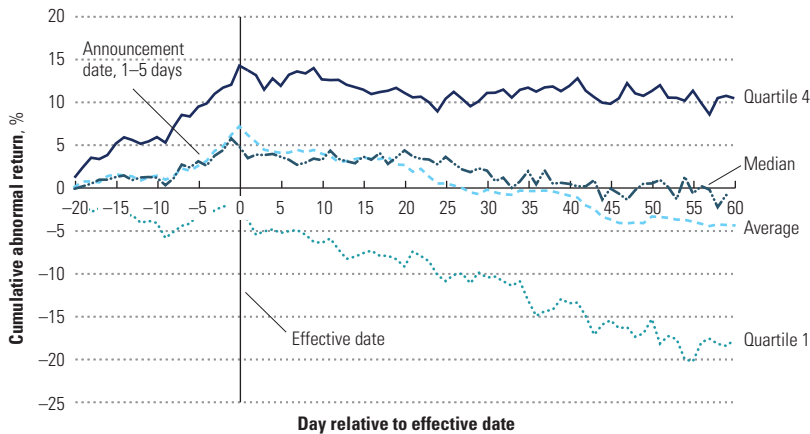
Becoming a member of a leading stock market index such as the S&P 500 or FTSE 100 appeals to managers because many large institutional investors track these indexes. Managers believe that when institutional investors rebalance their portfolios to reflect the change of index membership, demand will shift dramatically, boosting the share price. Anecdotal evidence appears to confirm this view. In 2001, Nortel, Shell, Unilever, and four other companies based outside the United States were removed from the S&P 500 index and replaced with the same number of U.S. corporations. The departing companies lost, on average, nearly 7.5 percent of their value in the three days after the announcement. The stock prices of the new entrants—including eBay, Goldman Sachs, and UPS—increased by more than 3 percent in the same period.

But empirical evidence shows that such changes are typically short-lived. On average, share prices of companies excluded from a major stock index do indeed decrease after the announcement. But this fall is fully reversed within one or two months.³¹ Surprisingly, the evidence on the impact of index inclusions appears less conclusive; several publications report that price increases occurring immediately after an inclusion are only partly reversed over time.³² We analyzed the effect

³¹ H. Chen, G. Noronha, and V. Singal, "The Price Response to S&P 500 Index Additions and Deletions: Evidence of Asymmetry and a New Explanation," *Journal of Finance* 59, no. 4 (August 2004): 1901–1929.

³² See also, for example, L. Harris and E. Gurel, "Price and Volume Effects Associated with Changes in the S&P 500: New Evidence for the Existence of Price Pressures," *Journal of Finance* 41 (1986): 815–830; and R. A. Brealey, "Stock Prices, Stock Indexes, and Index Funds," *Bank of England Quarterly Bulletin* (2000): 61–68.

EXHIBIT 7.14 Effects of Inclusion Disappear after 45 Days



on share price of 103 inclusions and 41 exclusions from the S&P 500 between December 1999 and March 2004.³³ As Exhibit 7.14 shows, new entrants to the index experienced only a short-lived increase in share price: statistically significant positive returns disappeared after only 20 days, and all effects largely disappeared after 45 days. As investors adjust their portfolios to changes in the index, share prices of new entrants initially increase but then revert to normal once portfolios are rebalanced. For 41 companies ejected from the S&P 500 over the same period, we found similar patterns of temporary price change. The pressure on their prices following exclusion from the index lifted after two to three weeks.

Cross-Listing

For years, many academics, executives, and analysts believed companies cross-listing their shares on exchanges in the United States, London, and Tokyo could realize a higher share price and a lower cost of capital.³⁴ Cross-listed shares would benefit from more analyst coverage, a broader shareholder base, improved liquidity, higher governance standards, and better access to capital.

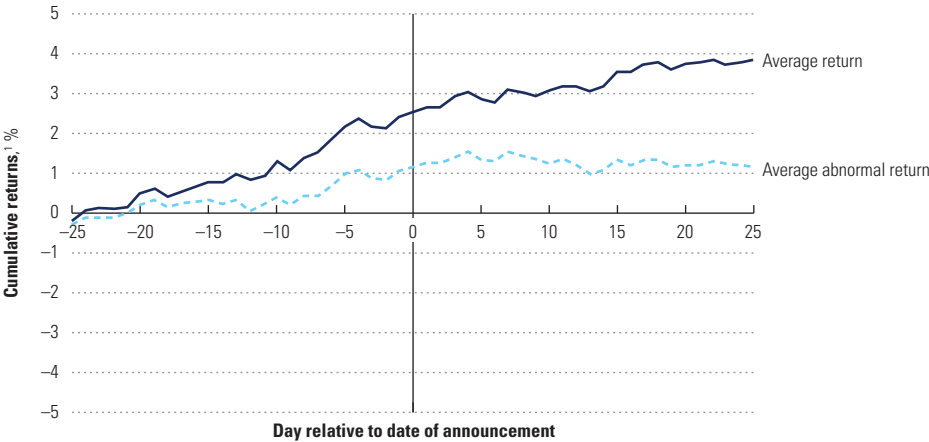
But our analysis does not find any significant impact on shareholder value from cross-listings for companies in the developed markets of North America, Western Europe, Japan, and Australia.³⁵ We found no decline in share price when companies announced a delisting from U.S. and UK stock exchanges

³³ For further details, see M. Goedhart and R. Huc, "What Is Stock Membership Worth?" *McKinsey on Finance*, no. 10 (Winter 2004): 14–16.

³⁴ See, for example, C. Doidge, A. Karolyi, and R. Stulz, "Why Are Foreign Firms That List in the U.S. Worth More?" *Journal of Financial Economics* 71, no. 2 (2004): 205–238; and M. King and U. Mittoo, "What Companies Need to Know about International Cross-Listing," *Journal of Applied Corporate Finance* 19, no. 4 (Fall 2007): 60–74.

³⁵ For further details, see R. Dobbs and M. Goedhart, "Why Cross-Listing Shares Doesn't Create Value," *McKinsey on Finance*, no. 29 (Autumn 2008): 18–23.

EXHIBIT 7.15 **Delisting from U.S./UK Exchanges: No Value Impact on Companies from Developed Markets**



¹ Sample of 229 delistings from New York Stock Exchange, NASDAQ, or London International Main Market. Announcement dates between December 31, 2002, and December 31, 2007.

Source: Reuters; Bloomberg; Datastream.

(Exhibit 7.15).³⁶ In fact, most announcements in our sample produced hardly any reaction from analysts and investors. Neither did we find any valuation premium for companies with cross-listings in New York or London relative to companies without any cross-listing, once we corrected for differences in return on invested capital (Exhibit 7.16).

In fact, we did not find evidence for any of the deemed benefits from cross-listings. After correcting for size, cross-listed European companies have only marginally more analyst coverage than those not cross-listed.³⁷ Institutional investors from the United States do not require the foreign companies in which they want to invest to be listed in the United States.³⁸ There is no impact on liquidity, as cross-listed shares of European companies in the United States—American depositary receipts (ADRs)—typically account for less than 3 percent of these companies' total trading volumes. Corporate governance standards across the developed world have converged with those in the United States and the United Kingdom. There is hardly any benefit from better access to capital, given that three-quarters of the U.S. cross-listings of companies from the European Union have never involved raising any new capital in the United States.³⁹

³⁶ We analyzed the stock market reactions to 229 voluntary delistings between 2002 and 2008.

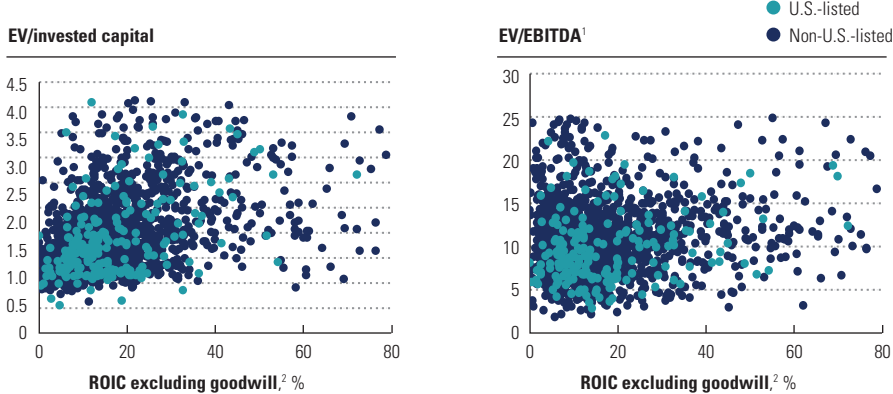
³⁷ See, for example, M. Lang, K. Lins, and D. Miller, "ADRs, Analysts, and Accuracy: Does Cross Listing in the U.S. Improve a Firm's Information Environment and Increase Market Value?" *Journal of Accounting Research* 41, no. 2 (May 2003): 317–345.

³⁸ For example, CalPERS, a large U.S. investor, has an international equity portfolio of around 2,400 companies, but less than 10 percent of them have a U.S. cross-listing.

³⁹ Based on 420 depositary receipt issues on the New York Stock Exchange, NASDAQ, and American Stock Exchange from January 1970 to May 2008. Data from the Bank of New York Mellon Corporation, www.adrbnymellon.com.

EXHIBIT 7.16 **U.S. Cross-Listing: No Impact on Valuation of Developed-Market Companies**

U.S.-listed and non-U.S.-listed companies in Western Europe, Japan, Canada, Australia, and New Zealand



¹ Enterprise value at year-end 2006 divided by 2006 EBITDA.

² Average ROIC for 2004–2006.

Source: New York Stock Exchange, NASDAQ, Bloomberg, Datastream, Corporate Performance Analytics by McKinsey.

For companies from the emerging world, however, the story might be different. These companies might benefit from access to new equity and more stringent corporate governance requirements through cross-listings in U.S. or UK equity markets.⁴⁰

Stock Splits

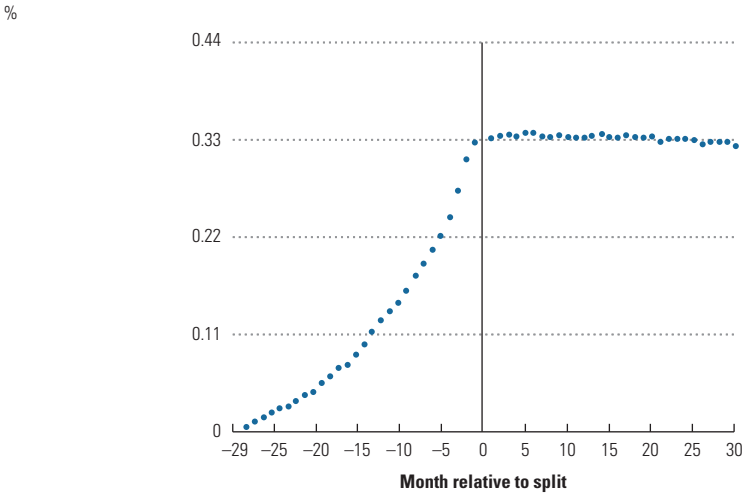
Although their numbers have come down significantly over the past decade, each year some listed companies in the United States increase their number of shares through a stock split to bring a company's share price back into the "optimal trading range."⁴¹ But fundamentally, stock splits can't create value, because the size of the pie available to shareholders does not change. For example, after a two-for-one stock split, a shareholder who owned two shares worth \$5 apiece ends up with four shares, each worth \$2.50. But some managers and academics claim that the lower price should make the stock more attractive for capital-constrained investors, thereby increasing demand, improving liquidity, and leading to higher returns for shareholders.⁴²

⁴⁰ See R. Newell and G. Wilson, "A Premium for Good Governance," *McKinsey Quarterly*, no. 3 (2002): 20–23.

⁴¹ R. D. Boehme and B. R. Danielsen report over 6,000 stock splits between 1950 and 2000: "Stock-Split Post-Announcement Returns: Underreaction or Market Friction?" *Financial Review* 42 (2007): 485–506. D. Ikenberry and S. Ramnath report over 3,000 stock splits between 1988 and 1998: "Underreaction to Self-Selected News Events: The Case of Stock Splits," *Review of Financial Studies* 15 (2002): 489–526.

⁴² There is ample evidence to show that this is not the case: after a split, trading volumes typically decline, and brokerage fees and bid-ask spreads increase, indicating lower liquidity, if anything. See T. Copeland, "Liquidity Changes Following Stock Splits," *Journal of Finance* 34, no. 1 (March 1979): 115–141.

EXHIBIT 7.17 Cumulative Average Abnormal Returns around Stock Splits



Source: E. Fama, L. Fisher, M. Jensen, and R. Roll, "The Adjustment of Stock Prices to New Information," *International Economic Review* 10 (1969): 1–21.

In many cases, a stock split is indeed accompanied by positive abnormal returns to shareholders in the months prior to the split (see Exhibit 7.17).⁴³ The abnormal returns have nothing to do with the split as such but are simply a function of self-selection and signaling. Self-selection is the tendency of companies to split their stocks into lower denominations because of a prolonged rise in their share price.

More insightful is the abnormal return for the three days around the date of the stock split announcement, at about 3 percent.⁴⁴ When managers announce a stock split, they are also signaling that they expect further improvement in economic fundamentals. Indeed, two-thirds of companies reported higher-than-expected earnings and dividends in the year following a stock split. When performance improvements followed the split, the stock market did not react, indicating that investors had already factored them into their decisions at the time of the stock split announcement. Consistent with this pattern, companies that did not improve performance as expected in the year after a stock split saw their share prices fall.⁴⁵

⁴³ E. Fama, L. Fisher, M. Jensen, and R. Roll, "The Adjustment of Stock Prices to New Information," *International Economic Review* 10 (1969): 1–21.

⁴⁴ Some researchers have reported positive abnormal returns not only in the days around but in the entire year following a split announcement. They conclude that the market is inefficient by underreacting to stock splits; see Ikenberry and Ramnath, "Underreaction to Self-Selected News Events." Others find that these abnormal returns do not lead to any arbitrage opportunities and that the market is efficient; see Boehme and Danielsen, "Stock-Split Post-Announcement Returns"; and J. Conrad and G. Kaul, "Long-Term Market Overreaction or Biases in Computed Returns?" *Journal of Finance* 48 (1993): 39–63.

⁴⁵ See Fama et al., "Adjustment of Stock Prices."

MYTHS ABOUT VALUE DISTRIBUTION

Another common misconception among executives is that share repurchases and dividends create value for shareholders. This view is often reinforced by both private and public demands from investors for companies to return more cash to shareholders, particularly as share repurchases. If you dig deeper into understanding investor demands, though, you will typically find that investors want more cash distributed not because the cash distribution itself creates value, but because investors are concerned that companies will squander excess cash and debt capacity on value-destroying investments. They view cash distributions as a way to impose discipline on the company's use of its cash.⁴⁶

More important, companies create value when they generate cash flows. Distributing those cash flows to shareholders cannot create additional value. That would be double-counting and akin to violating principles like the conservation of matter in physics.

So why do companies' share prices often increase on the announcement of share repurchases or dividend increases? In some cases, investors interpret dividend increases as a sign that management is confident enough about future cash flow generation to commit to a higher dividend level. In other cases, investors are relieved that management is less likely to squander cash on value-destroying investments. The result is that investors raise their expectations of future cash flows. If these expectations are not met, the companies' share prices will decline later.

Dividends and share repurchases are merely instruments for distributing cash generated by the company's operations. Furthermore, as we discuss in Chapter 33, decisions about cash distributions should not drive a company's investment decisions; they should be an integral part of a company's capital allocation that matches its investment needs, financing opportunities, and desired level of risk.

SUMMARY

Dramatic swings in share prices sometimes lead finance practitioners to suggest that established valuation theories are irrelevant and that stock markets lead lives of their own, detached from the realities of economic growth and business profitability. We disagree. There is compelling evidence that valuation levels for individual companies and the stock market as a whole clearly reflect the underlying fundamental performance in terms of return on capital and growth. Yes, there are times when valuations deviate from fundamentals, but these typically do not last long. Evidence also shows that some widespread

⁴⁶ See, for example, M. Goedhart and T. Koller, "How to Attract Long-Term Investors: An Interview with M&G's Aled Smith," *McKinsey on Finance* 46 (Spring 2013): 8–13.

beliefs espoused by managers and finance professionals are inconsistent with the fundamental principles of valuation and are erroneous.

We also find that executives are often overly focused on earnings and earnings growth. Earnings don't drive value in their own right; only cash flows do. Companies with attractive growth and returns on invested capital will also generate good earnings. The market sees through earnings that aren't backed up by solid fundamentals, such as earnings increases from share repurchases or from mergers and acquisitions that don't earn adequate returns on capital. Managers should also not be concerned about noneconomic events that reduce earnings, such as asset write-downs or the effects of changes in accounting rules. Nor should they be concerned about delivering smooth earnings or meeting short-term consensus earnings forecasts.

Finally, myriad myths have grown up about how the market values companies based on measures unrelated to the companies' economic performance. None stand up to scrutiny. There is no value premium from diversification, from cross-listing, or from size for size's sake. Conversely, there is no conglomerate discount, only a performance discount for many diversified companies. Dividends and share repurchases don't create value, but markets react positively when management signals it will be disciplined about future investments.

REVIEW QUESTIONS

1. Explain how long-term price-to-earnings ratios in the U.S. stock market of around 15 times are consistent with long-term expected stock returns of around 6 to 7 percent a year in real terms.
2. Many corporate executives focus on earnings per share (EPS) and attempt to manage reported earnings in order to meet analysts' expectations. Can managers succeed in protecting the stock price of their company by managing these accounts? Why or why not?
3. If a company's value is not driven by its short-term earnings, why do investors spend so much time analyzing a company's annual or even quarterly earnings announcements?
4. Empirical research shows that goodwill impairments have no impact on a company's share price. But these impairments do reflect an auditor's best estimate of the value lost in an acquisition by the company. Does the stock market not care about such losses? Explain.
5. In inflationary periods for input prices, what happens to earnings when firms change from first-in first-out (FIFO) to last-in first-out (LIFO) inventory accounting? How does the market react and why?
6. As a rule, cross-listings for companies with a home listing in a mature capital market do not offer material benefits. Discuss how and why this might be different for companies based in emerging capital markets.
7. Over the past five years, the highest share price for HighTechCo, a 15-year-old software company, was around \$650, and its lowest price was around \$150. BrandCo, a consumer products firm with a 70-year history, had share prices ranging from \$85 to \$40 over the same period. Do such wide ranges mean that the stock market is reflecting emotions more than fundamentals? Explain.
8. Why is it much riskier to take a short position in a stock than a long position? What does that mean for the likelihood of over- versus undervaluation of a company's share price?
9. Why do executives spend so much time and effort on communicating with noise traders if intrinsic investors ultimately drive a company's share price?
10. Would a company's share price benefit from having fewer traders and more fundamental investors among the company's shareholders? Why or why not?
11. Why do noise traders have limited impact on a company's share price even when they make the largest volume of trades in the company's stock over a given time period?

12. Certain agencies often categorize company shares as growth stocks or value stocks. What is the difference between growth and value stocks? Why are these labels misleading?
13. What do executives believe are the benefits of issuing EPS guidance? Do companies actually realize these benefits?
14. Some firms chose not to expense employee stock options until accounting rules required all firms to do so, whereas other firms chose to expense options even before being required to do so. What was the impact on financials for firms that chose to expense options? What was the impact on their share prices?

Return on Invested Capital

As Chapter 3 explains, the higher a company can raise its return on invested capital (ROIC), and the longer it can earn a rate of return on that capital greater than its cost of capital, the more value it will create. So it is critical to every strategic and investment decision to be able to understand and predict what drives and sustains ROIC.

Why do some companies develop and sustain much higher returns on capital than others? Consider a classic example from the days of the tech boom at the turn of the millennium. Two newcomers at the height of the boom in 2000 were the companies eBay and Webvan. In November 1999, eBay's market capitalization was \$23 billion, while Webvan's was \$8 billion. Over the years that followed, eBay continued to prosper, reaching a market capitalization of more than \$70 billion in 2015, when it spun off its subsidiary PayPal. By mid-2018, the combined market capitalization of eBay and PayPal was more than \$160 billion. Webvan, in contrast, disappeared into bankruptcy and liquidation after just a few years. To understand why, we can look at what these companies' underlying strategies meant for their respective returns on invested capital.

The core business of eBay is an online marketplace that collects a small amount of money for each transaction between a buyer and a seller. The business needs no inventories or accounts receivable, and it requires little invested capital. Once the service started and a growing number of buyers used eBay, more sellers were attracted to it, in turn drawing in still more buyers. Moreover, the marginal cost of each additional buyer or seller is close to zero. Economists say that a business in a situation like eBay's exhibits *increasing returns to scale*. In such a business, the first competitor to grow big can generate a very high ROIC and will usually create the bulk of value in its market. If, as in eBay's case, the business easily expands across borders, the potential for value creation becomes even greater.

Webvan was an online grocery-delivery business based in California. In contrast to eBay, it had a capital-intensive business model involving substantial warehouses, trucks, and inventory. In addition, Webvan was competing with local grocery stores in selling products at very thin margins. The complexity and costs of making physical deliveries to customers within precise time frames more than offset Webvan's savings from not having physical stores. Finally, Webvan's business did not enjoy increasing returns to scale; as demand increased, it needed more food pickers, trucks, and drivers to serve customers.

From the outset, it was clear that eBay's business model had a sound and sustainable competitive advantage that permitted high returns. Webvan had no such advantage over its grocery store competitors. Whereas eBay's strategy was primed for success, Webvan's foreshadowed doom. In general, success in the online grocery business has since proven to be far more elusive than in other forms of online retail. For example, Amazon Fresh has faced challenges expanding beyond the most densely populated metropolitan areas. Amazon's 2017 acquisition of Whole Foods was one signal that in grocery, competition from traditional stores is hard to overcome.

The importance of ROIC is universal: it applies to companies as well as to businesses within companies. For example, within its retail business model, Amazon creates substantial revenues from third-party sellers using its online platform. Platform sales by third parties generate increasing returns to scale, more so than Amazon's direct sales. Platform sales require little invested capital, and Amazon's marginal cost of additional transactions is minimal. As a result, platform sales have become an important driver of Amazon's overall value creation.

This chapter explores how rates of return on invested capital depend on competitive advantage. We examine how strategy drives competitive advantage, which when properly fitted to industry structure and competitive behavior can produce and sustain a superior ROIC. This explains why some companies earn only a 10 percent ROIC while others earn 50 percent. The final part of the chapter presents 55 years of ROIC data by industry over time. This analysis shows how ROIC varies by industry and how rates of ROIC fluctuate or remain stable over time.

WHAT DRIVES ROIC?

To understand how strategy, competitive advantage, and return on invested capital are linked, consider the following representation of ROIC:

$$\text{ROIC} = (1 - \text{Tax Rate}) \frac{\text{Price per Unit} - \text{Cost per Unit}}{\text{Invested Capital per Unit}}$$

This version of ROIC simply translates the typical formula of net operating profit after taxes (NOPAT) divided by invested capital into a per unit calculation: price per unit, cost per unit, and invested capital per unit.¹ To earn a higher ROIC, a company needs a competitive advantage that enables it to charge a price premium or produce its products more efficiently (at lower cost, lower capital per unit, or both). A company's competitive advantage depends on its chosen strategy and the industry in which it operates.

The strategy model that underlies our thinking about what drives competitive advantage and ROIC is the structure-conduct-performance (SCP) framework. According to this framework, the structure of an industry influences the conduct of the competitors, which in turn drives the performance of the companies in the industry. Originally developed in the 1930s by Edward Mason, this framework was not widely influential in business until Michael Porter published *Competitive Strategy* (Free Press, 1980), applying the model to company strategy. While there have been extensions and variations of the SCP model, such as the resource-based approach,² Porter's framework is probably still the most widely used for thinking about strategy.

According to Porter, the intensity of competition in an industry is determined by five forces: the threat of new entry, pressure from substitute products, the bargaining power of buyers, that of suppliers, and the degree of rivalry among existing competitors. Companies need to choose strategies that build competitive advantages to mitigate or change the pressure of these forces and achieve superior profitability. Because the five forces differ by industry, and because companies within the same industry can pursue different strategies, there can be significant variation in ROIC across and within industries.

Exhibit 8.1 underlines the importance of industry structure to ROIC. It compares the median return on invested capital over more than 20 years in two sectors: branded consumer goods and extraction industries (such as mining and oil and gas). Consumer goods have earned consistently higher ROICs than extraction companies. In addition, the returns of extraction-based companies have been highly volatile.

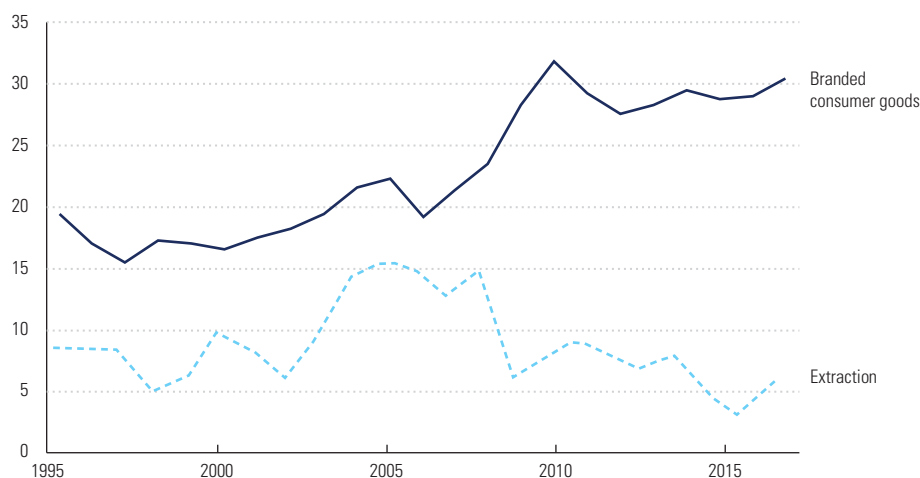
The reason for this difference in the industries' performances lies mainly in differences between their competitive structures. In the branded-consumer-goods industry, companies such as Nestlé, Procter & Gamble, and Unilever developed long-lasting brands with high consumer loyalty that made it difficult for new competitors to gain a foothold. Building on these advantages, these companies were able to increase their returns on capital from around 20 percent in the mid-1990s to roughly 30 percent two decades later, despite challenges to traditional brands from new market entrants. One example is

¹ We introduce *units* to encourage discussion regarding price, cost, and volume. The formula, however, is not specific to manufacturing. Units can represent the number of hours billed, patients seen, transactions processed, and so on.

² See, for example, J. Barney, "Resource-Based Theories of Competitive Advantage: A Ten-Year Retrospective on the Resource-Based View," *Journal of Management* 27 (2001): 643–650.

EXHIBIT 8.1 **Company Profitability: Industry Matters**

Industry median ROIC excluding goodwill, %



Source: Corporate Performance Analytics by McKinsey.

the competition faced by Procter & Gamble's Gillette shaving business from challengers such as Harry's and Dollar Shave Club.³

In extraction industries, one company's products are the same as another's (iron ore is iron ore, with minor quality differences), so prices are the same across the industry at any point in time. In addition, the companies use the same capital-intensive processes to extract their products. As a result, the median company in the industry doesn't have a competitive advantage, and returns are low, averaging only 9 percent during this 20-year period. It is worth noting that imbalances in supply and demand can lead to cycles in product price and ROIC, as was the case with a long run-up in commodity prices in the years leading up to 2005. In the end, though, competition leads to low ROIC on average.

Industry structure is by no means the only determinant of ROIC, as the significant variation among companies within industries shows. Consider the global automotive industry, which has been plagued by overcapacity for years. Still, the industry's low returns do not deter new entrants, whether from different geographies (such as South Korean automakers' entry into the U.S. and European markets) or due to the emergence of new technologies (such as electric-vehicle producers, including Tesla). Add in the difficulties that some manufacturers encounter in trying to close unionized plants, and it's easy to see how overcapacity keeps returns across the sector low. Only a few manufacturers, such as BMW, can parlay their premium brands and higher quality into higher prices and superior returns on capital compared with other

³ Unilever acquired Dollar Shave Club in 2016.

manufacturers. Or consider the highly competitive European airline industry, where most players typically generate returns very close to their cost of capital—and occasionally below it. Nevertheless, Ryanair earns superior returns, thanks to its strategy of strictly point-to-point connections between predominantly secondary airports at the lowest cost in the industry.

Finally, industry structure and competitive behavior aren't fixed; they're subject to shocks from technological innovation, changes in government regulation, and competitive entry—any or all of which can affect individual companies or an entire industry. We show in this chapter's final section that the software and pharmaceutical industries, for example, consistently earn high returns. However, the leading companies may not be the same in 20 years, just as many of today's leaders were not major players or didn't even exist 20 years ago.

COMPETITIVE ADVANTAGE

Competitive advantage derives from some combination of ten sources, defined in Exhibit 8.2. Of these, five allow companies to charge a price premium, four contribute to cost and capital efficiency, and one (often referred to as “network economies”) combines price and cost advantages to produce increasing returns to scale. It is important to understand that competitive advantage drawn from these sources is enjoyed not by entire companies but by particular business units and product lines. This is the only level of competition at which the concept of competitive advantage affords you any real traction in strategic thinking; even if a company sells soup or dog food exclusively, it may still have individual businesses and product lines with very different degrees of competitive advantage and therefore different returns on invested capital.

EXHIBIT 8.2 **Sources of Competitive Advantage**

Price premium	Cost and capital efficiency
Innovative products: Difficult-to-copy or patented products, services, or technologies	Innovative business method: Difficult-to-copy business method that contrasts with established industry practice
Quality: Customers willing to pay a premium for a real or perceived difference in quality over and above competing products or services	Unique resources: Advantage resulting from inherent geological characteristics or unique access to raw material(s)
Brand: Customers willing to pay a premium based on brand, even if there is no clear quality difference	Economies of scale: Efficient scale or size for the relevant market
Customer lock-in: Customers unwilling or unable to replace a product or service they use with a competing product or service	Scalable product/process: Ability to add customers and capacity at negligible marginal cost
Rational price discipline: Lower bound on prices established by large industry leaders through price signaling or capacity management	
Increasing returns to scale: Scalable products that offer increasing value to customers with scale	

On balance, price premiums offer any business the greatest scope for achieving an attractive ROIC, but they are usually more difficult to achieve than cost efficiencies. Also, the businesses or products with the highest returns are often those that weave together more than one advantage.

Price Premium Advantages

In commodity markets, companies are typically price takers, meaning they must sell at the market price to generate sales, because the products are hard to differentiate. To sell its products at a price premium, a company must find a way to differentiate its products from those of competitors. We distinguish five sources of price premiums: innovative products, quality, brand, customer lock-in, and rational price discipline.

Innovative Products Innovative goods and services yield high returns on capital if they are protected by patents, are difficult to copy, or both. Absent these protections, even an innovative product won't do much to generate high returns.

Pharmaceutical companies earn high returns because they produce innovative products that, although often easy to copy, are protected by patents for up to 20 years. The business can charge a price premium during the protected period, after which generics will enter the market and drive the price down. Even after the patent expires, the holder may enjoy some price "stickiness."

An example of an innovative product line that is not patent protected but still difficult to copy was Apple's series of iPod MP3 players. MP3 players had been on the market for several years before Apple introduced the iPod, and the core technology was the same for all competitors. The iPod was more successful, however, because of its appealing design and ease of use afforded by its user interface and integration with iTunes. Apple followed a similar approach with the iPhone and iPad; once again, the design and user interface were core drivers of the price premium. Although not patent protected, good design can be difficult to copy.

Quality A term used as broadly as *quality* requires definition. In the context of competitive advantage and ROIC, quality means a real or perceived difference between one product or service and another for which consumers are willing to pay a higher price. In the car business, for example, BMW enjoys a price premium because customers perceive that its cars handle and drive better than comparable automobiles that cost less. The cost of providing the extra quality is less than the price premium. Hence, BMW has often been able to earn higher returns than many other carmakers. Appliance makers such as Weber can price their products at a premium over those of their competitors because customers perceive their reliability and durability to be superior.

Sometimes the perception of quality lasts significantly longer than any actual difference in quality. This has been the case with Honda and Toyota, relative to many automakers (at least until Toyota had to make product recalls in 2009). While American and Japanese cars have been comparable in terms of quantifiable quality measures, such as the J.D. Power survey, Japanese companies have enjoyed a price premium for their products. Even when American and Japanese sticker prices on comparable vehicles were the same, American manufacturers were often forced to sell at a \$2,000 to \$3,000 discount, whereas Japanese cars sold for nearer the asking price.

Brand Price premiums based on brand are sometimes hard to distinguish from price premiums based on quality, and the two are highly correlated (as in the example of BMW). While the quality of a product may matter more than its established branding, sometimes the brand itself is what matters more—especially when the brand has lasted a very long time, as in the cases of Heineken, Coca-Cola, Perrier, and Mercedes-Benz.

Packaged food, beverages, and durable consumer goods are good examples of sectors where brands earn price premiums for some but not all products. In some categories, such as bottled water and breakfast cereals, customers are loyal to brands like Perrier and Cheerios despite the availability of high-quality branded and private-label alternatives. In other categories, including meat, branding has not been successful. Because of their strong brands, beverage and cereal companies can earn returns on capital of around 30 percent, while meat processors earn returns of around 15 percent.

Customer Lock-In When replacing one company's product or service with another's is relatively costly (relative to the price of the product) for customers, the incumbent company can charge a price premium—if not for the initial sale, then at least for additional units or for subsequent generations and iterations of the original product. Gillette's shaving products offer a classic example: the manufacturer realizes its margin not on the starter pack but on replacement razor blades. In consumer electronics, wireless-audio product manufacturers such as Sonos also create a form of lock-in: once customers have one or more loudspeakers installed, they are not likely to switch to other brands when replacing or adding units, as these would lack compatibility with their existing Sonos units.

High switching costs, relative to the price of the product or service, create the strongest customer lock-in. Medical devices, such as artificial joints, can lock in the doctors who purchase them, because doctors need time to train and become proficient in the procedures for using and/or implanting those devices. Once doctors are up to speed on a device, they won't switch to a competing product unless there is a compelling reason to invest the necessary effort. Similarly, bankers and traders who have invested considerable time in

learning how to work with particular brands of financial terminals are often reluctant to learn another system. An installed base can be a powerful driver of competitive advantage.

Rational Price Discipline In commodity industries with many competitors, the laws of supply and demand will drive down prices and ROIC. This applies not just to obvious commodities—say, chemicals and paper—but also to more recently commoditized products and services, such as airline seats. It would take a net increase of only 5 to 10 percent in airline ticket prices to turn the industry's aggregate loss to an aggregate profit. But each competitor is tempted to get an edge in filling seats by keeping prices low, even when fuel prices and other costs rise for all competitors. In the past several years, the airline sector in the United States has rapidly consolidated and become more cautious about adding seat capacity. That has allowed U.S. airlines to operate at more attractive price levels and earn healthy returns on capital. In contrast, most European airlines still face strong price competition and are rarely able to earn returns on capital above their cost of capital.

Occasionally, we find an example such as the U.S. airline industry that manages to overcome the forces of competition and set its prices at a level that earns its companies reasonable returns on capital (though rarely more than 15 percent) without breaking competition law. For example, for many years, almost all real estate agents in the United States charged a 6 percent commission on the price of each home they sold. In other cases, government sanctions disciplined pricing in an industry through regulatory structures. Until the late 1990s, airline fares in Europe were high because in most national markets, foreign competitors faced restrictions when competing with domestic airlines. Prices collapsed when the European airline markets were fully deregulated in 1997.

Rational, legitimate pricing discipline typically works when one competitor acts as the leader and others quickly replicate its price moves. In addition, there must be barriers to new entrants, and each competitor must be large enough that a price war will surely reduce the profit on its existing volume by more than any extra profit gained from new sales. If there are smaller competitors with more to gain from extra volume than they would lose from lower prices, then price discipline will be very difficult to maintain.

Most attempts by industry players to maintain a floor price fail. Take the paper industry, for example. Its ROIC averaged less than 10 percent from 1990 to 2013. The industry created this problem for itself because the companies all tended to expand at once, after demand and prices had risen. As a result, a large chunk of new capacity came on line at the same time, upsetting the balance of supply and demand and forcing down prices and returns.

Even cartels (which are illegal in most of the world) find it difficult to maintain price levels, because each cartel member has a great incentive to

lower prices and attract more sales. This so-called free-rider issue makes it difficult to maintain price levels over long periods, even for the Organization of Petroleum Exporting Countries (OPEC), the world's largest and most prominent cartel.

Cost and Capital Efficiency Advantages

Theoretically, cost and capital efficiency are two separate competitive advantages. Cost efficiency is the ability to deliver products and services at a lower cost than the competition. Capital efficiency is about delivering more products per dollar of invested capital than competitors. In practice, both tend to share common drivers and are hard to separate. (Is a company's outsourcing of manufacturing to Asia a source of cost efficiency or capital efficiency?) Consequently, we treat the following four sources of competitive advantage as deriving from both the cost and capital efficiencies they achieve.

Innovative Business Method A company's business method is the combination of its production, logistics, and pattern of interaction with customers. Most production methods can be copied, but some are difficult to copy at some times. For example, early in its life, Dell developed a new way of making and distributing personal computers. Dell sold directly to its customers, made its machines to order with almost no inventory (by assembling machines with standardized parts that could be purchased from different suppliers at different times at very low cost), and received payments from customers as soon as products shipped. In contrast, Hewlett-Packard and Compaq, Dell's dominant competitors at that time, were producing in large batches and selling through retailers. Dell's cost and capital efficiency enabled the company initially to generate a much higher ROIC than its competitors, who couldn't switch quickly to a direct-sales model without angering their retailers and reengineering their production processes.

Notably, Dell's success formula eroded over time as its sales shifted from desktop to notebook computers. Notebook computers are built to much tighter part specifications, often using parts from vendors made expressly for Dell. Since everything must fit together just right, Dell needed more support from its vendors and saw its leverage over them diminished.

Sweden's IKEA provides another example of advantage gleaned from an innovative business method. IKEA transformed the home furniture business around the world, driving innovations in all steps of the business chain from design to manufacturing and distribution to sales. Its concept of self-assembly furniture reduces production and storage costs. Close collaboration in design and manufacturing minimizes product development costs and time. Manufacturing costs are kept low by using a limited range of raw materials. Its automated distribution centers are highly efficient because its product meets

standard packaging requirements. Its retail stores are highly standardized and operate at low labor costs because customers pick up their furniture, still in packages, directly from storage. By making sure all these steps in the chain also stay carefully aligned with customer preferences, IKEA has become the largest furniture retailer in the world, operating more than 400 stores in more than 50 markets as of 2018.

Unique Resources Sometimes a company has access to a unique resource that cannot be replicated. This provides a significant competitive advantage. For example, in general, gold miners in North America earn higher returns than those in South Africa because the northern ore is closer to the surface, so extracting it is easier and costs less. These lower extraction costs are a primary driver of higher returns from North American mines (though partially offset by higher investment costs).

Another example is Nornickel's nickel mine in northern Siberia. The content of precious metals (e.g., palladium) in the mine's nickel ore is significantly higher than in the ore from Canadian and Indonesian mines. In other words, Nornickel extracts not only nickel from its ore but also some high-priced palladium. As a result, Siberian mines earn higher returns than other nickel mines.

Geography often plays a role in gaining advantage from unique resources. Obviously, most leading seaports and airports owe their success to their specific location. The Port of Rotterdam Authority operates the largest seaport of Europe, benefiting from a location that connects the Rhine River (Europe's busiest waterway) and the continent's largest economy (Germany) to the North Sea and global shipping routes. But geography is important not only for infrastructure companies. In general, whenever the cost of shipping a product is high relative to the value of the product, producers near their customers have a unique advantage. China is the largest consumer of iron ore. South American mines, therefore, face a distinct transportation cost disadvantage compared with Australian iron mines, and this contributes to the South American mines' lower returns compared with Australian competitors.

Economies of Scale The notion of economies of scale is often misunderstood to mean that there are automatic economies that come with size. Scale can indeed be important to value, but usually only at the regional or even local level, not in the national or global market. For example, for many retail businesses in dry cleaning, funeral services, or workspace rentals, it's much more important to be large in one city than large across the entire country, because local costs for facilities and advertising are either lumpy or fixed. Buying advertising airtime and space in Chicago is the same whether you have one store or a dozen. Likewise, a key element that determines the profitability of health insurers in the United States is their ability to negotiate prices with providers (hospitals and doctors), who tend to operate locally rather than nationally. The

insurer with the highest market share in a local market will be in a position to negotiate the lowest prices, regardless of its national market share. In other words, it's better to have the number-one market share in ten states than to be number one nationwide but number four in every state.

Another aspect of scale economies is that a company derives benefit only if competitors cannot easily achieve similar scale. Sometimes the required investments are large enough to deter competitors. Anyone who wants to compete with United Parcel Service (UPS), for instance, must first pay the enormous fixed expense of installing an international network and then operate at a loss for quite some time while drawing customers away from the incumbent. Even though UPS continually must add new costs for planes, trucks, and drivers, these costs are variable—in contrast to the fixed cost of building the international network—and are incurred in stepwise fashion. That does not mean the industry is completely safe from competition. Over the past few years, Amazon has been building its own shipping network. Scale is less effective as a barrier to entry for Amazon: the company can rapidly reach sufficient scale thanks to its internal demand, and it has shown itself prepared and able to incur significant upfront investments.

Scalable Product or Process Having products or processes that are scalable means the cost of supplying or serving additional customers is very low at almost any level of scale. Businesses with this advantage usually deliver their products and services using information technology (IT). Consider a company that provides standardized software (in other words, a product that requires little customization). Once the software is developed, it can be sold to many customers with no incremental development costs. So the gross margin on incremental sales could be as high as 100 percent. As sales rise the only costs that increase are typically for selling, marketing, and administration.

For scalable software businesses, the upfront investments are not the only hurdle that competitors must deal with. Customers face costs of switching to other software providers, so competitors cannot easily achieve a similar scale as the incumbent player. That does not mean such competitive advantages last indefinitely, however; ongoing technological innovations in IT create opportunities for new competitors. For example, in financial and payments services, new entrants such as PayPal or Ayden have secured leading positions by starting new business models built on innovative technology platforms. Incumbent players, strapped with heritage organizations, systems, and processes, have found it difficult to copy the innovations.

Other examples of scalable businesses include media companies that make and distribute movies or TV shows. Making the movie or show requires an initial outlay for the crew, sets, actors, and so on. But those costs are fixed regardless of how many people end up viewing and paying for the show. There may be some incremental advertising costs and very small costs associated

with putting the movie on DVD or streaming it. But overall, costs do not rise as customer numbers increase. In this case, it is the access to unique resources—namely, media content—that holds off competitors from capturing similar scale economies.

Most IT-based or IT-enabled businesses offer some form of scalability, especially given recent developments in cloud-based computing. But what counts is whether all critical elements of a business system are scalable. Take, for example, online food delivery businesses. These businesses can easily scale up in terms of number of registered restaurants, customers, and orders, but they still incur incremental costs for each individual order delivery, if only for transportation. Such costs still mount with the number of clients, which presents some limits on scalability and reduction of costs to serve as the business grows.

Network Economies

Some scalable businesses models provide extraordinarily high returns on capital because they exhibit network economies that lead to increasing returns to scale. As the business gains customers and grows, the cost of offering the products decreases, and their value to customers increases. The eBay example we related at the beginning of this chapter illustrates this. Other examples are online lodging and travel platforms such as Airbnb and Booking.com. These models feature scalable products where the marginal cost of additional transactions is minimal. In addition, with scale, these platform services also become more valuable to both end customers and lodging providers. As a result, Airbnb and Booking.com can realize competitive advantages both in price and in cost and capital efficiencies.

Such sources of competitive advantage become even more powerful when customers face high switching costs. Consider a company like Microsoft. Its Office software benefits from scalable operations on the cost side because it can supply online products and services at extremely low marginal cost. Office has also become more valuable as the customer base has expanded over time. Microsoft has been able to lock in customers who want to easily exchange documents with other Office users and who are not keen to spend time and effort switching to alternative software. Some social-media business models, such as Facebook's, offer similar customer lock-in combined with increasing returns on scale.

Although many new digital business models for social media, digital marketplaces, and e-commerce like to claim such increasing returns to scale, they occur in rare circumstances only. Economists Carl Shapiro and Hal Varian popularized this concept in their 1998 book *Information Rules*.⁴

⁴ C. Shapiro and H. Varian, *Information Rules: A Strategic Guide to the Network Economy* (Boston: Harvard Business School Press, 1998).

The management implication of this insight was that in a business with increasing returns, it is important to get big faster than anyone else. Shapiro and Varian also explained the rare conditions under which it is possible to increase returns to scale. Sadly, executives who ignored that part of the book and pursued “network effects” faced disaster. For example, many U.S. electric-power producers tried to get big fast by buying up everything they could. Most collapsed, because there are no increasing returns from scaling electric-power production. Perhaps more important, such scale effects lead to superior lasting returns only if a company can prevent competitors from achieving similar scale.

SUSTAINING RETURN ON INVESTED CAPITAL

The longer a company can sustain a high ROIC, the more value management will create. In a perfectly competitive economy, ROIC higher than the cost of capital get competed away. Whether a company can sustain a given level of ROIC depends on the length of the life cycles of its businesses and products, the length of time its competitive advantages can persist, and its potential for renewing businesses and products.

Length of Product Life Cycle

The longer the life cycle of a company’s businesses and products, the better its chances of sustaining its ROIC. To illustrate, while the products of companies such as Coca-Cola or Mars may not seem as exciting as the latest flashy electronics items, culturally entrenched, branded soft drinks and snacks are likely to have a market for far longer than many new gadgets. Similarly, a unique resource (like palladium-rich nickel ore) can be a durable source of advantage if it is related to a long product life cycle but will be less so if it isn’t (this appears to be the case for lignite and coal today).

A business model that locks customers into a product with a short life cycle is far less valuable than one that locks customers in for a long time. Once users of Microsoft’s Windows have become well versed in the platform, they are unlikely to switch to a new competitor. Even Linux, a low-cost alternative to Windows, has struggled to gain market share as system administrators and end users remain wary of learning a new way of computing. Microsoft’s success in extending the life cycle of Windows has been a huge source of value to the company. Contrast this with a company like BlackBerry, which had an impressive customer base until the life cycle of its early smartphones was cut short by the introduction of the iPhone and other next-generation devices.

Persistence of Competitive Advantage

If a company cannot prevent competition from duplicating its business, high ROIC will be short-lived, and the company's value will diminish. Consider two major cost improvements that airlines implemented over the past decade. The self-service kiosk and, more recently, the smartphone app allow passengers to purchase a ticket and to print or download a boarding pass from anywhere in the world without waiting in line. From the airlines' perspective, fewer ground personnel and equipment are needed to handle even more passengers. So why has this cost improvement not translated into high ROIC for the airlines?⁵ Since every company has access to the technology, any cost improvements are passed directly to the consumer in the form of lower prices. A similar example comes from robotic automation's ongoing effect on productivity improvements in automotive manufacturing: all players adopt the new technology and pass on the cost reductions to customers. In general, advantages that arise from brand and quality on the price side and scalability on the cost side tend to have more staying power than those arising from more temporary sources of advantage, such as an innovation that will tend to be superseded by subsequent innovations.

Potential for Product Renewal

Few businesses or products have life cycles as long as Coca-Cola's. Most companies need to find renewal businesses and products where they can leverage existing advantages or build new ones. This is an area where brands prove their value. Consumer goods companies excel at using their brands to launch new products: think of Apple's success with the iPhone, Bulgari moving into fragrances, Mars entering the ice cream business, Netflix switching from DVD rentals by mail to video streaming online, John Deere offering information services to farmers, and Signify (the former Philips Lighting) developing connected lighting solutions such as Hue. Being good at innovation also helps companies renew products and businesses. Thus, pharmaceutical companies exist because they can discover new drugs, and semiconductor technology players such as ASML and Intel rely on their technology innovation to launch new products and stay ahead of competitors.

Some companies, such as Procter & Gamble and Alphabet's Google subsidiary, are able to maintain their primary product lines while simultaneously expanding into new markets. Google built new advertising and subscription businesses around, for example, YouTube and G Suite (which comprises Gmail, Calendar, and Google+) to complement the original advertising business that its search engine powers. Procter & Gamble has a strong record of

⁵ Although ROIC in the U.S. airline industry has increased over recent years, credit for this improvement goes not to cost reduction from new technology but to earnings gains from ongoing consolidation and lower fuel prices.

continuing to introduce successful new products, including Swiffer, Febreze, and Crest Whitestrips. It also anticipated the strong growth in beauty products in the early 2000s with a number of acquisitions that increased its revenues in the category from \$7.3 billion to \$20 billion from 1999 to 2013. Product development and renewal have enabled the company to advance from owning just a single billion-dollar brand (by sales) in 1999 to 23 such brands in 2013. Underlining its competitive strength in managing very large brands, Procter & Gamble announced in 2014 that it would discontinue or divest 90 to 100 smaller brands from a total of 180 brands in its portfolio.

As the next section of this chapter indicates, empirical studies show that over the past five decades, companies have been generally successful in sustaining their rates of ROIC. It appears that when companies have found a strategy that creates competitive advantages, they are often able to sustain and renew these advantages over many years. This also holds for the relatively new digital business models with which Amazon, Google, Microsoft, and others have retained and renewed their competitive advantages for two decades and more. While competition clearly plays a major role in driving down ROIC, managers can sustain a high rate of return by anticipating and responding to changes in the environment better than their competitors do.

AN EMPIRICAL ANALYSIS OF RETURNS ON INVESTED CAPITAL

Several key findings concerning ROIC emerge from a study of 1963–2017 returns on invested capital at U.S.-based nonfinancial companies with (inflation-adjusted) revenues greater than \$1 billion:⁶

- The median ROIC was stable at about 10 percent until the turn of the century and then increased to 17 percent after 2010, where it has remained since. Important drivers of this effect were a general increase in profitability across sectors, combined with a shift in the mix of U.S.-based companies to higher-returning sectors in life sciences and technology. These sectors not only significantly increased their ROIC, but also grew faster.
- Returns on invested capital differ by industry. Industries such as pharmaceuticals and branded consumer goods that rely on patents and brands for their sustainable competitive advantages tend to have high median ROIC, whereas companies in basic industries, such as oil and gas, mining, and utilities, tend to earn low ROIC.

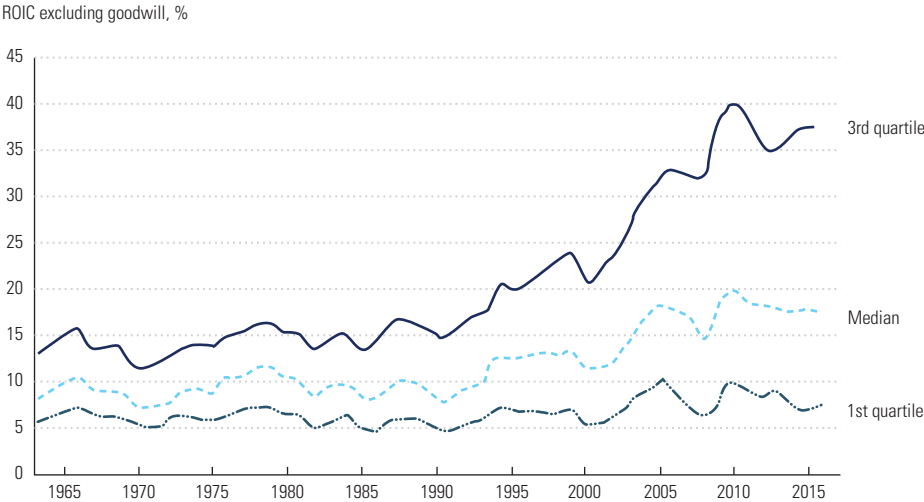
⁶ The results come from Corporate Performance Analytics by McKinsey, which relies on financial data provided by Standard & Poor's Compustat and Capital IQ. The number of companies in the sample varies from year to year and excludes financial institutions and industrial companies with significant financial businesses. In 2017, the sample included 1,095 companies.

- There are large variations in rates of ROIC within industries. Some companies earn attractive returns in industries where the median return is low (e.g., Walmart), and vice versa.
- Relative rates of ROIC across industries are generally stable, especially compared with rates of growth (discussed in the next chapter). Industry rankings by median ROIC do not change much over time, with only a few industries making a clear aggregate shift upward or downward. These shifts typically reflect structural changes, such as the widespread consolidation in the defense and airline industries over the past two decades and the maturing of the biotech industry. Individual company returns gradually tend toward their industry medians over time but are generally persistent. Even the 2008 financial crisis did not upset this trend.

ROIC Trends and Drivers

Relatively stable ROIC levels from the early 1960s to the early 2000s are evident in Exhibit 8.3, which plots median ROIC between 1963 and 2017 for U.S.-based nonfinancial companies.⁷ In that exhibit, the measure of ROIC excludes goodwill and acquired intangibles, which allows us to focus on the underlying economics of companies without the distortion of premiums paid for acquisitions (discussed later in the chapter).

EXHIBIT 8.3 ROIC of U.S.-Based Nonfinancial Companies, 1963–2017



Source: Corporate Performance Analytics by McKinsey.

⁷ The numbers in this section are based on U.S. companies because longer-term data for non-U.S. companies are not readily available.

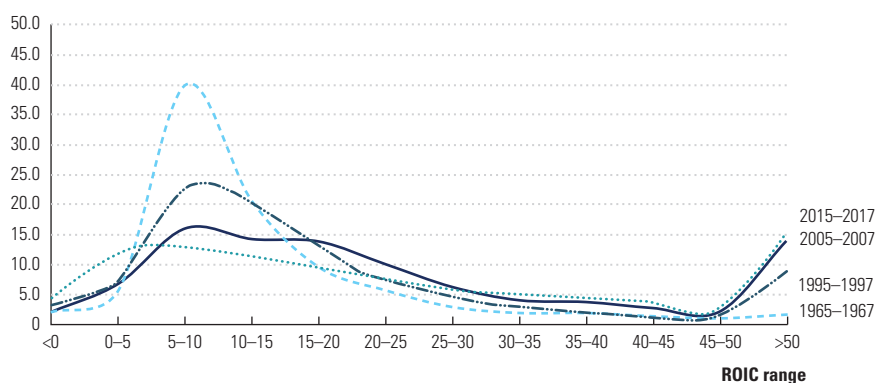
Until the 2000s, the median ROIC without goodwill was about 10 percent. Furthermore, annual medians oscillated in a tight range, with higher returns in high-GDP-growth years and lower returns in low-growth years. Since the 2000s, however, median ROIC without goodwill has increased to what appears to be a new level of about 17 percent in 2010 and beyond. Notice also that the spread between the first and third quartiles has widened. The first-quartile company earned around 5 to 7 percent during the entire period, while the third-quartile company's return has increased from the midteens to over 35 percent.

In fact, the entire distribution of ROIC has widened as more and more companies earn high returns on capital. Exhibit 8.4 shows the distribution of ROICs over different eras. In the 1965–1967 period, only 14 percent of companies earned more than a 20 percent ROIC, compared with 45 percent in 2015–2017. At the same time, the share of companies earning less than 10 percent has declined from 53 percent to 30 percent.

One factor powering the shift in the median ROIC is the steady increase of operating margins across sectors since the mid-1990s. As shown in Exhibit 8.5, the median operating margin (NOPAT over sales) has risen by two percentage points. When higher margins combine with improved capital productivity (lower invested capital over sales), returns on capital rise. The U.S. economy's changing industry mix serves as an even more powerful driver and helps to explain the widening dispersion of returns. Among U.S.-based nonfinancial companies, the share of operating profits from companies in the life science and technology sectors increased from 19 percent of total operating profits in 1995 to 38 percent in 2017.⁸ This impressive increase has been driven by the

EXHIBIT 8.4 **Distribution of ROIC**

% of companies with given annual ROIC excluding goodwill, by time period



Source: Corporate Performance Analytics by McKinsey.

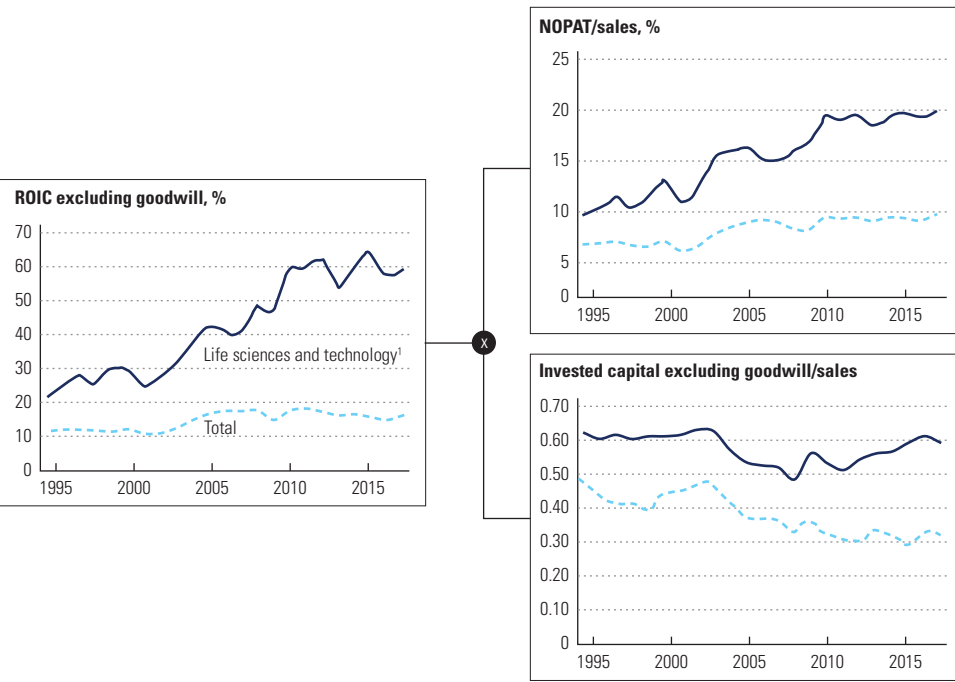
⁸ We defined life science and technology sectors to comprise pharmaceuticals, biotechnology, health-care equipment and supplies, information services and software, and technology hardware, storage, and peripherals.

faster growth of these sectors relative to the rest of the economy, these sectors' generally higher margins and returns on capital, and increases in these sectors' margins and returns on capital. As a result, the life science and technology sectors generate six percentage points of the total 16 percent return on capital for the U.S. economy as a whole, compared with only two percentage points of a 12 percent overall return in 1995 (see Exhibit 8.6).⁹

ROIC by Industry

To see how differences in ROIC across industries and companies relate to likely differences in drivers of competitive advantage, we examined variations in ROIC by industry over the past two decades. Our findings are in line with results from prior editions of this book, in which we tracked profitability going back to the 1960s. Exhibit 8.7 shows the median returns on invested

EXHIBIT 8.5 Disaggregating ROIC of U.S.-Based Nonfinancial Companies, 1995–2017



Note: ROIC, NOPAT/sales, and invested capital/sales are aggregate metrics.

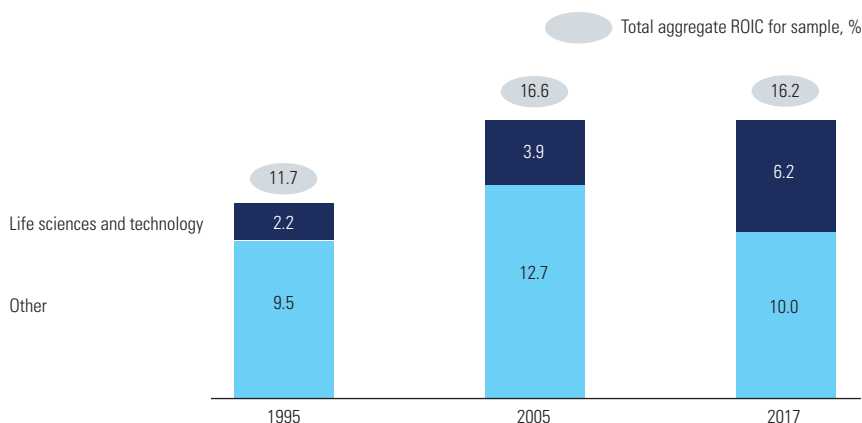
¹ Life sciences and technology sectors comprise pharmaceuticals, biotechnology, health-care equipment and supplies, information services and software, and technology hardware, storage, and peripherals.

Source: Corporate Performance Analytics by McKinsey.

⁹ The ROIC contribution is calculated as the total NOPAT of the life science and technology sectors divided by the invested capital of all sectors. Note that the aggregate ROIC for the total sample is close to the median ROIC of 12.4% in 1995, 16.7% in 2005, and 17.4% in 2017.

EXHIBIT 8.6 **Contribution of Life Sciences and Technology Industries to the Broader Economy, 1995–2017**

Contribution to aggregate ROIC excluding goodwill for total sample, %



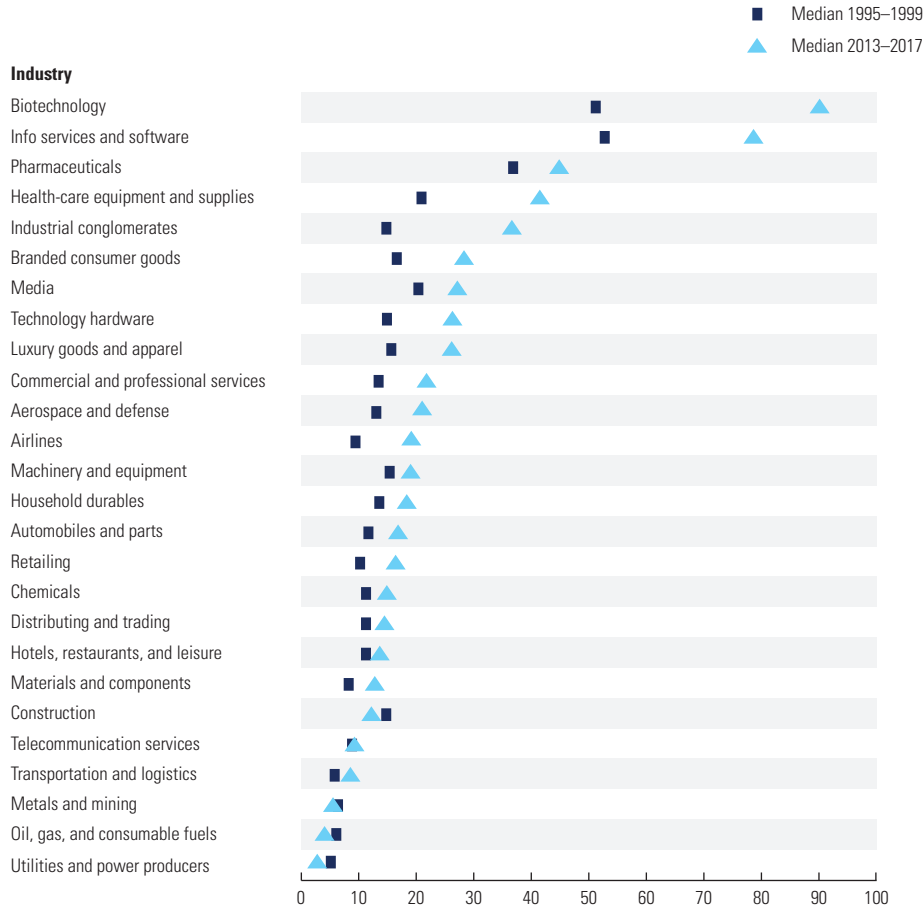
Source: Corporate Performance Analytics by McKinsey.

capital for a range of industries during the periods 1995–1999 and 2013–2017. The exhibit reveals large differences in median ROIC across industries. Not surprisingly, industries with the highest returns, such as pharmaceuticals, health-care equipment, and technology-related businesses, are those with sustainable competitive advantages. In the case of pharmaceuticals and health-care equipment, this is due to patent-protected innovation. In technology-related businesses, advantage typically flows from increasing returns to scale and customer lock-in. The branded consumer goods and luxury goods sectors have high returns thanks to customer loyalty based on brand strength. The industries at the bottom of the chart tend to be those where it is difficult to achieve a price premium or cost advantage—often commodity-based industries, including oil, gas, metals, and mining.

Industries typically recognized for having higher returns have often been the ones that also deliver the clearest improvements in ROIC over time. The reasons vary by sector. For example, leading aerospace and defense companies tend to focus on government contracts that include advance payments. This keeps the companies' invested capital at low levels relative to revenues. The biotechnology sector matured over the past 20 years with large companies such as Amgen and Gilead generating outstanding returns from the successful development and marketing of innovative blockbuster drugs. Finally, technology players benefited from growth and innovation in hardware (via semiconductors, servers, and smartphones) and in software and services. As markets became increasingly global, the shift to more scalable online software and information services also contributed to higher margins and returns; consider Facebook's and YouTube's growth in social media or Microsoft and Oracle in software. One outlier from this high-return club: the airline sector, which has

EXHIBIT 8.7 ROIC by Industry, 1995–2017

ROIC excluding goodwill, median, %



Source: Corporate Performance Analytics by McKinsey.

delivered low ROIC historically but managed to increase returns in recent years, thanks to ongoing consolidation in the United States and significantly lower fuel prices.

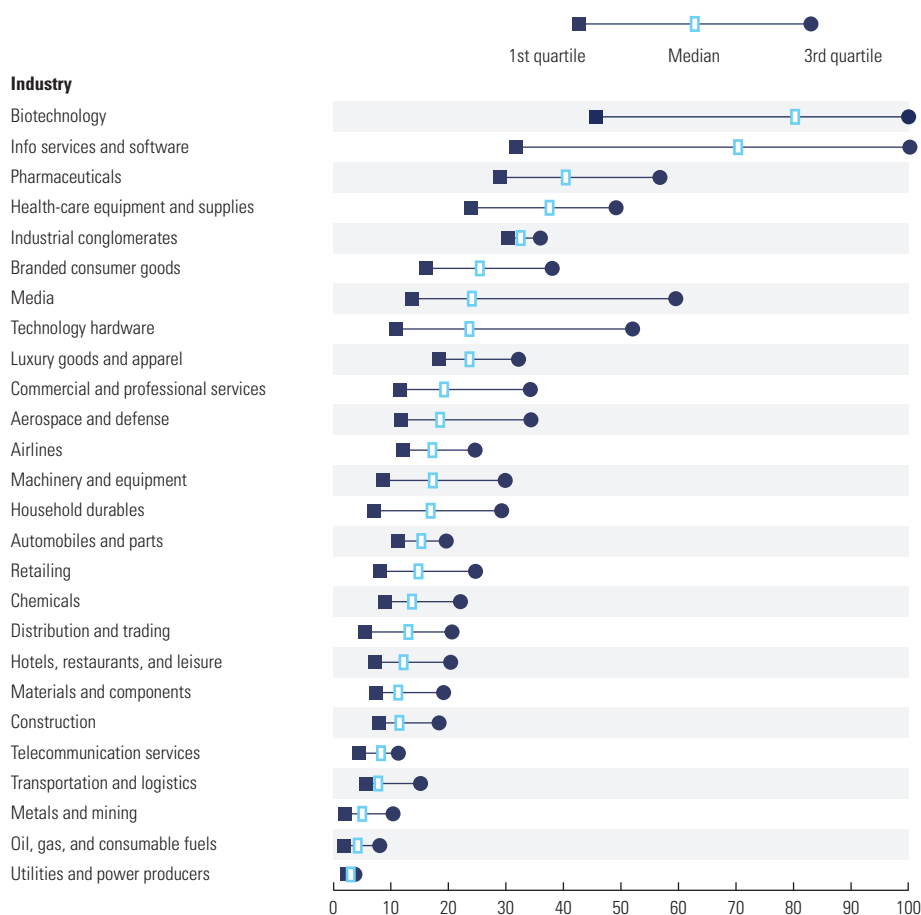
To some extent, the increases in ROIC reflect a trend across industries to lower capital intensity, as we observed in Exhibit 8.5. This could be interpreted as U.S. companies simply reducing their capital base—for example, by outsourcing operations without necessarily creating value.¹⁰ This is not the case, however. Total economic profit for our sample of the largest U.S. companies increased from \$31 billion in 1995 to \$560 billion in 2017. Moreover, economic profit increased for most sectors over the same period, with similar patterns as for ROIC.

¹⁰ A ROIC increase from a reduction in invested capital from outsourcing does not necessarily indicate value creation. As Chapter 24 notes, the change in economic profit provides a reliable indication.

Differences in ROIC within industries can be considerable. Exhibit 8.8 shows the variation between the first and third quartiles for the same industries. Note the wide range of returns in information services and software. Some of the companies in the sector earn low returns because they are capital intensive, and low margins because their business model is not scalable, as in the case of running data centers. Other companies provide services that are based on standardized and scalable software, where the incremental cost to serve a new customer is small, leading to high ROIC. In some industries, the largest players also generate the highest returns, and median ROIC does not reflect the aggregated ROIC for the sector as a whole (defined as NOPAT for the sector divided by its total invested capital). An example is the technology hardware sector, where players like Apple drive the aggregate ROIC to almost 70 percent, versus a median of 27 percent in 2015–2017.

EXHIBIT 8.8 **Variation in ROIC within Industries, 2015–2017**

ROIC,¹ excluding goodwill, %



¹ Scale limited to 100% for presentation purposes.

Source: Corporate Performance Analytics by McKinsey.

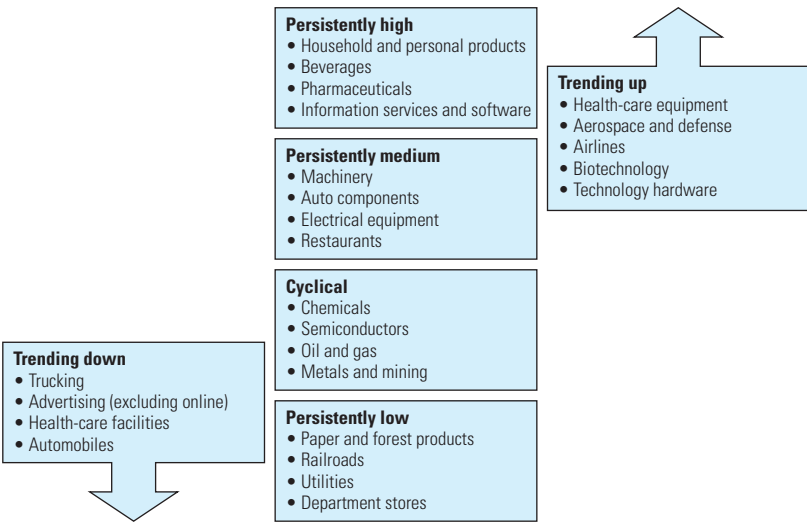
This chart also shows that the best performers in a weaker or mediocre industry may outperform the median performer in a stronger industry. Consider, for example, retailing, shown in the bottom half of the chart. The stronger retailers (like Walmart) outperform the weaker companies in the media industry, which appears in the top half of the chart.

Stability of ROIC

While industries often exhibit variations in their respective ROIC, many industries tend to remain fairly stable over time. We can see this by grouping the industry-level returns on invested capital according to whether they are relatively high, medium, or low. As shown in Exhibit 8.9, most industries stayed in the same group over the period we studied, starting in the early 1960s. In addition, some industries are cyclical, having high and low returns at different points in the cycle but demonstrating no clear trend up or down over time.

Persistently high-return industries included household and personal products, beverages, pharmaceuticals, and information services and software. These industries have consistently high returns because they are scalable or are protected by brands or patents. Persistently low returns characterize paper and forest products, railroads, and utilities. These are industries in which price premiums are difficult to achieve because of, for example, low barriers to entry, commodity products, or regulated returns. Perhaps surprisingly, this group also includes department stores. Like commodity industries, department stores can achieve little price differentiation, so as a rule, they realize persistently low returns.

EXHIBIT 8.9 Persistence of Industry ROIC



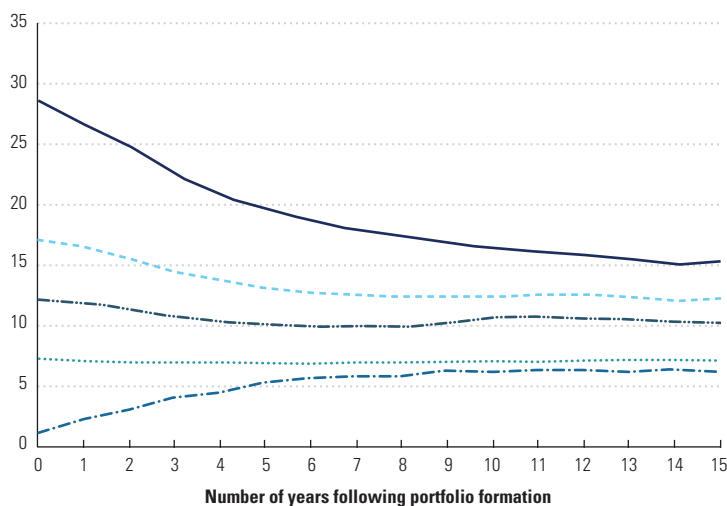
In several industries, there was a clear downward trend in returns. These included trucking, health care facilities, and automobiles. Competition in trucking, advertising, and automobiles has increased substantially over the past five decades. Health-care facilities have had their prices squeezed by the government, insurers, and competition with nonprofits.

Industries where returns on invested capital clearly are trending up are rare. Examples are health-care equipment, airlines, and aerospace and defense. Innovation in health-care equipment has enabled the industry to produce higher-value-added, differentiated products such as artificial joints, as well as more commoditized products, including syringes and forceps. As mentioned earlier, the U.S. airlines industry benefited from consolidation, and companies in aerospace and defense reduced their capital intensity as governments provided up-front funding for many more contracts.

There is similar evidence of sustained rates of return at the company level. We measured the sustainability of company ROIC in our database of nonfinancial corporations by ranking companies based on their ROIC in each year and dividing the group into quintiles. We treated each quintile as a portfolio and tracked the median ROIC for the portfolio over the following 15 years, as shown in Exhibit 8.10. The results indicate some mean reversion: companies earning high returns tended to see their ROIC fall gradually over the succeeding 15 years, and companies earning low returns tended to see them rise over time. Only in the portfolio containing companies generating returns between 5 and 10 percent (mostly regulated companies) do rates of return

EXHIBIT 8.10 **ROIC Decay Analysis**

Median ROIC of portfolios (without goodwill), by quintile,¹ %



¹ At year 0, companies are grouped into one of five portfolios, based on ROIC.

Source: Corporate Performance Analytics by McKinsey.

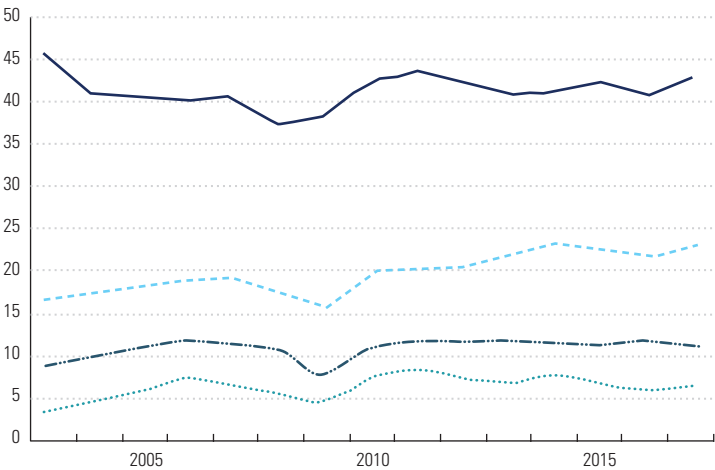
remain constant. However, an important phenomenon is the persistence of superior performance beyond ten years. The returns of the best-performing companies do *not* decline all the way to the aggregate median over 15 years. High-performing companies are in general remarkably capable of sustaining a competitive advantage in their businesses and/or finding new business where they continue or rebuild such advantages. The pattern is stable over time—even over the most recent 15 years, which included the 2008 credit crisis (see Exhibit 8.11).

Since a company's continuing value is highly dependent on long-run forecasts of ROIC and growth, this result has important implications for corporate valuation. Basing a continuing value on the economic concept that ROIC will approach the weighted average cost of capital (WACC) is overly conservative for the *typical* company generating high ROIC. (Continuing value is the focus of Chapter 14.)

Keeping this range of performance in mind, it is important when benchmarking the historical decay of company ROIC to segment results by industry, especially if industry is a proxy for sustainability of competitive advantage. As an example, Exhibit 8.12 plots the ROIC decay rates for branded consumer goods, again sorting the companies into five portfolios based on their starting ROICs. Here, the top-performing companies don't show much reversion to the mean. Even after 15 years, the original class of best performers still outperforms the bottom quintile by more than 13 percentage points.

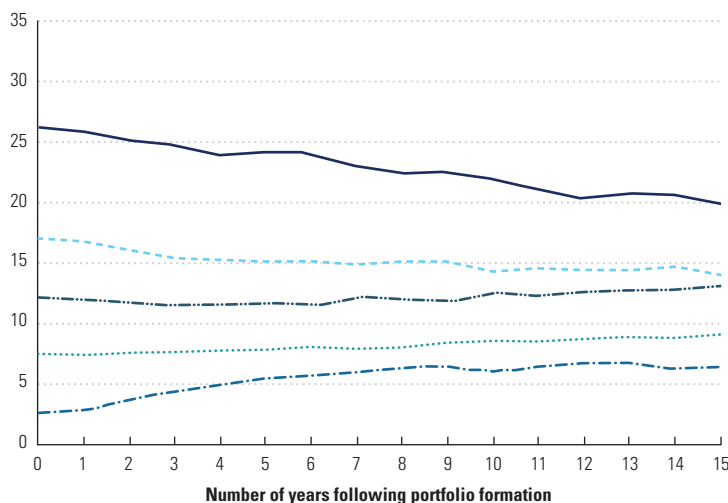
EXHIBIT 8.11 **ROIC Decay through Economic Crisis and Recovery**

Median ROIC of portfolios (excluding goodwill), by 2003 quartile,¹ %



¹ As of 2003, companies are grouped into quartiles, based on ROIC.

Source: Corporate Performance Analytics by McKinsey.

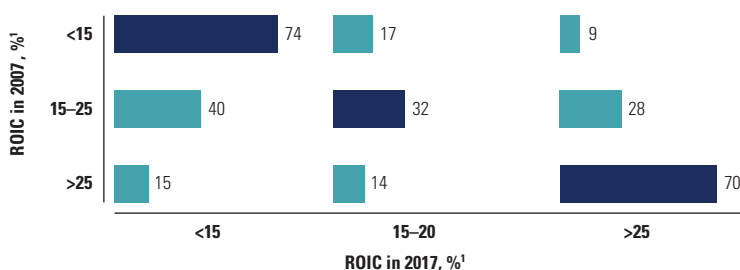
EXHIBIT 8.12 **ROIC Decay for Branded Consumer Goods**Median ROIC of portfolios (without goodwill), by quintile,¹ %¹ At year 0, companies are grouped into one of five portfolios, based on ROIC.

Source: Corporate Performance Analytics by McKinsey.

Although decay rates examine the *rate* of regression toward the mean, they present only aggregate results and tell us nothing about the spread of potential future performance. Does every company generating returns greater than 20 percent eventually migrate to 15 percent, or do some companies actually go on to generate higher returns? Conversely, do some top performers become poor performers? To address this question, we measured the probability that a company will migrate from one ROIC grouping to another in ten years. The results are presented in Exhibit 8.13. Read each row from left to right.

EXHIBIT 8.13 **ROIC Transition Probability**

Probability of achieving ROIC in 2017, %

¹ ROIC excluding goodwill.

Source: Corporate Performance Analytics by McKinsey.

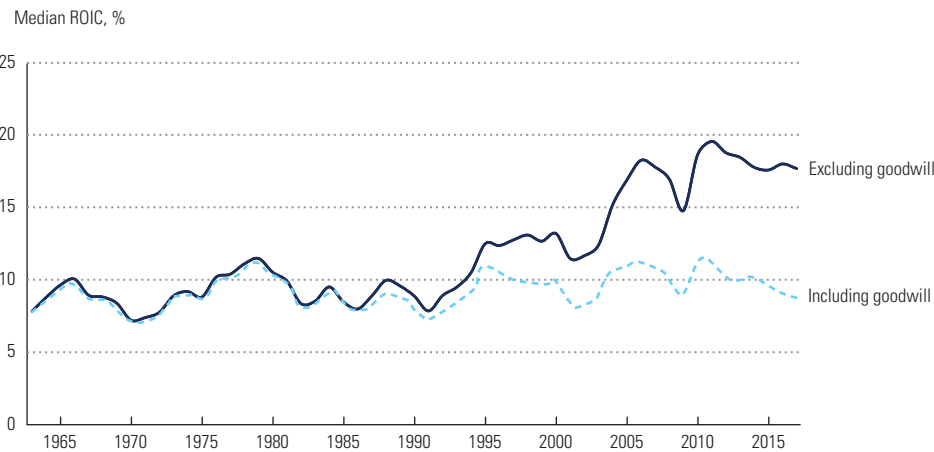
Both high and low performers demonstrate significant stability in their performance. Companies with high or low ROIC are most likely to stay in the same grouping. A company whose ROIC was below 15 percent in 2007 had a 74 percent chance of earning less than 15 percent in 2017. For companies with a ROIC above 25 percent, the probability of maintaining that high performance was 70 percent. Among companies whose ROIC was between 15 and 25 percent in 2007, there was no clear tendency for companies to increase or decrease their ROIC ten years later.

Effect of Acquisitions on ROIC

While returns on invested capital without goodwill have been increasing, returns on invested capital with goodwill have been flat, as shown in Exhibit 8.14. Companies paid high prices for their acquisitions, so much of the value the deals created was transferred to the shareholders of the target company. (Acquisitions and value creation are discussed in Chapter 31.) It does not mean that companies have failed to create value from acquisitions: returns on capital including goodwill above the cost of capital, combined with ongoing growth, indicate that they have created value above and beyond the price paid for these acquisitions. Increasing returns without goodwill indicates that companies have captured significant synergies to improve the performance of the acquired businesses.

For some industries, the differences in return with and without goodwill are even bigger than shown here. For the life science and technology sectors, for example, returns on capital including goodwill were around 25 percent, versus 65 percent without goodwill. Companies in this sector have created more value than any other sector, but shareholders of acquired companies captured much of it.

EXHIBIT 8.14 ROIC Including and Excluding Goodwill, 1995–2017



Source: Corporate Performance Analytics by McKinsey.

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SUMMARY

There is much to learn about returns on invested capital. First, these returns are driven by competitive advantages that enable companies to realize price premiums, cost and capital efficiencies, or some combination of these. Second, industry structure is an important—but not an exclusive—determinant of ROIC. Certain industries are more likely to earn either high, medium, or low returns, but there is still significant variation in the rates of return for individual companies within each industry. Third, and most important, if a company finds a formula or strategy that earns an attractive ROIC, there is a good chance it can sustain that attractive return over time and through changing economic, industry, and company conditions, especially in the case of industries that enjoy relatively long product life cycles. Of course, the converse also is true: if a company earns a low ROIC, that is likely to persist as well.

REVIEW QUESTIONS

1. From a value creation perspective, is it more important for a company to know *where* to compete or *how* to compete? That is, is it more important to play in the right markets or to be the best player in your current markets?
2. What are the sources of competitive advantage that can enable a firm to have a high ROIC?
3. Identify and discuss real examples of companies with a competitive advantage based on customer lock-in as opposed to product innovation. Which type of competitive advantage do you expect to sustain a high ROIC for a longer time?
4. Why do companies operating within the pharmaceutical and biotechnology industries typically sustain higher ROICs than firms in the hardware and equipment industries?
5. Why are competitive advantages based on brands, as in the consumer goods industry, often more important for long-term value creation than advantages based on product quality or innovation?
6. Discuss potential explanations for the increase in median ROIC and the widening of the distribution of ROIC across all companies over recent decades.
7. Explain the difference between ROIC excluding and ROIC including goodwill for U.S. companies: What does this difference imply, and why has it increased so much over the past two decades?
8. In Exhibit 8.10, the gradual decline in ROIC of some of the top-performing companies can be explained by gradual erosion of competitive advantages. What could be the explanation for the gradual increase in ROIC of the bottom-performing companies?
9. Exhibit 8.10 shows that firms with high ROICs tend to decline on average. What does Exhibit 8.12 tell us about differences in ROIC decline within this high ROIC group?
10. Discuss why, within the broader health-care sector, ROIC can be declining for health-care facility companies but increasing for health-care equipment companies.

Growth

Growth and its pursuit grip the business world. The popular view is that a company must grow to survive and prosper. There is certainly some truth to this. Slow-growing companies present fewer interesting opportunities for managers and so may have difficulty attracting and retaining talent. They are also much more likely to be acquired than faster-growing firms. Over the past 25 years, most of the companies that have disappeared from the S&P 500 index were acquired by larger companies or went private.

However, as discussed in Chapters 2 and 3, growth creates value only when a company's new customers, projects, or acquisitions generate returns on invested capital (ROIC) greater than its cost of capital. And as companies grow larger and their industries become ever more competitive, finding good, high-value-creating projects becomes increasingly difficult. Striking the right balance between growth and return on invested capital is critically important to value creation. Our research shows that for companies with a high ROIC, shareholder returns are affected more by an increase in revenues than an increase in ROIC.¹ Indeed, we have found that if such companies let their ROIC drop a bit (though not too much) to achieve higher growth, their returns to shareholders are higher than for companies that maintain or improve their high ROIC but grow more slowly. Conversely, for companies with a low ROIC, increasing it will create more value than growing the company will.

The previous chapter explored why executives need to understand whether their strategies will lead to high returns on invested capital. Similarly, they also need to know which growth opportunities will create the most value. This chapter discusses the principal strategies for driving revenue growth, the ways in which growth creates value, and the challenges of sustaining growth. It ends by analyzing the data on corporate growth patterns over the past 55 years.

¹ See T. Koller and B. Jiang, "How to Choose between Growth and ROIC," *McKinsey on Finance*, no. 25 (Autumn 2007): 19–22.

DRIVERS OF REVENUE GROWTH

When executives plan for growth, a good starting point is for them to disaggregate revenue growth into its three main components:²

1. *Portfolio momentum*. This is the organic revenue growth a company enjoys because of overall expansion in the market segments represented in its portfolio.
2. *Market share performance*. This is the organic revenue growth (or reduction) a company earns by gaining or losing share in any particular market.
3. *Mergers and acquisitions (M&A)*. This represents the inorganic growth a company achieves when it buys or sells revenues through acquisitions or divestments.

Baghai, Smit, and Viguerie showed that for large companies, the most important source of growth by far was portfolio momentum.³ In other words, being in fast-growing markets was the largest driver of growth. Least important was market share growth. Yet managers tend to focus most of their attention on gaining share in their existing product markets. While it's necessary to maintain and sometimes increase market share, changing a company's exposure to growing and shrinking market segments should be a major focus.

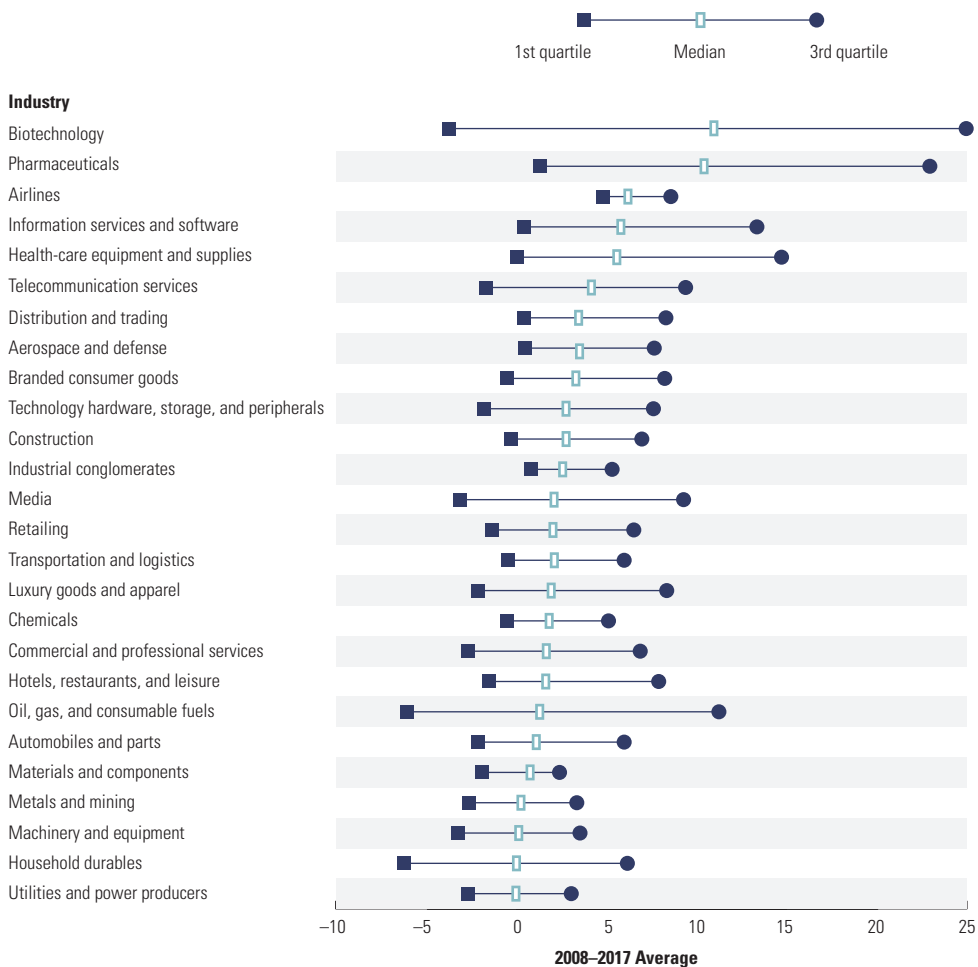
To see the effect of portfolio momentum, consider how the median growth from 2008 to 2017 differs by industry (Exhibit 9.1). Not surprisingly, the fastest-growing sector over this period was biotechnology, which in 2008 was still a small industry that fueled impressive growth from a wave of innovative, blockbuster drugs. Makers of traditional pharmaceuticals delivered the second-highest growth, but mostly driven by consolidation rather than innovation, as many of the highest-selling drugs from the 1990s came off patent. For the same reason, airlines stand high on the list, with some of the largest U.S. players having merged in the past decade. Compared with the prior decade, oil and gas companies dropped to a spot near the bottom of the list, primarily because of significant oil price decreases since 2014. These underlined the industry's cyclical nature in terms of growth as well as ROIC, as discussed in the prior chapter. Note how a sector with high volume growth, such as technology hardware, nonetheless fails to beat many other sectors in terms of revenue growth (in contrast to other technology sectors, such as information services and software). Despite tremendous increases in volume, lower prices have kept total revenue growth relatively modest.

² This section draws on P. Viguerie, S. Smit, and M. Baghai, *The Granularity of Growth* (Hoboken, NJ: John Wiley & Sons, 2008).

³ M. Baghai, S. Smit, and P. Viguerie, "The Granularity of Growth," *McKinsey on Finance*, no. 24 (Summer 2007): 25–30.

EXHIBIT 9.1 Variation in Revenue Growth by Industry

Annual revenue growth rate, adjusted for inflation, %



Source: Compustat; Corporate Performance Analytics by McKinsey.

Exhibit 9.1 also shows widely varied growth within industries. For some sectors, such as pharmaceuticals and airlines, part of the variation can be explained by growth from mergers and acquisitions. In this instance, some players benefited, but not all. But for many other sectors, M&A cannot explain the wide variation in growth. If a company's growth depends mainly on the dynamics of the sector markets in which it operates, why should there be such big differences in growth among different companies operating in the same sector? The most important reason is that the median growth rate of companies competing in any sector masks big differences in growth across the sector's market segments and subsegments.

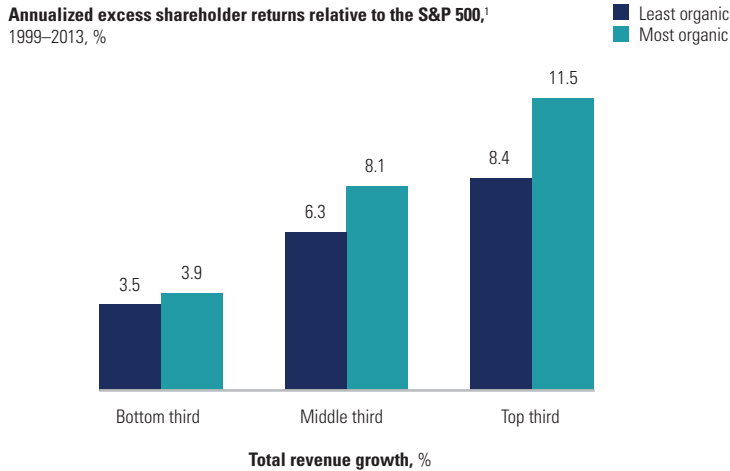
To understand markets in this fine-grained way and the differences in companies' revenue growth, Baghai, Smit, and Viguerie analyzed market growth at the level of individual product and geographical segments with around \$50 million to \$200 million in sales, rather than at the company, divisional, or business unit level.⁴ Their example of a large European manufacturer of personal-care products shows why such analysis is revealing. The company has three divisions with apparently low prospective growth rates ranging from 1.6 percent to 7.5 percent a year. However, the range of forecast growth rates for individual product lines within the divisions is much wider. For instance, the division with the lowest expected growth rate has one product line growing at 24 percent, one of the company's best growth opportunities. At the same time, the division with the highest growth rate has several product lines that are shrinking fast and may warrant divestment.

GROWTH AND VALUE CREATION

While managers typically strive for high growth, the highest growth will not necessarily create the most value. The reason is that the three drivers of growth (portfolio momentum, acquisitions, and market share gains) do not all create value in equal measure. To understand why not, consider who loses under alternative scenarios for revenue growth and how effectively losers can retaliate:

- Growth from increases in market share, particularly in slow- and moderate-growth markets, rarely creates much value for long, because established competitors typically retaliate to protect their market shares. Lasting value creation could only occur in situations where smaller competitors are pushed out of the market entirely or where the company introduces differentiated products or services that are hard for competitors to copy.
- Growth driven by price increases comes at the expense of customers, who are likely to react by reducing consumption and seeking substitute products, so new value created by price increases may not last long either.
- Growth driven by general market expansion comes at the expense of companies in other industries, which may not even know to whom they are losing market share. This category of victim is the least able to retaliate, which makes product market growth the driver likely to create the most value.

⁴ See M. Baghai, S. Smit, and P. Viguerie, "Is Your Growth Strategy Flying Blind?" *Harvard Business Review* (May 2009): 86–96.

EXHIBIT 9.2 **Value Creation from Organic Growth Higher Than from Acquisitions**

¹ Excludes banks, insurance companies, extraction companies, and cyclical commodities.

- The value of growth from acquisitions is harder to characterize, because it depends so much on the price of the acquisition (as discussed in Chapter 31). However, as shown in Exhibit 9.2, a sample of 550 U.S. and European companies reveals that, in general, growth from acquisitions creates less value than organic growth.⁵ The main reason is that companies don't have to invest as much up front for organic growth. In growing through acquisitions, companies typically must pay for the stand-alone value of an acquired business plus a takeover premium. This results in a lower return on invested capital and lower value creation compared with growing organically.

A Hierarchy of Growth Scenarios

It is possible to rank different growth scenarios that fall within the three overall growth strategies according to their potential for creating value (see Exhibit 9.3). This ranking may not be exactly the same for all industries, but it works well as a starting point. The scenarios with the highest potential to create value are all variations on entering fast-growing product markets that take revenues from distant companies, rather than from direct competitors or customers via price increases.

⁵ See M. Goedhart and T. Koller, "The Value Premium of Organic Growth," *McKinsey on Finance*, no. 61 (Autumn 2017): 14–15.

EXHIBIT 9.3 Value of Major Types of Growth

Value created ¹	Type of growth	Rationale
 Above average 	• Create new markets through new products • Convince existing customers to buy more of a product • Attract new customers to the market	• No established competitors; diverts customer spending • All competitors benefit; low risk of retaliation • All competitors benefit; low risk of retaliation
 Average 	• Gain market share in fast-growing market • Make bolt-on acquisitions to accelerate product growth	• Competitors can still grow despite losing share; moderate risk of retaliation • Modest acquisition premium relative to upside potential
 Below average 	• Gain share from rivals through incremental innovation • Gain share from rivals through product promotion and pricing • Make large acquisitions • Increase prices	• Competitors can replicate and take back customers • Competitors can retaliate quickly • High premium to pay; most value diverted to selling shareholders • Unless demand has low price elasticity; customers likely to reduce or divert consumption

¹ Per dollar of revenue.

Developing *new products* or services that are so innovative as to create entirely new product categories has the highest value-creating potential. The stronger the competitive advantage a company can establish in the new-product category, the higher will be its ROIC and the value created. For example, the coronary stent commercialized in the early 1990s reduced the need for heart surgery, lowering both the risk and cost of treating cardiac problems. Owing to this innovation's overwhelming competitive advantage over traditional treatments, as well as over subsequent products entering the market,⁶ neither type of competitor could retaliate, so the innovators created large amounts of value. (As the stent market became highly competitive over the past decade, however, returns on capital have declined considerably.) Similarly, traditional music retailers have been all but competed away, first by online music sales giants such as iTunes and Amazon, and more recently as consumers have taken up online streaming services for mobile devices offered by Spotify, Amazon Music, Apple Music, and others. However, competition in the new digital-entertainment category is itself fierce, so the value created per dollar of revenue in this sector is unlikely to reach the levels that coronary stents once generated.

Next in the pecking order of value-creating growth tactics comes *persuading existing customers to buy more* of a product or related products. For example, if Procter & Gamble convinces customers to wash their hands more frequently,

⁶ Products that entered the market at a later stage were less successful because of high switching costs for customers (see Chapter 8).

the market for hand soap will grow faster. Similarly, if antivirus software provider McAfee convinces computer owners that they need better protection against hackers and viruses, total demand for antivirus software and services will grow faster. Direct competitors will not respond, because they benefit as well. The ROIC associated with the additional revenue is likely to be high, because the companies' manufacturing and distribution systems can typically produce the additional products at little additional cost. Clearly, the benefit will not be as large if the company has to increase costs substantially to secure those sales. For example, offering bank customers insurance products requires the expense of an entirely new sales force, because the products are too complex to add to the list of products the bankers are already selling.

Attracting new customers to a market also can create substantial value. Consumer packaged-goods company Beiersdorf accelerated growth in sales of skin-care products by convincing men to use its Nivea products. Once again, competitors didn't retaliate because they also gained from the category expansion. Men's skin-care products aren't much different from women's, so much of the research and development, manufacturing, and distribution cost could be shared. The major incremental cost was for marketing and advertising.

The value a company can create from increasing market share depends on both the market's rate of growth and the way the company goes about gaining share. There are three main ways to grow market share, and these don't fall next to each other in our pecking order shown in Exhibit 9.3. When a company *gains market share in a fast-growing market*, the absolute revenues of its competitors may still be growing strongly, too, so the competitors may not retaliate. However, gaining share in a mature market is more likely to provoke retaliation by competitors.

Gaining share from *incremental innovation*—for example, through incremental technology improvements that neither fundamentally change a product nor create an entirely new category and that are possible to copy—won't create much value or maintain the advantage for long. From a customer's viewpoint, hybrid and electric vehicles aren't fundamentally different from gas or diesel vehicles, so they cannot command much of a price premium to offset their higher costs. The total number of vehicles sold will not increase, and if one company gains market share for a while, competitors will try to take it back, as competitors can copy each other's innovations before the innovator has been able to extract much value, if any. All in all, auto companies, whether new or incumbent, may not create much value from hybrid or electric vehicles; competition will likely transfer most benefits to consumers.

Gaining share through *product pricing and promotion* in a mature market rarely creates much value, if any. Huggies and Pampers dominate the disposable-diaper market and are financially strong, and each can easily respond if the other tries to gain share. Therefore, any growth arising from, say, an intense campaign to reduce prices that hits directly at the other competitor will provoke a response. And as Amazon continued expanding into the U.S.

consumer electronics retail market in 2009, Walmart reduced prices on key products such as top-selling video games and game consoles, even though Amazon's \$20 billion in sales in 2008 were a fraction of Walmart's \$406 billion sales in the same year. Although Walmart's competitive reaction could not stop Amazon from surpassing Walmart as the largest U.S. electronics retailer by 2014, it drove down margins across the segment and rewrote the competitive dynamics of the electronics category.

In concentrated markets, share battles often lead to a cycle of market share give-and-take but rarely to a permanent share gain for any one competitor, unless that competitor changes the product or its delivery enough to create what is effectively a new product. The possible exception, as with the Amazon example in the preceding paragraph, is stronger companies gaining share from smaller, weaker competitors and forcing the weaker players out of the market entirely.

Price increases, over and above cost increases, can create value as long as any resulting decline in sales volume is small. However, they tend not to be repeatable: if a company or several competitors get away with a price increase one year, they are unlikely to have the same good fortune the next. Furthermore, the first increase could be eroded fairly quickly. Otherwise, you would see some companies increasing their profit margins year after year, while in reality, long-term increases in profit margins are rare. There was an exception among packaged-goods companies in the mid-1990s. They passed on increases in commodity costs to customers but did not lower prices when their commodity costs subsequently declined. But the prospect of higher margins made it more attractive for retailers to enter the packaged-goods segments with offerings of private-label brands, sometimes via online sales channels.

There are two main approaches to growing through acquisitions. Growth through *bolt-on acquisitions* can create value if the premium paid for the target is not too high. Bolt-on acquisitions make incremental changes to a business model—for example, by completing or extending a company's product offering or filling gaps in its distribution system. In the 2000s, IBM was very successful in bolting on smaller software companies and subsequently marketing their applications through its existing global sales and distribution system, which could absorb the additional sales without too much extra investment. Because such acquisitions are relatively small, they boosted IBM's growth but added little cost and complexity.

In contrast, creating growth through *large acquisitions*—say, one-third the size or more of the acquiring company—tends to create less value. Large acquisitions typically occur when a market has begun to mature and the industry has excess capacity. While the acquiring company shows revenue growth, the combined revenues often do not increase, and sometimes they decrease because customers prefer to have multiple suppliers. Any new value comes

primarily from cost cutting, not from growth. Furthermore, integrating the two companies requires significant investments and involves far more complexity and risk than integrating small, bolt-on acquisitions.

Choosing a Growth Strategy

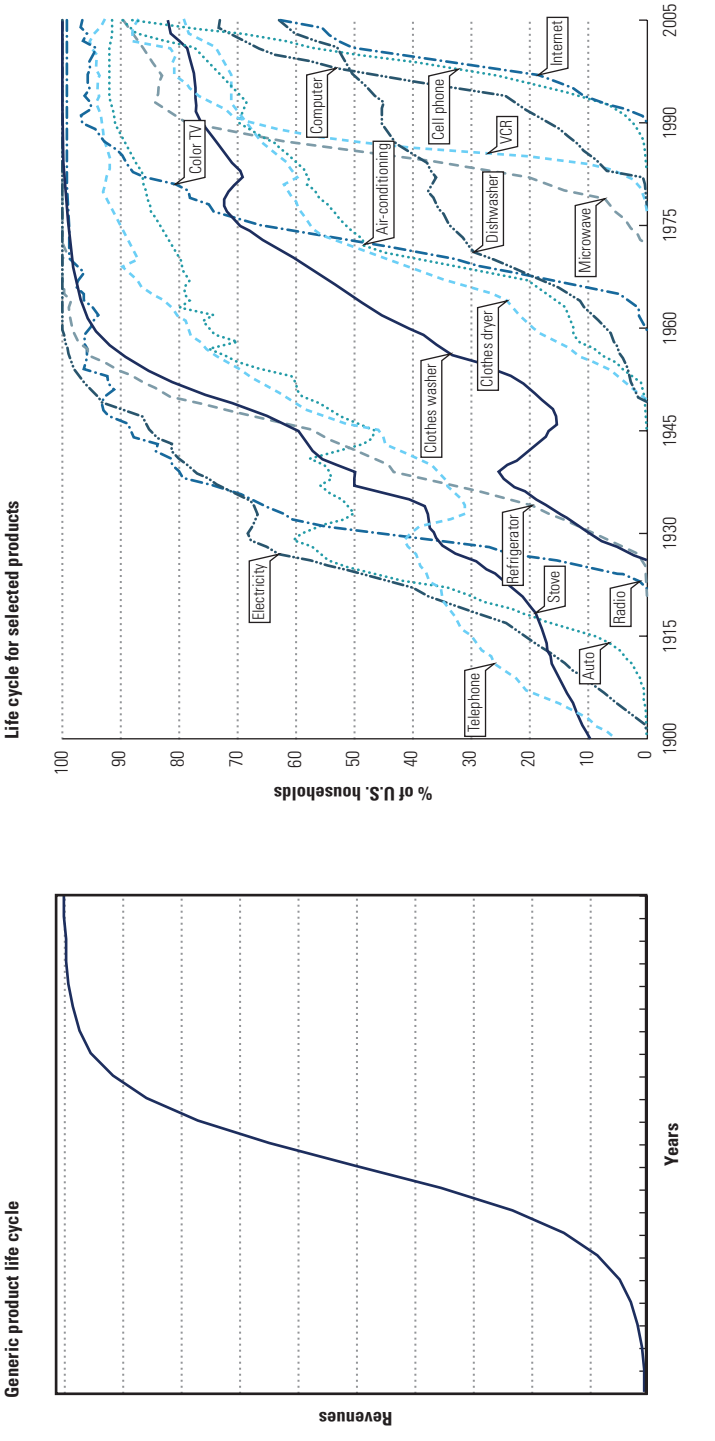
The logic explaining why growth from product market expansion creates greater and more sustainable value than growth from taking share is compelling. Nevertheless, the dividing line between the two types of growth can be fuzzy. For instance, some innovations prevent existing competitors from retaliating, even though the innovator's products and services may not appear to be that new. Walmart's innovative approach to retailing in the 1960s and 1970s offered an entirely new shopping experience to its customers, who flocked to the company's stores. One could argue that Walmart was merely taking share away from small local stores. But the fact that its competitors could not retaliate suggests that Walmart's approach constituted a truly innovative product. However, if Walmart were to grow by winning customers from Target, that would count as market share gain, because Target and Walmart offer their retailing product in a similar fashion. Notably, over the past decade, Walmart itself has been facing competition from innovative offerings in direct and platform sales by Amazon, which has taken significant market share from Walmart in many retail categories.

In general, underlying product market growth tends to create the most value. Companies should aim to be in the fastest-growing product markets, so they can achieve growth that consistently creates value. If a company is in the wrong markets and can't easily get into the right ones, it may do better by sustaining growth at the same level as its competitors while finding ways to improve and sustain its ROIC.

WHY SUSTAINING GROWTH IS HARD

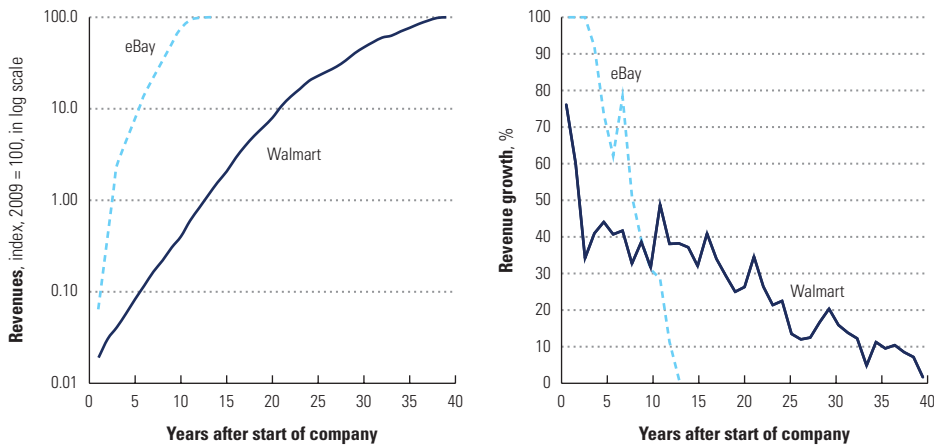
Sustaining high growth is much more difficult than sustaining ROIC, especially for larger companies. The math is simple. Suppose your core product markets are growing at the rate of the gross domestic product (GDP)—say, 5 percent nominal growth—and you currently have \$10 billion in revenues. Ten years from now, assuming you grow at 5 percent a year, your revenues will be \$16.3 billion. Assume you aspire to grow organically at 8 percent a year. In ten years, your revenues will need to be \$21.6 billion. Therefore, you will need to find new sources of revenues that can grow to more than \$5.3 billion per year by the tenth year. Adjusting for inflation of 1 to 2 percent, you need an extra \$4.3 billion to \$4.8 billion per year in today's dollars. Another way to think of it is that to find such revenues, you would need to reinvent a

EXHIBIT 9.4 **Variation in Growth over Product Life Cycle**



Source: W. Cox and R. Alm, "You Are What You Spend," *New York Times*, February 10, 2008.

EXHIBIT 9.5 Walmart and eBay: Growth Trajectories



business almost half your current size and close to a Fortune 500 company.⁷ If your product markets are growing at only 5 percent, how can you possibly achieve that magnitude of growth?

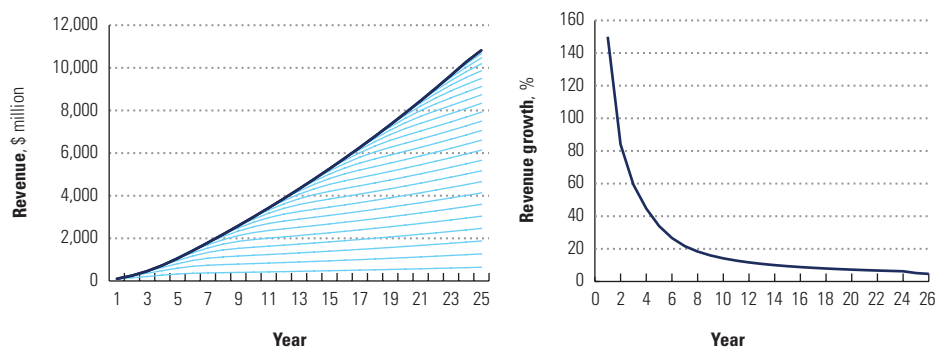
Given this difficulty, the growth targets that some companies embrace are simply unrealistic. One with sales already in excess of \$5 billion announced organic growth targets of more than 20 percent a year for the next 20 years. Since annual world economic growth is typically less than 4 percent in real terms and many companies are competing for a share of that growth, such growth targets are hardly achievable.

Sustaining growth is difficult because most product markets have natural life cycles. The market for a product—which means the market for a narrow product category sold to a specific customer segment in a specific geography—typically follows an S-curve over its life cycle until maturity, as shown on the left side of Exhibit 9.4. The right side shows the growth curves for various real products, scaled to their relative penetration of U.S. households. First, a product has to prove itself with early adopters. Growth then accelerates as more people want to buy the product, until it reaches its point of maximum penetration. After this point of maturity, and depending on the nature of the product, either sales growth falls back to the same rate of growth as the population or the economy, or sales may start to shrink. To illustrate, autos and packaged snacks have continued to grow in line with economic growth for half a century or more, while videocassette recorders (VCRs) lasted less than 20 years before they started to decline and then disappeared.

While the pattern of growth is usually the same for every product and service, the amount and pace of growth will vary for each one. Exhibit 9.5

⁷ The cutoff point for the Fortune 500 in terms of revenues was around \$5.5 billion in 2018.

EXHIBIT 9.6 The Challenge of Sustaining High Growth



compares Walmart and eBay. While both have some activities outside their core business, they are largely one-product companies. Walmart's growth did not dip below 10 percent until the end of the 1990s, some 35 years after it was founded. In contrast, eBay saw its growth fall to below 10 percent after only 12 years, having grown very rapidly to reach maturity early. Because eBay is an Internet-based auction house, it doesn't need to add many more staff members in order to grow. In contrast, Walmart, as a physical retailer, has to add people as quickly as it adds stores and sales. The speed at which Walmart can hire and train people limits its rate of growth relative to eBay. But Walmart's core market is much larger than eBay's. In 2018, Walmart generated \$500 billion of revenues, mostly from its core discount and supercenter stores, whereas eBay generated only about \$10 billion of revenues because its core addressable market is so much smaller.⁸

Sustaining high growth presents major challenges to companies. Given the natural life cycle of products, the only way to achieve consistently high growth is to consistently find new product markets and enter them successfully in time to enjoy their more profitable high-growth phase. Exhibit 9.6 illustrates this by showing the cumulative sales for a company that introduces one new product in one market (geographic or customer segment) in each year. All products are identical in terms of sales volume and growth; their growth rates are very high in the beginning and eventually slow to 3 percent once the market is fully penetrated. Although the company continues to launch new products that are just as successful as their predecessors, aggregate sales growth slows down rapidly as the company gets bigger. In the long term, growth approaches 3 percent, equal to the long-term growth rate of the markets for the company's products. Ultimately, a company's growth and size are constrained by the growth and size of its product markets and the number of product markets in which it competes.

⁸ The comparison is somewhat distorted: eBay spun off its subsidiary PayPal in 2012, reducing its revenues by \$6 billion from \$14 billion at that time.

To sustain high growth, companies need to overcome this “portfolio treadmill” effect: for each product that matures and declines in revenues, the company needs to find a similar-size replacement product to stay level in revenues—and even more to continue growing. Think of the pharmaceutical industry, which showed unprecedented growth from the mid-1990s, thanks to so-called blockbuster drugs such as Lipitor and Celebrex. Then growth plummeted as these drugs came off patent and the next generation of drugs didn’t deliver the same outsize sales as the blockbusters. Finding sizable new sources of growth requires more experimentation and a longer time horizon than many companies are willing to invest in. Royal Philips’s health technology business was a small corporate division in 1998, when it generated around 7 percent of total company revenues. It took 15 years of ongoing investments and acquisitions to become Philips’s largest business unit, generating half of its total revenues. After the carve-out of its lighting business and other divestitures, health technology has now become Philips’s core business.

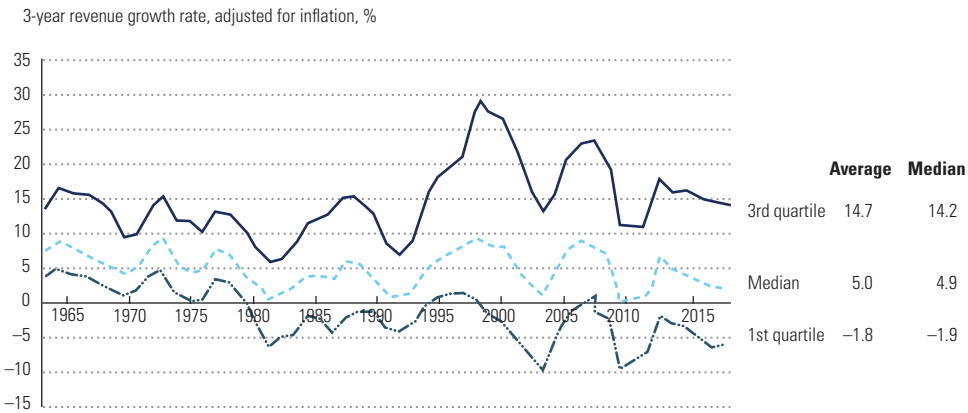
EMPIRICAL ANALYSIS OF CORPORATE GROWTH

The empirical research backs up the principles we have been discussing. This section presents our findings on the level and persistence of corporate growth for U.S.-based nonfinancial companies with revenues greater than \$1 billion (inflation-adjusted) from 1963 to 2017. (The sample size for each year is different but amounts to 1,095 companies in 2017.) The analysis of their revenue growth follows the same procedure as the analysis of ROIC data in Chapter 8, except here we use three-year rolling averages to moderate distortions caused by currency fluctuations and M&A activity. We also use real, rather than nominal, data to analyze all corporate growth results, because even mature companies saw a dramatic increase in revenues during the 1970s as inflation increased prices. Ideally, we would report statistics on *organic* revenue growth, but current reporting standards do not require companies to disclose the effects of currencies and M&A on their revenues.

The overall findings concerning revenue growth are as follows:

- The median rate of revenue growth between 1965 and 2017 was 4.9 percent in real (inflation-adjusted) terms. Real revenue growth fluctuated significantly, ranging from around 0 percent to 9 percent, with significant cyclicalities.
- High growth rates decayed very quickly. Companies growing faster than 20 percent in real terms typically grew at only 8 percent within five years and at 5 percent within ten years.

EXHIBIT 9.7 Long-Term Revenue Growth for Nonfinancial Companies, 1965–2017



Source: Compustat; Corporate Performance Analytics by McKinsey.

Growth Trends

Let's begin by examining aggregate levels and trends of corporate growth. Exhibit 9.7 presents median revenue growth rates in real terms between 1965 and 2017. The average median revenue growth rate for that period equals 4.9 percent per year and oscillates between roughly 0 percent and 9 percent. Median revenue growth demonstrates no trend over time, but over the past five years, growth rates declined to around 2 percent in real terms.

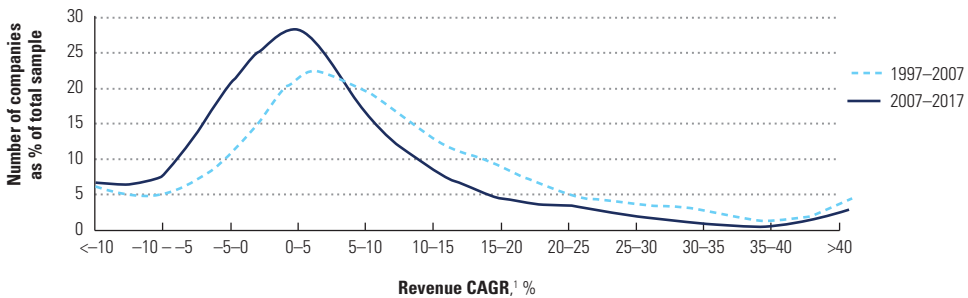
Real revenue growth of 4.9 percent is quite high when compared with real GDP growth in the United States, which was at 3.0 percent during the same period. Why the difference? Possible explanations abound. The first is self-selection: companies with good growth opportunities need capital to grow. Since public markets are large and liquid, high-growth companies are more likely to be publicly traded than privately held ones. We measure only publicly traded companies, so these growth results are likely to be higher.

Second, as companies become increasingly specialized and outsource more services, firms providing services will grow and develop quickly without affecting the GDP figures. Consider Jabil Circuit, a contract electronics manufacturer. When a company like Apple or IBM has Jabil manufacture products or components on its behalf, GDP, which measures aggregate output, will not change. Yet Jabil's growth will influence our sample.

A third explanation is global expansion. Many of the companies in the sample create products and generate revenue outside the United States, so they can grow faster than U.S. GDP without gaining sales in the United States. Finally, although we use rolling averages and medians, these cannot eliminate but only dampen the effects of M&A and currency fluctuations, which do not reflect organic growth.

EXHIBIT 9.8 Distribution of Growth Rates

Revenue growth rate, inflation-adjusted

¹ Compound annual growth rate.

Source: Compustat; Corporate Performance Analytics by McKinsey.

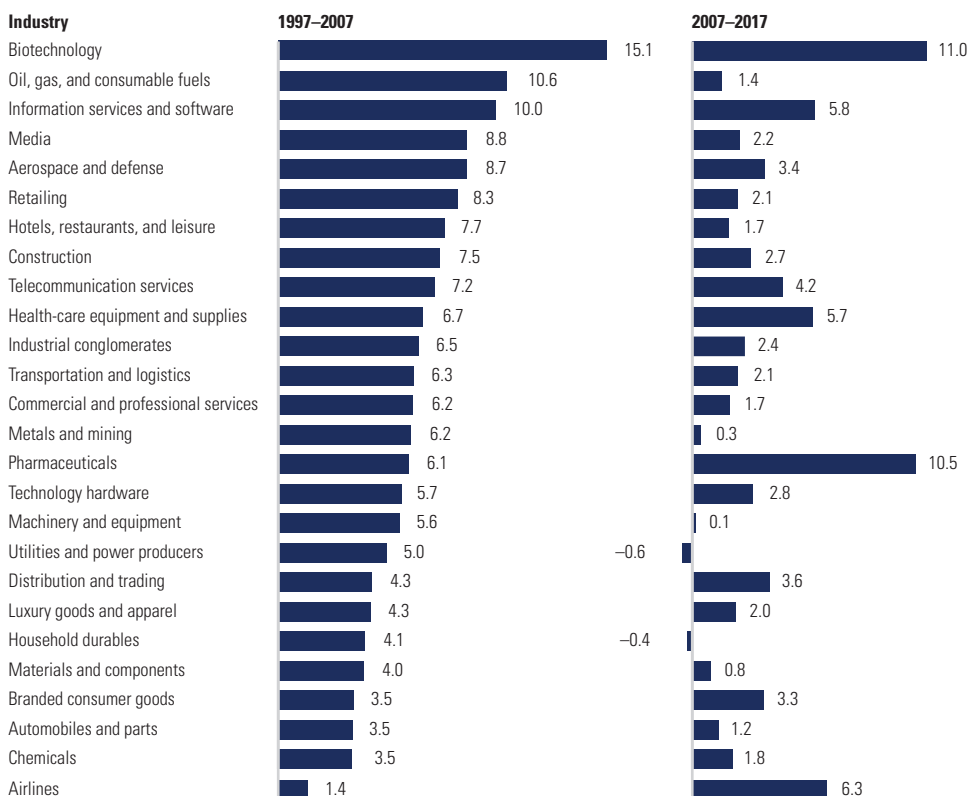
In addition to mapping median growth, Exhibit 9.7 also reveals that from the mid-1970s to 2017, at least one-quarter of all companies shrank in real terms almost every year. Thus, although most companies project healthy growth over the next years in their public communications or even analyst guidance, the reality is that many mature firms will shrink. This underlines the need to exercise caution before projecting strong growth for a valuation, especially for large companies in mature sectors.

Exhibit 9.8 shows the distribution of three-year real revenue growth for two periods, 1997–2007 (before the 2008 financial crisis) and 2007–2017. Not surprisingly, the distribution became wider and shifted to the left in the latter period. From 2007 to 2017, almost two-thirds of companies in the sample grew at an annual real rate of less than 5 percent. Only 21 percent grew faster than 10 percent. (This includes the effect of acquisitions, so fewer companies grew faster than 10 percent just through organic growth.)

Growth across Industries

As Exhibit 9.1 illustrated, growth rates vary widely across and within industries. In addition—unlike ROIC, where the industry ranking tends to be stable—the industry growth ranking varies significantly over time, as shown in Exhibit 9.9 for the decades 1997–2007 and 2007–2017. Some of the variation is explained by structural factors, such as the saturation of markets (the declining growth in hotels and restaurants and in chemicals) or the effect of technological innovation in creating entirely new markets (the strong growth in biotechnology and information services). In other cases, growth is more cyclical. Growth in the oil and gas sector varied from more than 10 percent in the first decade to just 1 percent over the past ten years, as oil prices plummeted after 2014. Similarly, the construction industry is subject to cycles, with growth

EXHIBIT 9.9 Volatile Growth by Industry

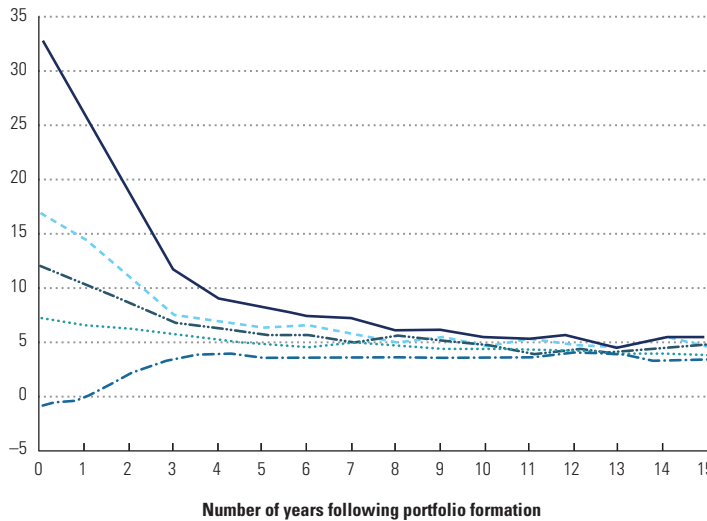
10-year revenue growth rate,¹ industry median adjusted for inflation, %¹ Compound annual growth rate.

Source: Compustat; Corporate Performance Analytics by McKinsey.

at much lower levels since the 2008 credit crisis. Telecommunications service providers enjoyed a burst of growth in the 2000s, when mobile phones became ubiquitous. But revenue growth rates over the past decade ended significantly lower due to strong price pressure.

Despite this high degree of variation, some sectors have consistently been among the fastest growing, not only during the 30 years covered in this sample, but also for earlier periods. These include life sciences and technology, such as information services and software, technology hardware, pharmaceuticals, biotechnology, and health care, where demand has remained strong for three decades. Others, such as automobile parts, chemicals, and branded consumer goods, have consistently registered lower growth rates, as their markets had already reached maturity well before the 1990s.

EXHIBIT 9.10 Revenue Growth Decay Analysis

Median growth of portfolios, by quintile,¹ %¹ At year 0, companies are grouped into one of five portfolios, based on revenue growth.

Source: Compustat; Corporate Performance Analytics by McKinsey.

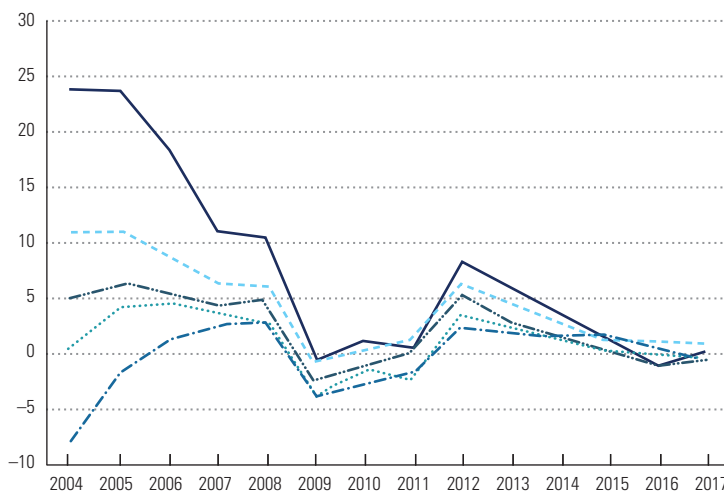
Sustaining Growth

Understanding a company's potential for growing revenues in the future is critical to valuation and strategy assessment. Yet developing reasonable projections is a challenge, especially given the upward bias in growth expectations demonstrated by equity research analysts and the media. Research shows that analyst forecasts of one-year-out aggregate earnings growth for the S&P 500 are systematically overoptimistic, exceeding actual earnings growth by five percentage points or more.⁹

To put long-term corporate growth rates in their proper perspective, we analyzed historical rates of growth decay since 1963. Companies were segmented into five portfolios, depending on their growth rate in the year the portfolio was formed. Exhibit 9.10 plots how each portfolio's median company grows over time. As the exhibit shows, growth decays very quickly; high growth is not sustainable for the typical company. Within three years, the difference across portfolios narrows considerably, and by year 5, the highest-growth portfolio outperforms the lowest-growth portfolio by less than five percentage points. Within ten years, this difference drops to less than two percentage points.

⁹ See, for example, M. Goedhart, B. Raj, and A. Saxena, "Equity Analysts: Still Too Bullish," *McKinsey on Finance*, no. 35 (Spring 2010): 14–17.

EXHIBIT 9.11 Revenue Growth Decay through Economic Crisis and Recovery

Median growth of portfolios,¹ %¹ As of 2002, companies are grouped into one of five portfolios, based on their 2002–2004 revenue growth.

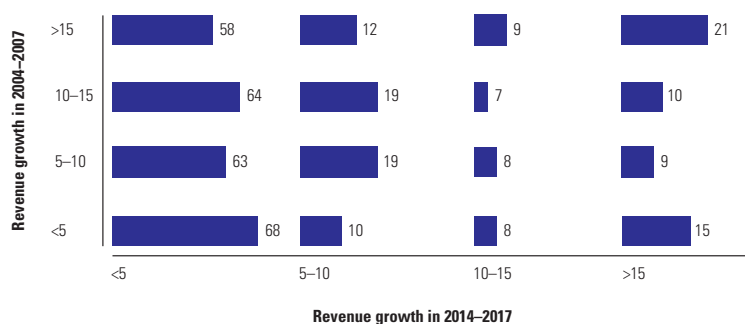
Source: Compustat; Corporate Performance Analytics by McKinsey.

We also analyzed the decay rates for the most recent 15 years and found similar patterns of rapid convergence to 5 percent and lower (Exhibit 9.11). Note how the 2008 credit crisis caused a temporary decline of growth rates overall but without changing the typical decay pattern from the long-term data in Exhibit 9.10. Comparing the decay of growth to that of ROIC shown in the previous chapter, it is possible to see that although companies' rates of return on invested capital generally remain fairly stable over time—top companies still outperform bottom companies by more than ten percentage points after 15 years—rates of growth do not.

As discussed earlier in this chapter, companies struggle to maintain high growth because product life cycles are finite and growing becomes more difficult as companies get bigger. Do any companies counter this norm? The short answer: very few. Exhibit 9.12 shows what happened to the growth rates of companies grouped by their 2004–2007 growth rates. Reading across each row, the percentages indicate the share of companies in each group that fell into each of the growth categories one decade later. Clearly, maintaining high growth is much less common than being stuck with slow growth. Of the companies reporting less than 5 percent revenue growth from 2004 to 2007, 68 percent continued to report growth below 5 percent ten years later. In contrast, only 21 percent of high-growth companies maintained better than 15 percent real growth ten years later. Even more concerning for high-growth companies, 58 percent of the companies that grew faster than 15 percent from 2004 to 2007 were growing at real rates below 5 percent a decade later. Sustaining high growth is very difficult—much more difficult than sustaining high ROIC.

EXHIBIT 9.12 Revenue Growth Transition Probability

% Growth rate, given rate in 2004–2007



SUMMARY

To maximize value for their shareholders, companies should understand what drives growth and how it creates value. For large companies, the growth of the markets in which they operate largely drives long-term revenue growth. Although gains in market share contribute to revenues in the short term, these gains are far less important for long-term growth.

Revenue growth is not all that matters for creating value; the value created per dollar of additional revenues is the crucial point. In general, this depends on how easily competitors can respond to a company's growth strategy. The growth strategy with the highest potential in this respect is true product innovation, because entirely new product categories by definition have no established competition. Attracting new customers to an existing product or persuading existing customers to buy more of it also can create substantial value, because direct competitors in the same market tend to benefit as well. Growth through bolt-on acquisitions can add value, because such acquisitions can boost revenue growth at little additional cost and complexity. Typically, revenue growth from market share gains is much less attractive, because it comes at the expense of established direct competitors, who are likely to retaliate, especially in maturing markets.

Sustaining high growth is no less a challenge than initiating it. Because most products have natural life cycles, the only way to achieve lasting high growth is to continue introducing new products at an increasing rate—which is nearly impossible. Not surprisingly, growth rates for large companies decay much faster than do returns on invested capital; growth rates for even the fastest-growing companies tend to fall below 5 percent within ten years.

REVIEW QUESTIONS

1. Discuss the three generic sources of a company's growth, their relative importance for its growth, and their implications for a company's strategy.
2. For which type of company is additional growth likely to create more value: a high-ROIC company in a mature market or a low-ROIC company in a fast-growing market? Give reasons for your answer.
3. Why could growth through a series of bolt-on acquisitions create more value than growth through a single large acquisition? (Consider premium paid and synergies created for each individual transaction.)
4. Identify and discuss an example where growth in market share through a price war created long-term value for a company.
5. Why do fast-growing companies typically fail to sustain their high- growth rates?
6. Why do company growth rates typically converge much more quickly toward the average rate across all companies than their rates of ROIC do, given that both ultimately depend on the underlying product life cycles?
7. Discuss why the ranking of industries by growth varies more over time than their ranking by ROIC.
8. If growth from gaining market share through product promotion and pricing rarely creates much value, why do most consumer goods companies put so much effort into them?
9. How did the 2008 financial crisis impact growth rates for U.S. corporations during the crisis and post-crisis?